



MFU Food Guide and Review Application

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**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2025

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MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

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Abstract

This senior project presents the development of an interactive application designed to help MFU University students explore food options and make informed dining decisions. The application features two main components: Food Exploration and Review Analysis. Food Exploration allows users to discover dining choices through sorting by food type and location, while Review Analysis enables users to filter and sort reviews based on criteria such as ratings. With an intuitive interface and efficient filtering and search capabilities, the application allows users to seamlessly navigate and identify the most suitable dining options in a convenient and user-friendly manner.

Keywords: Food Recommendation System, User Reviews, Mobile Application, User Interface, Filtering, and Searching

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Food and beverage selection is an integral part of the university experience because dining is more than just satisfying hunger it's about fostering connections, exploring new cultures, and enhancing daily life in a vibrant academic environment. However, navigating the diverse dining options available at the University can be challenging.

This study proposes the development of an interactive application aimed at enhancing the dining experience for Mae Fah Luang University students. The application will feature user-friendly tools designed to help users efficiently select dining options. From our experience, choosing where to dine was sometimes more challenging than studying in class. Therefore, we decided to develop this application with the goal of helping users spend less time deciding what to eat. By seamlessly integrating these features, the application empowers users to make informed dining decisions, ultimately enriching their overall university dining experience.

1.2 Objective

- 1.2.1 To develop a mobile application for restaurant recommendations within Mae Fah Luang University.
- 1.2.2 To provide a platform that enhances the dining experience for students and staff at Mae Fah Luang University.
- 1.2.3 To help users easily find the food they want and allow university restaurants to receive valuable customer feedback to improve their services according to customer needs.

1.3 Scope

- 1.3.1 Develop a mobile application that will improve the dining experience for students and staff at Mae Fah Luang University.
- 1.3.2 Users are divided into three roles: Guest users, Registered users, and Admin
- 1.3.3 System features
 - 1.3.3.1 Guest users (Users that didn't log in)
 - Home page (Show all restaurants).
 - Sort restaurants with filters.
 - 1. Rating(Overall,Hygiene,Flavor,Service)
 - 2. Locations(E2, E1, D1, etc)
 - 3. Food type (Main dish, Dessert, Snack, Drinks)
 - View restaurant information.

- Name
- Address
- Operating hours
- Menu with prices
- Photos
- Contract
- View comment and overall rating.
- View the location of restaurants. (Google map)
- View leaderboard(Top 5 highest overall rating restaurants and 5 most liked in user comments).

1.3.3.2 MFU Students/Staff (Lamduan Email)

- Login/Log-out (Using @lamduan.mfu.ac.th and @mfu.ac.th)
- Manage Profile
- Home page (Show all restaurants).
 - Sort restaurants with filters.
 1. Rating(Overall, Hygiene, Flavor, Service)
 2. Locations(E2, E1, D1, etc)
 3. Food type (Main dish, Dessert, Snack, Drinks)
 - View restaurant information.
 - Name
 - Address
 - Operating hours
 - Menu with prices
 - Photos
 - Contract
 - locations of restaurants.
 - View comment and overall rating.
 - Give comments to restaurants.
 - Give a rating to restaurants.
 - Hygiene rating
 - Flavor rating
 - Service rating
- My History Page
 - Display all previous ratings and comments that the user has left for each restaurant.
- My Threads History Page
 - Display all previous Threads that the user created or participated in.
- View Leaderboard (Top 3 highest rating restaurants and 3 most liked in user comments).
- AI chat bot
 - Atlas (Generative AI) can ask general questions.
 - Nexus (Chat Bot AI) can ask for information in an

application and can ask for your own statistics.

- Participate in the Threads Community to discuss food-related topics
 - Post messages and interact with other users.
 - AI moderates messages in real-time to detect inappropriate content.
- Coin Exchange page.
- User Overview page.

1.3.3.3 Administrators

- Login/Logout
- Manage Profile
- Home page (Show all restaurants).
 - Sort restaurants with filters.
 1. Rating(Overall, Hygiene, Flavor, Service)
 2. Locations(E2, E1, D1, etc)
 3. Food type (Main dish, Dessert, Snack, Drinks)
 - View restaurant information.
 - Name
 - Address
 - Operating hours
 - Menu with prices
 - Photos
 - Contract
 - locations of restaurants.
 - View comment and overall rating.
 - Give comments to restaurants.
 - Give a rating to restaurants.
 - Hygiene rating
 - Flavor rating
 - Service rating
- Manage restaurant information.
- AI chat bot
 - Atlas (Generative AI) can ask general questions.
 - Nexus (Chat Bot AI) can ask for information in an application and can ask for your own statistics.
- Verify Threads Community by filtering out inappropriate language.
- View leaderboard (Top 3 highest rating restaurants and 3 most liked in user comments).
- Verify the user's reviews in case of inappropriate reviews.
- Manage users' accounts (Suspend or Ban).
- Coin Exchange Management page

1.4 Methodology

- 1.4.1 Requirement gathering and analysis.
- 1.4.2 Literature review
- 1.4.3 System analysis
- 1.4.4 Database Design
- 1.4.5 UI/UX Design
- 1.4.5 Senior project 1 document preparation
- 1.4.6 Senior Project 1 presentation
- 1.4.7 System development
- 1.4.8 System testing
- 1.4.9 Senior Project 2 Presentation

1.5 Working plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Requirement gathering and analysis.										
Literature review										
System analysis										
Database Design										
UI/UX Design										
Senior project 1 document preparation										
Senior Project 1 presentation										
System development										
System testing										
Senior Project 2 Presentation										

1.6 Expected Result

- 1.6.1 An application that allows users to view reviews and restaurant information within the university.
- 1.6.2 Users will be able to share their reviews with other users.
- 1.6.3 Helps students and staff find their restaurants quickly and easily.
- 1.6.4 Provides a seamless and enjoyable food discovery experience.

1.7 Place and Equipment

1.7.1 Office, Mae Fah Luang University

1.7.2 Software

1.7.2.1 FIGMA [1]

1.7.2.2 LUCIDCHART [2]

1.7.2.3 XAMPP (V3.3.0) [3]

1.7.2.4 VISUAL STUDIO CODE (V1.95.3) [4]

1.7.3 Hardware

1.7.3.1 Spec

- CPU: Intel Core I5-13420H
- GPU: NVIDIA RTX 3060
- Display: Full HD (1920x1080 resolution)

1.7.3.2 Operating system

- Windows 11 (64-bit)

1.7.3.3 Ram

- 16 GB DDR5 4800Mhz

1.7.3.4 Hard disk

- 512 GB SSD PCIe M.2 Gen 4

1.7.4 Programming Languages

1.7.4.1 Dart (Flutter) [5]

1.7.4.2 Node.js [6]

1.7.5 Database

1.7.5.1 MySQL [7]

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

The MFU Food Guide and Review app is designed to help students make informed food choices by leveraging the principles of Social Proof Theory [8]. Influence: The Psychology of Persuasion. New York: Harper Collins.]and Communication Theory [9]. *The Mathematical Theory of Communication*. Urbana: University of Illinois Press. According to Social Proof Theory, individuals often rely on the opinions and actions of others when deciding what to do, especially in uncertain situations [8]. In the app, users can read ratings, reviews, and comments from their peers, which influence their decisions on which food or restaurant to choose. At the same time, Communication Theory emphasizes the exchange of information between senders and receivers, where feedback and interaction facilitate understanding [9]. Through posting and reading reviews, students act as both senders and receivers, sharing experiences, providing recommendations, and creating a dynamic communication network that supports informed decision-making.

2.2 Related Work

Several related studies have explored mobile applications for food reviews and recommendations. Platforms such as "Zomato" [10] "Yelp" [11] and "Wongnai" [12] allow users to share reviews and ratings, providing insights into food quality, service, and ambiance. Wongnai, in particular, is popular in Thailand, focusing on local food reviews and offering features such as detailed menus and convenient search options. This has inspired the development of an application that emphasizes local usability.

However, some existing applications may have limitations in providing information about restaurants within Mae Fah Luang University. The available restaurant data may be incomplete or limited. In contrast, our application provides comprehensive and diverse information on restaurants within the university, effectively meeting the specific needs of students and university staff. This ensures an improved dining experience for users.

Table 2.1 Related Work Comparison

Below is a comparison of the MFU Food Guide and Review Application with existing platforms.

Feature	MFU Food Guide & Review	Yelp	Zomato	Wongnai
University-Focused	✓ Yes	✗ No	✗ No	✗ No
User Reviews & Ratings	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Filtering & Search	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Photo Uploads	✓ Yes	✓ Yes	✓ Yes	✓ Yes
University Restaurant Feedback	✓ Yes	✗ No	✗ No	✗ No
AI-Based Recommendations	✗ No (Future plan)	✓ Yes	✓ Yes	✗ No
Community Chat	✓ Yes	✗ No	✗ No	✓ Yes
Leaderboard	✓ Yes	✗ No	✓ Yes	✓ Yes
Profile Customization	✓ Yes	✓ Yes	✓ Yes	✓ Yes

From the Comparison Table 2.1, you can see that while Yelp[11] , Zomato[10] , and Wongnai[12] provide general food reviews, they don't focus on universities or offer features like Community Chat and University Restaurant Feedback. Our app is built specifically for the Mae Fah Luang University community, making sure all recommendations are relevant to students and staff.

Unlike global platforms, our app is dedicated to MFU restaurants, giving university members a space to share trusted reviews and recommendations without outside distractions. This makes our app more than just a food review platform it's a community hub that helps students and staff connect and share useful information tailored to university life.

2.3 Related Technology

2.3.1 Software

2.3.1.1 FIGMA

Figma [1] is a software tool for designing and creating UX/UI prototypes.

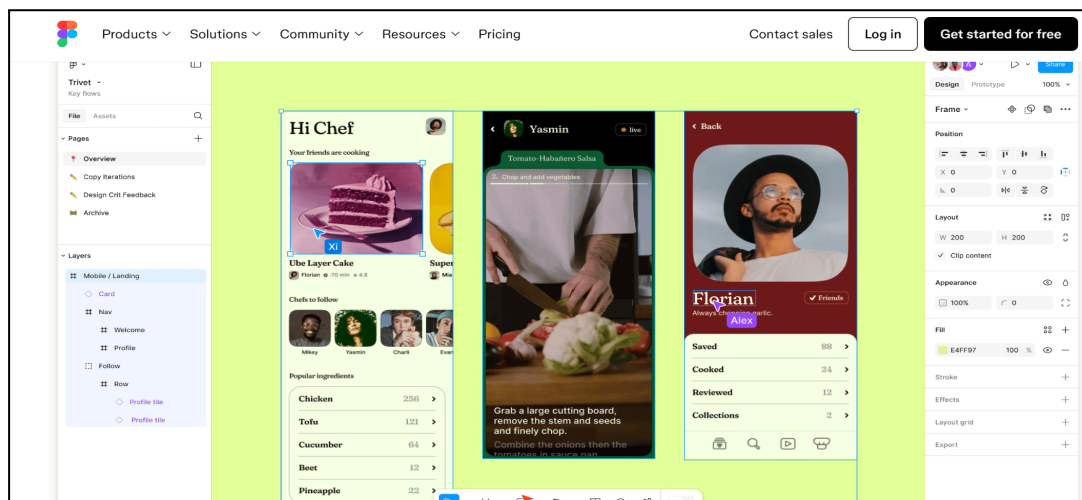


Figure 2.1 Figma

2.3.1.2 LUCIDCHART

Lucidchart [2] is a tool for creating diagrams for system planning, such as Flowcharts and ER Diagrams.

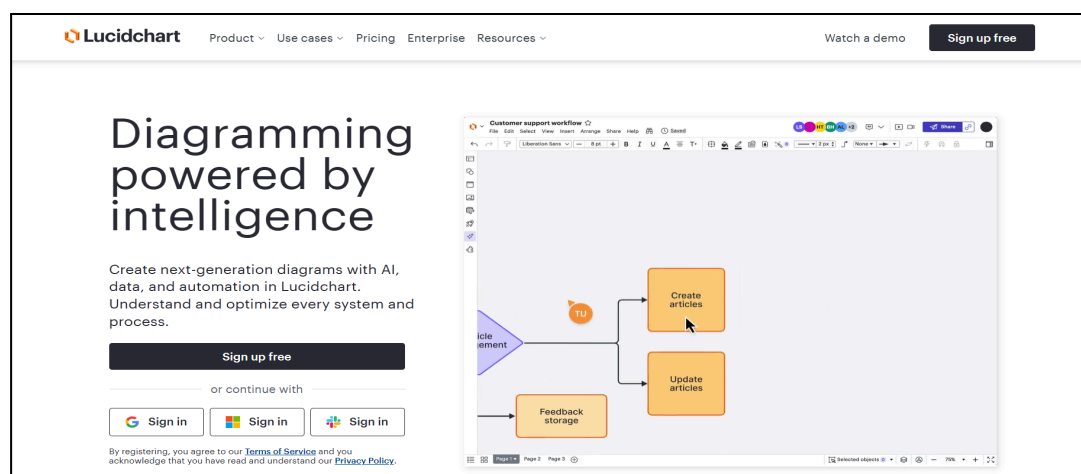


Figure 2.2 Lucidchart

2.3.1.3 XAMPP (v3.3.0)

Xampp [3] runs the server simulation tool

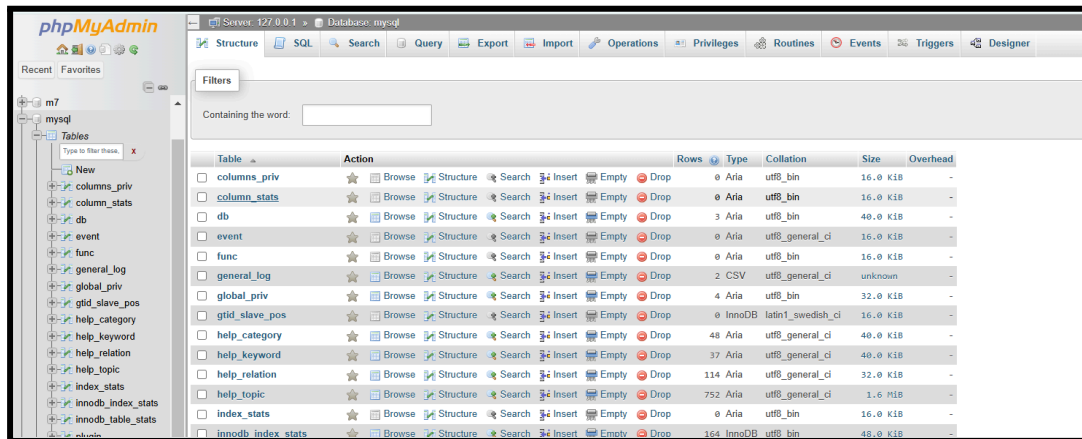


Figure 2.3 Xampp

2.3.1.4 Visual Studio Code (v1.95.3)

Visual Studio Code [4] for a code editor for application development that supports multiple programming languages.

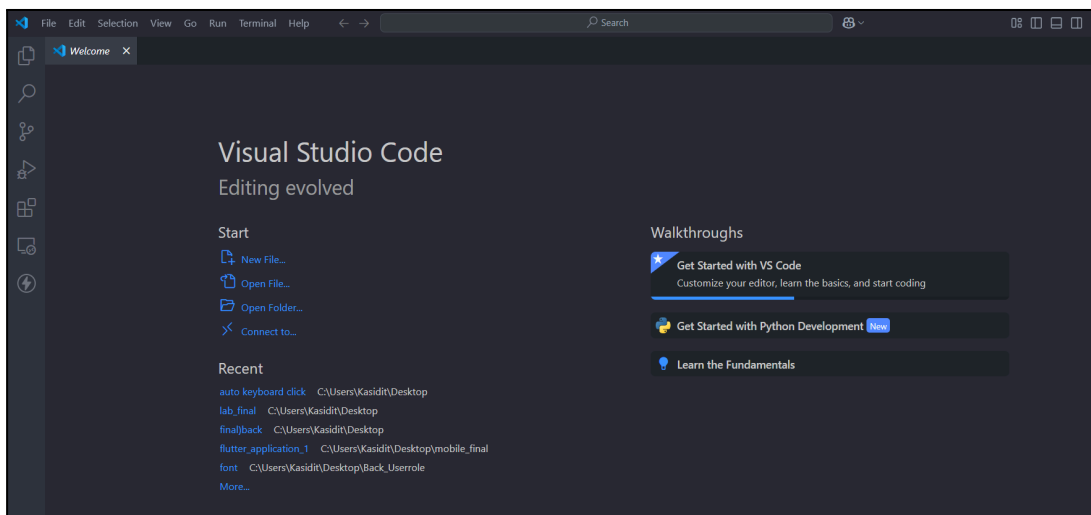


Figure 2.4 Visual Studio Code

2.3.2 Additional Tools for Development

2.3.2.1 Tools for Front-End Development

- **Flutter** [5] : A framework for developing cross-platform mobile applications using Dart.



Figure 2.5 Flutter language

-**Dart DevTools** [5] : A suite of debugging and performance tools for Flutter development.



Figure 2.6 Dart language

2.3.2.2 Tools for Back-End and API Development

- **Postman** [13] : A tool for testing and debugging REST APIs, useful for integrating Flutter with the Node.js backend.



Figure 2.7 Postman

- **Node.js** [6]: The primary runtime environment for building the back-end of the application.



Figure 2.8 Node.js

2.3.2.3 Tools for AI Moderation and Content Safety (Perspective API)

- **Perspective API [14]:** The Perspective API, developed by Google Jigsaw, provides AI-powered text analysis that helps our application detect and filter toxic, inappropriate, or unsafe language to ensure a safer user experience.

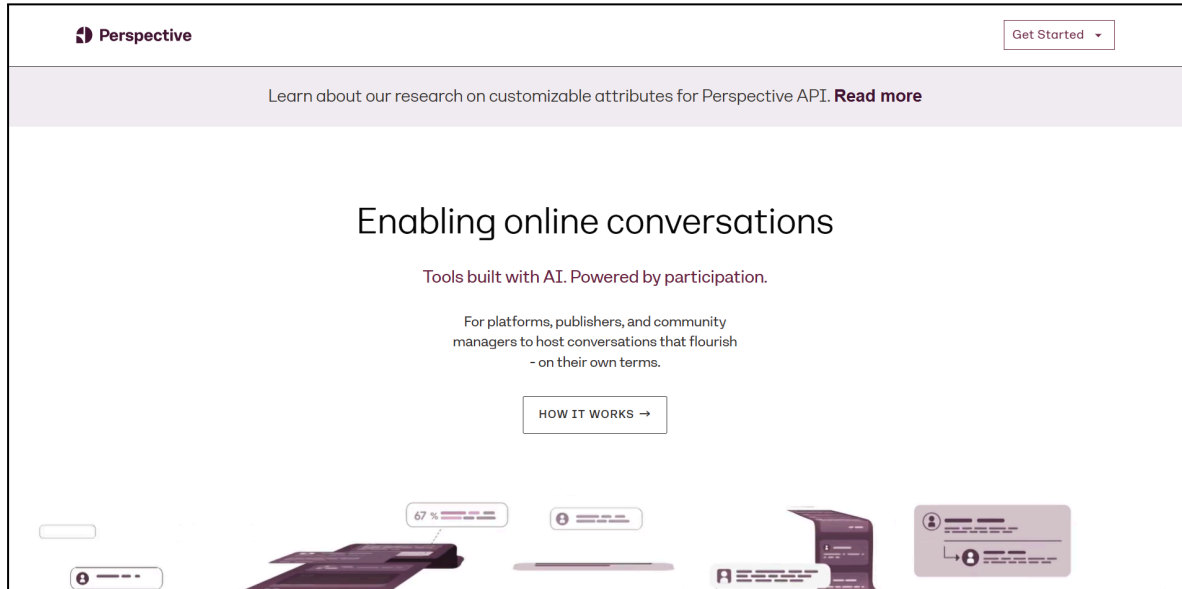


Figure 2.9 Perspective API

2.3.2.4. Tools for AI-Generated Text (Cohere API)

- **Cohere API [15]:** The Cohere API provides the text-generation capabilities used by our Atlas chatbot, enabling it to respond to users smoothly and naturally.

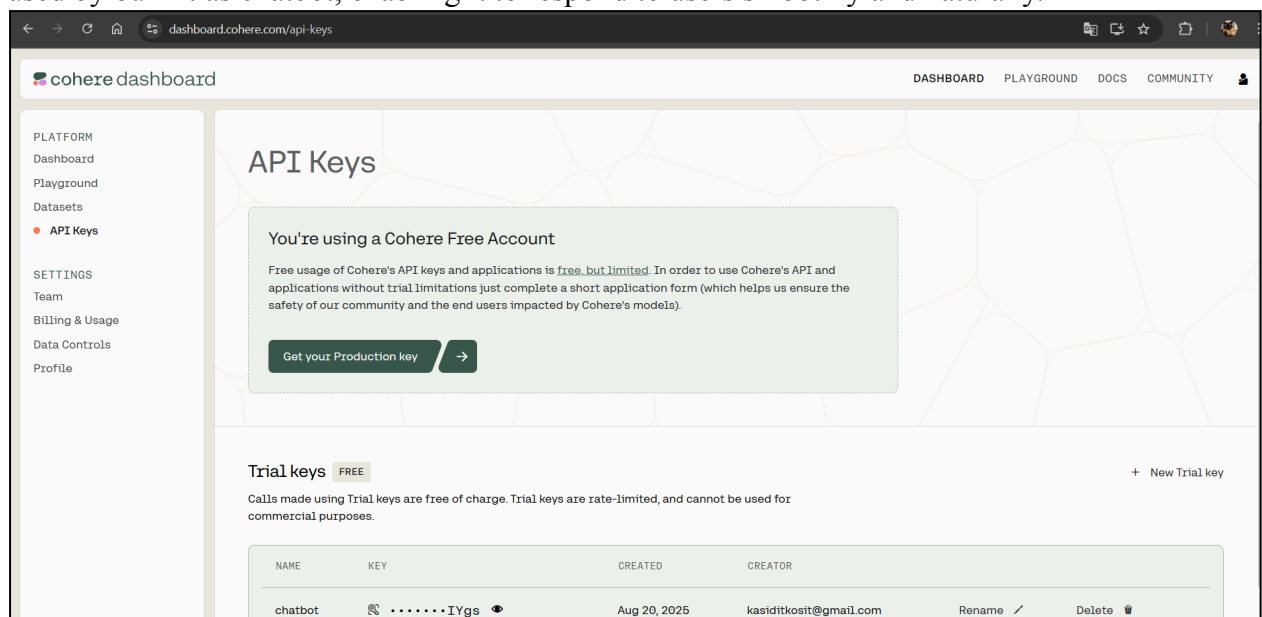


Figure 2.10 Cohere API

CHAPTER 3 ANALYSIS AND DESIGN

3.1 Use case diagram

After gathering user requirements, the new system could be represented by a use case diagram as shown in Figure 3.1

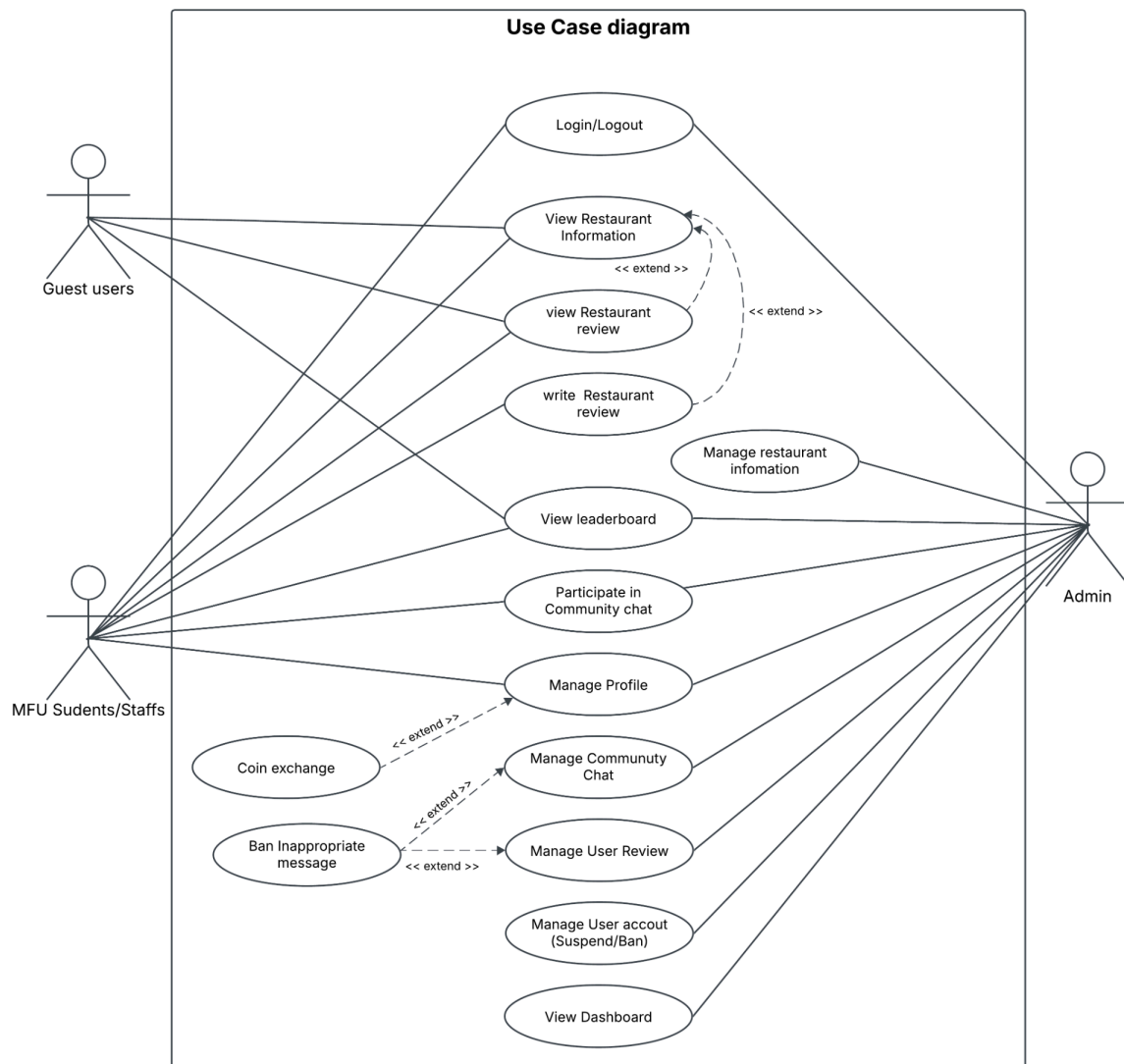


Figure 3.1 Use case diagram of the system

3.2 ER Diagram

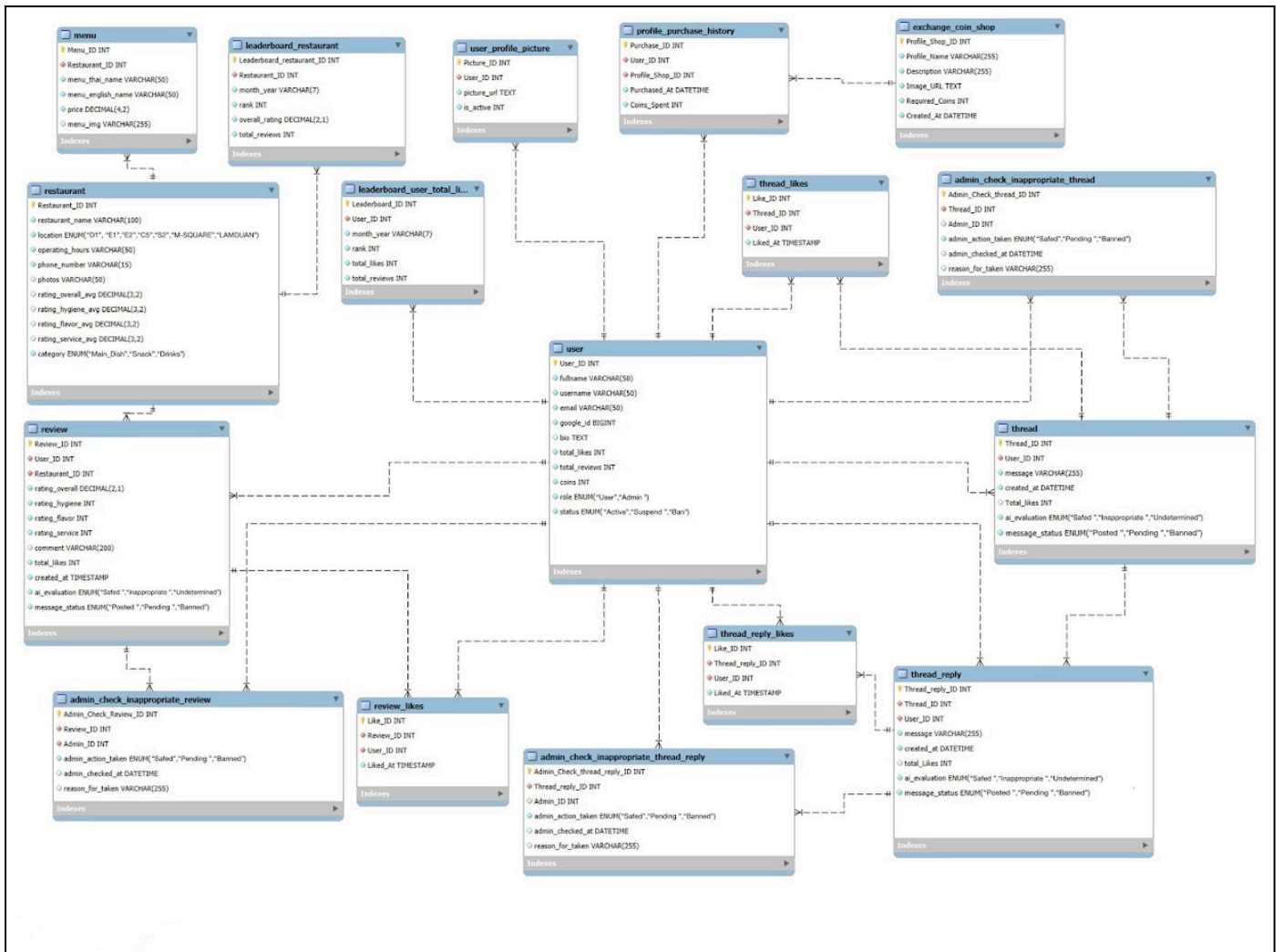


Figure 3.2 ER Diagram

3.3 Data Dictionary

3.3.1 User Table

Table 3.1 Data Dictionary of User Table

Field name	Data Type	Description	Nullable	Key	Example Data
User_ID	INT	Unique identifier for each user.	NO	PK	100001
fullname	VARCHAR (50)	Unique Full name for authentication.	No	-	'Kasidit Kosit'
username	VARCHAR (50)	Username for each User.	NO	-	'Kasidit01'
email	VARCHAR (50)	User's email address for each user.	NO	-	'6531501005@lamduan.mfu.ac.th'
google_id	BIGINT	Google User ID for each user.	NO	-	100000000000001
bio	TEXT	A short personal description or bio provided by the user.	yes	-	'Food lover'
total_likes	INT	Total number of likes the user has received on all their reviews.	NO (default : 0)	-	32
total_reviews	INT	Total number of reviews the user has posted.	NO (default : 0)	-	3
coins	INT	The number of coins the user earns from the leaderboard ranking	NO (default : 0)	-	5
role	ENUM ('User', 'Admin')	Defines the user type.	NO (default: 'User')	-	'User'
status	ENUM ('Active', 'Suspended', 'Ban')	Account status.	NO (default: 'Active')	-	'Active'

3.3.2 Profile_picture_user Table

Table 3.2 Data Dictionary of Profile_picture_user Table

Field name	Data Type	Description	Nullable	KEY	Example Data
Picture_ID	INT	Unique identifier for each profile record.	NO	PK	200001
User_ID	INT	Unique identifier for each User record.	NO	FK	100001
picture_url	TEXT	URL of each picture that is used on the profile based on their User_ID	NO	-	'profile_pic/user/user_10001_1.jpg'
is_active	INT	The number shows the active status on the picture that User_ID has and uses. 1 for activation on the profile. 0 means have but not in use.	NO (default :0)	-	1

3.3.3 Restaurant Table

Table 3.3 Data Dictionary Restaurant Table

Field name	Data Type	Description	Nullable	Key	Example Data
Restaurant_ID	INT	Unique identifier for each restaurant.	NO	PK	400001
restaurant_name	VARCHAR (100)	Restaurant name.	NO	-	'Ban Somtum'
location	VARCHAR (50)	Restaurant location.	NO	-	'E1'
operating_hours	Enum ('D1','E1','E2','C5','S2','M-SQUARE','LAMDUAN')	Restaurant open and close hours.	NO	-	'10.00-20.00'
phone_number	VARCHAR (15)	Restaurant Contact	Yes	-	0812345678
photos	VARCHAR (50)	Restaurant photo.	Yes	-	'photo1.png'
rating_overall_avg	DECIMAL (1,2)	Average restaurant overall rating.	Yes	-	4.7
rating_hygiene_avg	DECIMAL (1,2)	Average hygiene rating	Yes		
rating_flavor_avg	DECIMAL (1,2)	Average flavor rating	Yes	-	
rating_service_avg	DECIMAL (1,2)	Average service rating	Yes		
category	ENUM ('Main_dish','Snack','Drinks')	Categories of restaurants.	NO	-	main_dish

3.3.4 Menu Table

Table 3.4 Data Dictionary Menu Table

Field name	Data Type	Description	Nullable	Key	Example Data
Menu_id	INT	Unique identifier for each menu.	NO	PK	400001
Restaurant_id	INT	About the restaurant where the menu belongs to.	NO	FK	300001
menu_thai_name	VARCHAR (50)	Thai Name of the menu item.	NO	-	‘ส้มตำไทย’
menu_english_name	VARCHAR (50)	English Name of the menu item.	NO	-	‘Thai Papaya salad’
price	DECIMAL (4,2)	Price of that menu item.	NO	-	45.00
menu_img	VARCHAR (255)	image of menu item.	Yes	-	'somtum.jpg'

3.3.5 Review Table

Table 3.5 Data Dictionary Review Table

Field name	Data Type	Description	Nullable	Key	Example Data
Review_id	INT	Unique ID for each review.	NO	PK	600001
User_ID	INT	ID of the user who posted the review.	NO	FK	100001
Restaurant_id	INT	ID of the restaurant being reviewed.	NO	FK	400001
rating_overall	Double	overall rating (1-5)	NO	-	4.0
rating_hygiene	INT	cleanliness rating(1-5)	NO	-	5
rating_flavor	INT	flavor/taste rating(1-5)	NO	-	4
rating_service	INT	service experience rating. (1-5)	NO	-	3
comment	VARCHAR (200)	Review content	Yes	-	'Delicious Food'
total_likes	INT	Number of likes for the review.	NO (default : 0)	-	20
created_at	TIMESTAMP	Date and time the review was posted	NO	-	2025-03-02 14:30:00
ai_evaluation	ENUM ('Safe', 'Inappropriate', 'Undetermined')	AI evaluation of the message. If "Safe", the review automatically posted. If "Inappropriate", it requires admin verification..	NO (default: 'Undetermined')	-	'Safe'
message_status	ENUM ('Posted', 'Pending', 'Banned')	Admin decision on the message. If AI marks it as "Safe", the status changes to "Posted" automatically. If AI marks it as "Inappropriate", the status remains "Pending" until reviewed by an admin	NO (default: 'Pending')	-	'Posted'

3.3.6 review_likes Table

Table 3.6 Data Dictionary review_likes Table

Field name	Data Type	Description	Nullable	Key	Example Data
Like_ID	INT	Unique identifier for each review like record.	NO	PK	600001
Review_ID	INT	ID of the review.	NO	FK	500001
User_ID	INT	ID of User who Likes the review.	NO	FK	100018
Liked_At	TIMESTAMP	Date and time when User likes the review.	NO	-	2023-08-06 10:20:00

3.3.7 admin_check_inappropriate_review Table (Only in cases where AI reports the review as inappropriate or Undetermined)

Table 3.7 Data Dictionary admin_check_inappropriate_review Table

Field name	Data Type	Description	Nullable	Key	Example Data
Admin_Check_review_ID	INT	Unique identifier for each admin check record.	NO	PK	600001
Review_ID	INT	ID of the review.	NO	FK	500002
Admin_ID	INT	ID of Admin who is responsible for each review	YES	FK	100009
admin_action_taken	ENUM ('Pending', 'Safe', 'Banned')	Status of admin check on the message .	NO (default : 'Pending')	-	'Pending'
admin_Checked_at	DATE TIME	Date and time when the admin reviewed the message flagged by AI as inappropriate.	YES	-	NULL
reason_for_action	VARCHAR (255)	The reason for banning and posting the message.	YES	-	NULL

3.3.8 Leaderboard_user_most_like Table (Collect who have top 5 of most Total Likes of each month)

Table 3.8 Data Dictionary Leaderboard_user_most_like Table

Field name	Data Type	Description	Nullable	Key	Example Data
Leaderboard_user_ID	INT	Unique identifier for each leaderboard record.	NO	PK	800001
User_ID	INT	ID of the user	NO	FK	100001
month_year	VARCHAR (7)	The month and year for which the leaderboard is generated (format: YYYY-MM).	NO	UNIQUE (Composite: User_ID, month_year)	"2025-01"
rank	INT	Rank of the user based on the number of likes for the amount review in that month	NO	-	1
total_likes	INT	Total number of likes received for all reviews by the user in the period.	NO (Default: 0)	-	1300
total_reviews	INT	Total number of reviews written by the user in the period.	NO (Default: 0)	-	50

3.3.9 Leaderboard_restaurant_most_rating Table (Collect restaurant have top 5 of most Total average overall rating of each month)

Table 3.9 Data Dictionary Leaderboard_restaurant_most_rating Table

Field name	Data Type	Description	Nullable	Key	Example Data
Leaderboard_restaurant_ID	INT	Unique identifier for each leaderboard record.	NO	PK	900001
Restaurant_ID	INT	ID of the restaurant	NO	FK	400001
month_year	VARCHAR (7)	The month and year for which the leaderboard is generated (format: YYYY-MM).	NO	UNIQUE (Composite: User_ID, month_year)	"2025-01"
rank	INT	Rank of the restaurant based on average rating	NO	-	1
overall_rating	DECIMAL (2,1)	Average rating of all reviews during the month	NO (Default: 0)	-	4.9
total_reviews	INT	Total number of reviews received in the month	NO (Default: 0)	-	50

3.3.10 Threads Table

Table 3.10 Data Dictionary Threads Table

Field name	Data Type	Description	Nullable	Key	Example Data
Thread_ID	INT	Unique identifier for each threaded message.	NO	PK	110001
User_ID	INT	ID of the user who wrote the message.	NO	FK	100001
message	VARCHAR (255)	Message content.	NO	-	‘หาบ้านหรือย โกสสิน’
created_at	DATETIME	Timestamp of when the message was posted.	NO	-	2025-03-02 14:30:00
ai_evaluation	ENUM (‘Undetermined’, ‘Safe’, ‘Inappropriate’)	AI evaluation of the message. If "Safe", the threads automatically posted. If "Inappropriate", it requires admin review	NO (default: ‘Undetermined’)	-	‘Safe’
admin_decision	ENUM (‘Posted’, ‘Pending’, ‘Banned’)	Admin decision on the message. If AI marks it as "Safe", the status changes to "Posted" automatically.If AI marks it as "Inappropriate", the status remains "Pending" until reviewed by an admin.’;	NO (default: ‘Pending’)	-	‘Posted’

3.3.11 Threads_likes Table

Table 3.11 Data Dictionary Threads_likes Table

Field name	Data Type	Description	Nullable	Key	Example Data
Like_ID	INT	Unique identifier for each Threads like record.	NO	PK	120001
Thread_ID	INT	ID of the Threads.	NO	FK	110001
User_ID	INT	ID of User who Likes the Threads.	NO	FK	100015
Liked_At	DATETIME	Date and time when User likes the Threads	NO	-	2025-03-02 14:30:00

3.3.12 Threads_reply Table

Table 3.12 Data Dictionary Threads_reply Table

Field name	Data Type	Description	Nullable	Key	Example Data
Thread_reply_ID	INT	Unique identifier for each thread's reply message.	NO	PK	130001
Thread_ID	INT	Unique identifier for each threads reply message	NO	PK	110001
User_ID	INT	ID of the user who wrote the message of Threads reply..	NO	FK	10010
message	VARCHAR (255)	Message content.	NO	-	‘ลองดูร้านอื่น’
created_at	DATETIME	Timestamp of when the Threads reply was posted.	NO	-	2025-03-02 14:30:00
ai_evaluation	ENUM ('Undetermined', 'Safe', 'Inappropriate')	AI evaluation of the message. If "Safe", the Threads reply automatically posted. If "Inappropriate", it requires admin review	NO (default: 'Undetermined')	-	'Safe'
admin_decision	ENUM ('Posted', 'Pending', 'Banned')	Admin decision on the message. If AI marks it as "Safe", the status changes to "Posted" automatically.If AI marks it as "Inappropriate", the status remains "Pending" until reviewed by an admin.;	NO (default: 'Pending')	-	‘Posted’

3.3.13 Threads_reply_likes Table

Table 3.13 Threads_reply_likes Table

Field name	Data Type	Description	Nullable	Key	Example Data
Like_ID	INT	Unique identifier for each Threads reply like record.	NO	PK	140001
Thread_reply_ID	INT	ID of the Threads reply.	NO	FK	130001
User_ID	INT	ID of User who Likes the Threads reply.	NO	FK	100005
Liked_At	DATETIME	Date and time when User likes the Threads reply.	NO	-	2025-03-02 14:30:00

3.3.14 Admin_check_inappropriate_Threads Table (Only in cases where AI reports the Threads message as inappropriate or Undetermined)

Table 3.14 Data Dictionary Admin_check_inappropriate_Threads Table

Field name	Data Type	Description	Nullable	Key	Example Data
Admin_Review_thread_ID	INT	Unique identifier for each admin review record.	NO	PK	150001
Thread_ID	INT	ID of the Threads.	NO	FK	110006
admin_ID	INT	ID of Admin who is responsible for each check Threads.	YES	FK	NULL
admin_action_taken	ENUM ('Pending', 'Safe', 'Banned')	Status of admin check on the Threads message.	NO default ('Pending')	-	'Pending'
admin_checked_at	DATE TIME	Date and time when the admin checked the message flagged by AI as inappropriate.	YES	-	NULL
reason_for_taken	VARCHAR (255)	The reason for the action taken by the admin	YES	-	NULL

3.3.15 Admin_check_inappropriate_Threads_reply Table (Only in cases where AI reports the Threads reply message as inappropriate or Undetermined)

Table 3.15 Data Dictionary Admin_check_inappropriate_Threads reply Table

Field name	Data Type	Description	Nullable	Key	Example Data
Admin_Review_threads_reply_ID	INT	Unique identifier for each admin checked record.	NO	PK	160001
Thread_reply_ID	INT	ID of the Threads reply.	NO	FK	130011
admin_ID	INT	ID of Admin who is responsible for each check Threads reply.	YES	FK	NULL
admin_action_taken	ENUM ('Pending', 'Safe', 'Banned')	Status of admin check on the Threads reply message.	NO default ('Pending')	-	'Pending'
admin_checked_at	DATETIME	Date and time when the admin checked the message flagged by AI as inappropriate.	YES	-	NULL
reason_for_taken	VARCHAR (255)	The reason for the action taken by the admin	YES	-	NULL

3.3.16 Profile_Shop Table

Table 3.16 Data Dictionary Profile_Shop Table

Field name	Data Type	Description	Nullable	Key	Example Data
Profile_ID	INT	Unique identifier for each item in profile shop.	NO	PK	170001
Profile_Name	VARCHAR (255)	Name of profile.	NO	-	Standard Profile
Description	VARCHAR (255)	Description of each profile.	NO	-	โปรไฟล์พื้นฐานสำหรับผู้ให้
Image_URL	TEXT	URL of each profile that's used in this shop.	NO	-	http://example.com/images/profile1.png
Required_Coins	INT	The price of each profile that you can buy by using coins.	NO	-	50
Created_At	DATETIME	Date and time when this profile was made and available in the shop.	NO	-	2025-03-09 11:10:54

3.3.17 Profile_Purchase_History Table

Table 3.17 Profile_Purchase_History Table

Field name	Data Type	Description	Nullable	Key	Example Data
Purchase_ID	INT	Unique identifier for each purchase made in the profile shop.	NO	PK	180001
User_ID	VARCHAR (255)	Unique identifier for the user who made the purchase.	NO	FK	100001
Profile_shop_ID	TEXT	Unique identifier for the profile available in the shop.	NO	FK	170001
Purchased_At	DATETIME	The date and time when the profile was purchased.	NO	-	2025-03-09 11:10:54
Coin_Spent	INT	The amount of coins spent for the purchase.	NO	-	150

3.4 Interface Design

3.4.1 Guest Role (Person Does not Login)

3.4.1.1 Login Page



Figure 3.3 Login Page (Guest)

To allow users without an MFU email to quickly access the app as a Guest, without registration or login, by tapping the 'Stay as Guest' button. This feature is designed for external visitors, such as parents or guests, who wish to explore the app without the need for a university email account.

3.4.1.2 Home Page

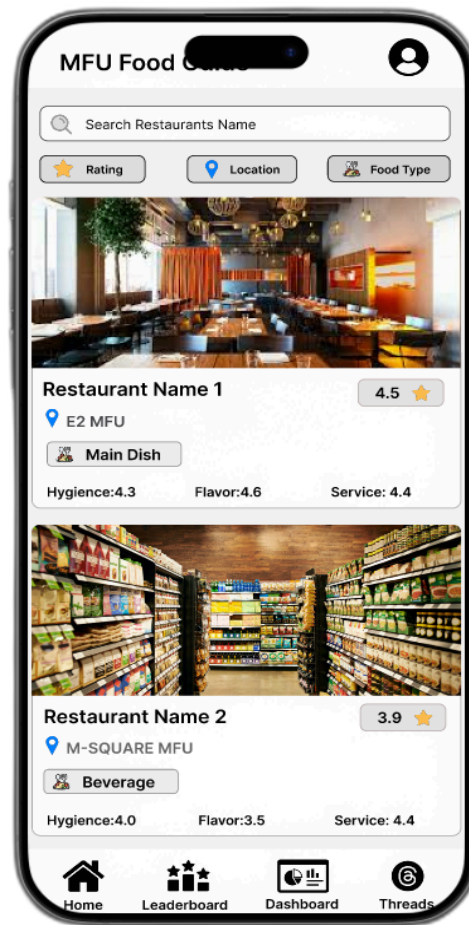


Figure 3.4 Home Page (Guest)

The Home page displays all available restaurants, allowing users to filter them by rating, location, or type of food for easier browsing and discovery. This helps users quickly find restaurants that match their preferences and explore new dining options more efficiently.

3.4.1.3 Restaurant Information Page

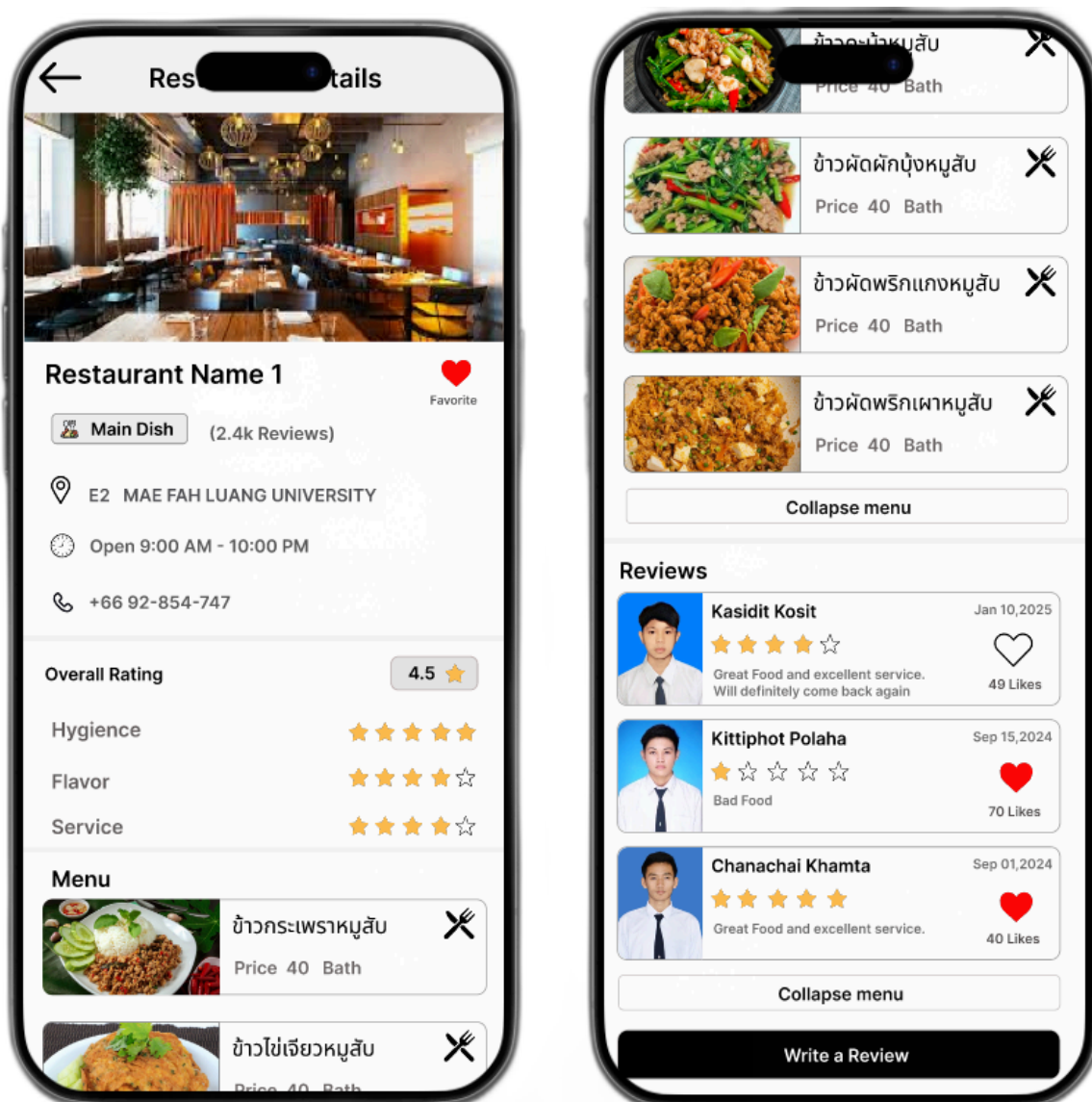


Figure 3.5 Restaurant Information Page (Guest)

When a user finds an interesting restaurant from the Home page and clicks on it, this page displays detailed restaurant information, including name, food type, location, opening hours, contact number, menu with images and prices, and all user reviews, with review submission restricted to MFU personnel only.

3.4.1.3 Write Review Page (Can not Access)

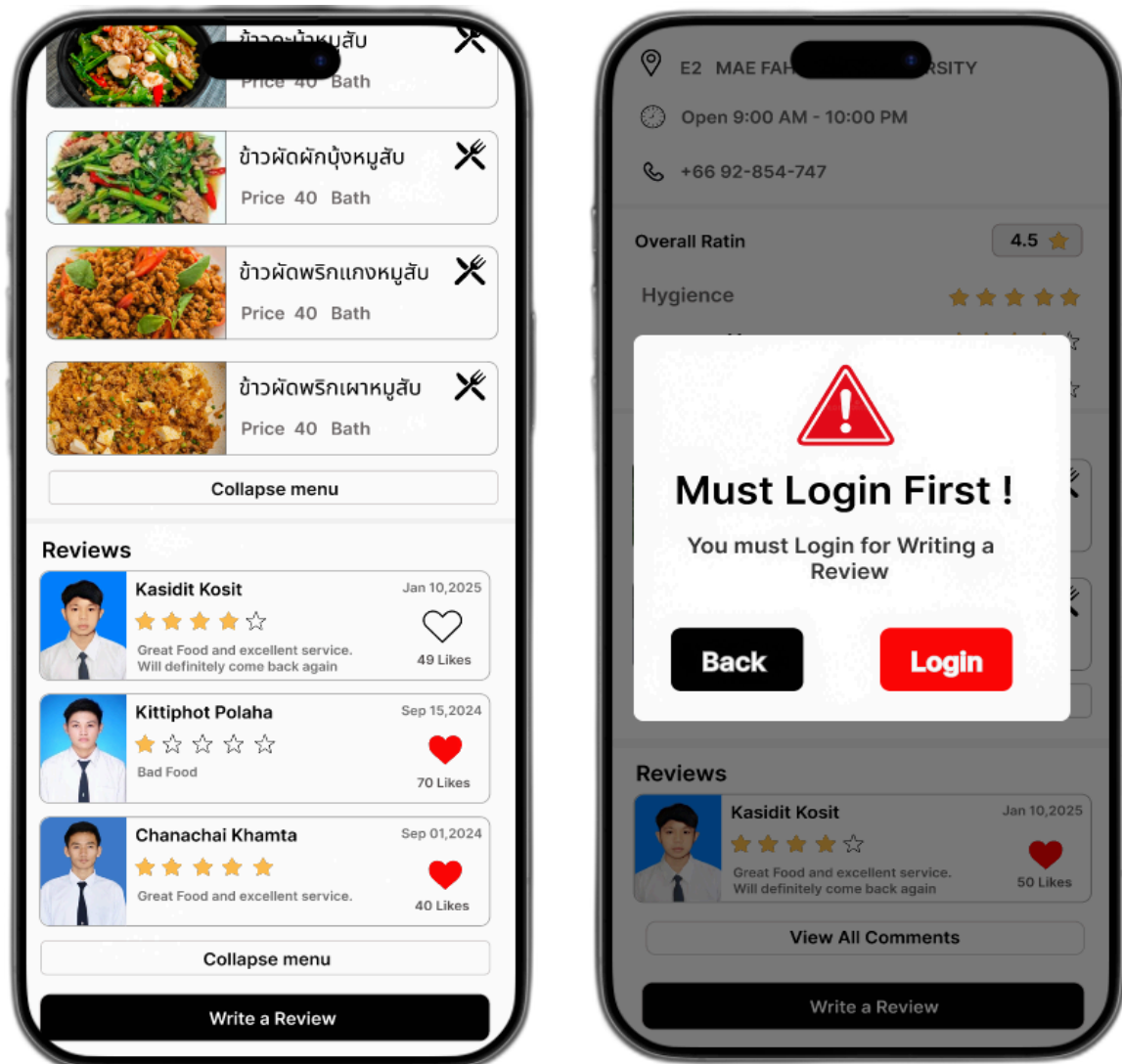


Figure 3.6 Write Review Page (Guest)

When a user taps the 'Write Review' button on the Restaurant Information page without being logged in, an alert will appear notifying them that access is restricted users must log in to write a review. This helps enhance the credibility of the reviews by ensuring they are written by MFU personnel.

3.4.1.4 Leaderboard Page

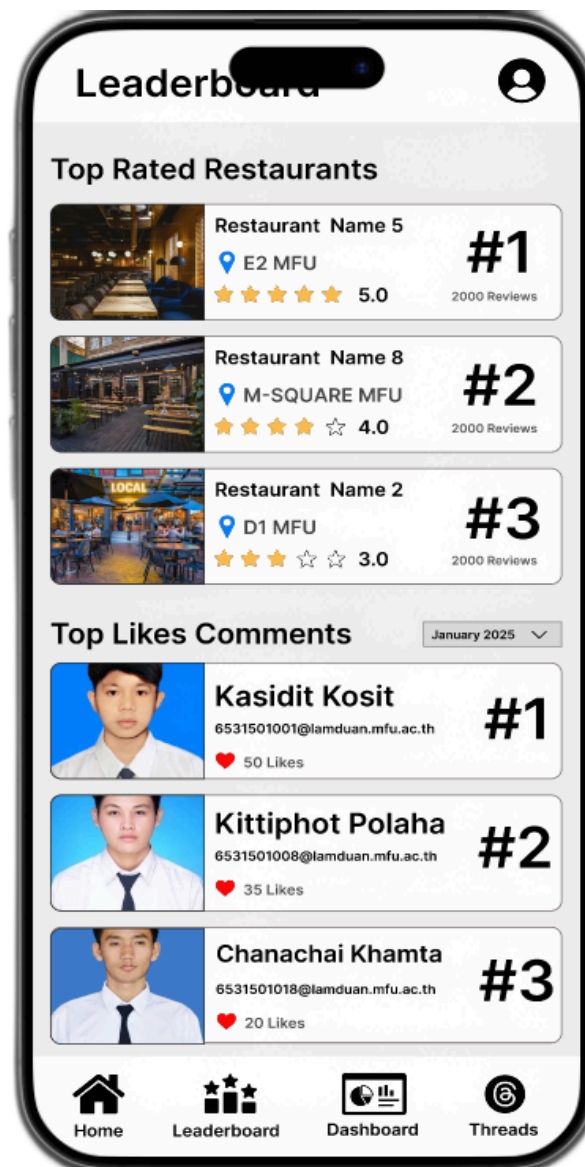


Figure 3.7 Leaderboard Page (Guest)

The Leaderboard page displays two sections: the top 3 restaurants with the highest overall ratings, and the top 3 users with the most total likes on their restaurant reviews for each month. Rankings reset on the 1st of every month, and users can also view results from previous months.

3.4.1.6 Threads Page (Cannot Access)

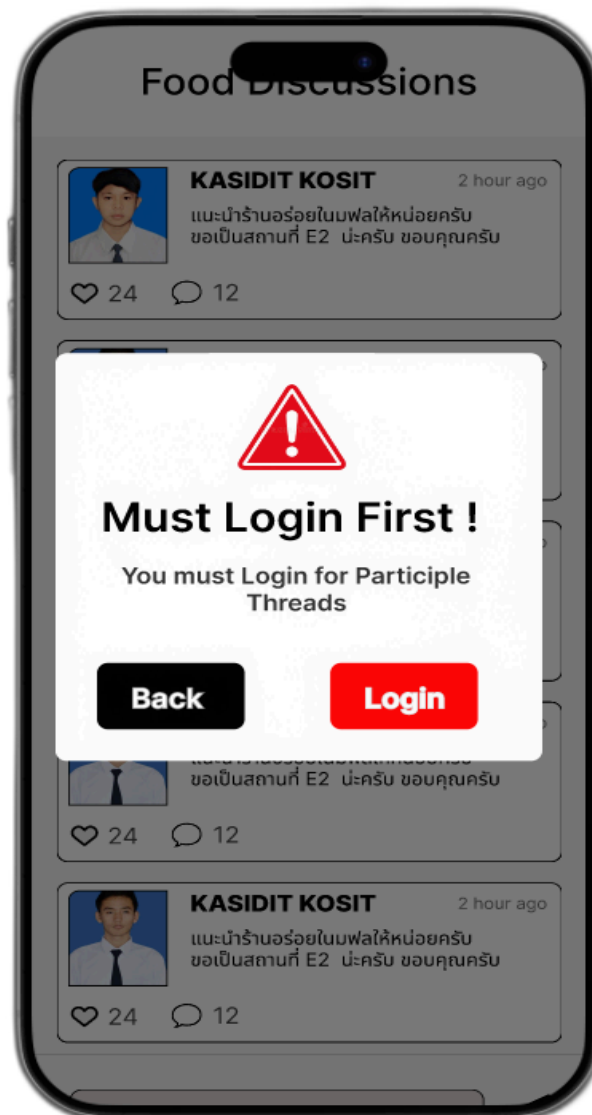


Figure 3.8 Threads Page (Guest)

When a user taps the 'Threads' button from the navigation bar of another page without being logged in, an alert will appear notifying them that access is restricted. Users must log in to participate in Threads. This is to ensure that only members of Mae Fah Luang University can join the discussions, maintaining a secure and trusted environment exclusive to the MFU community Threads.

3.4.2 User Login Role (Login with Lamduan Email or MFU Email)

3.4.2.1 Login Page

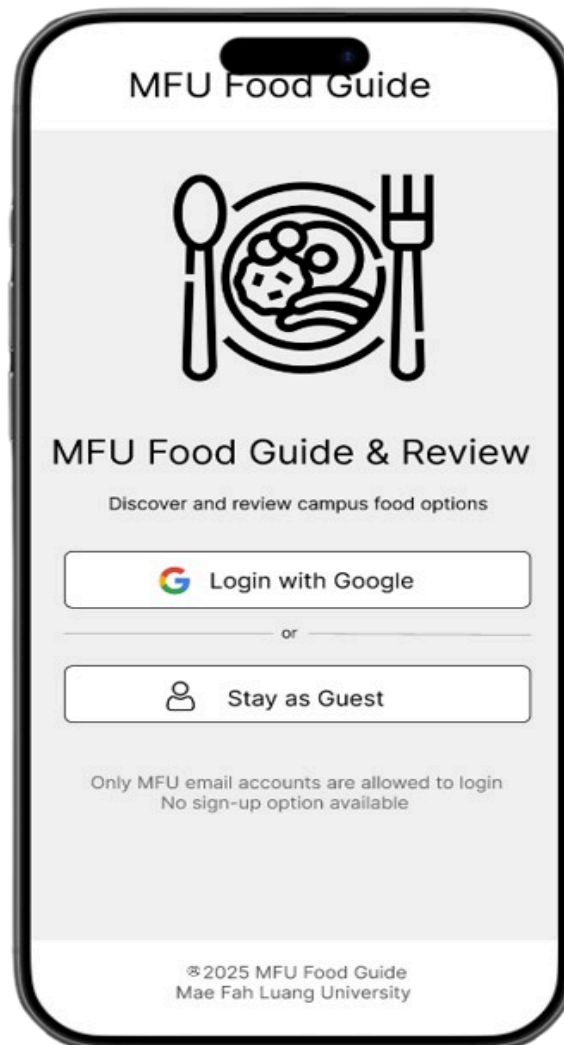


Figure 3.9 Login Page (MFU)

This page provides login access to two types of users: MFU students using an @lamduan.ac.th email and MFU staff members using an @mfu.ac.th email. The system verifies email domains before granting access to the platform's full features. Users log in via the "Login with Google" button.

3.4.2.2 Home Page

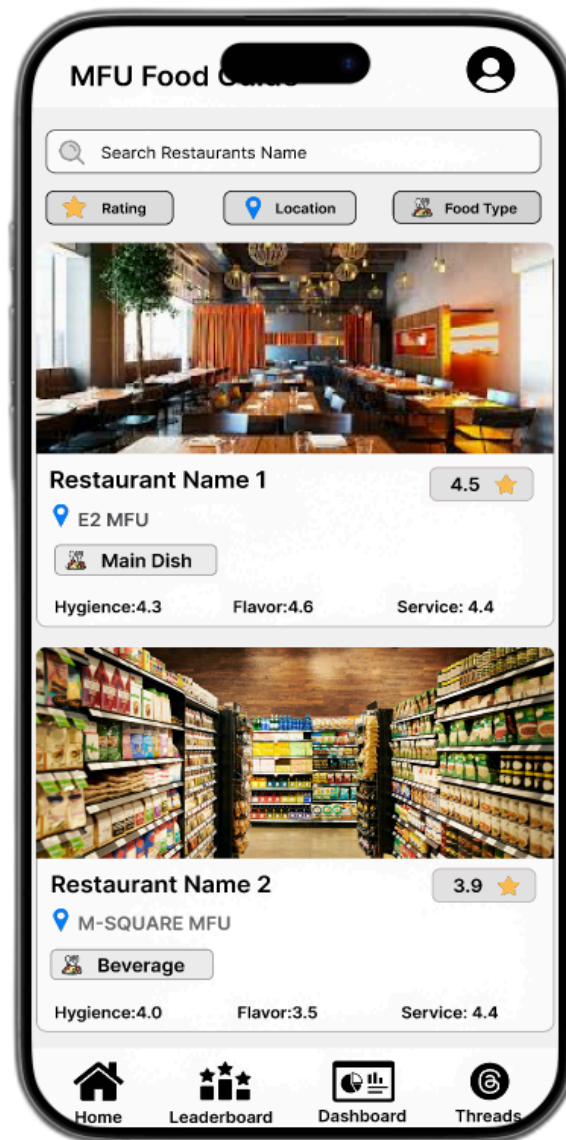


Figure 3.10 Home Page (MFU)

The Home page displays all available restaurants, allowing users to filter them by rating, location, or type of food for easier browsing and discovery. This helps users quickly find restaurants that match their preferences and explore new dining options more efficiently.

3.4.2.3 Restaurant Information Page

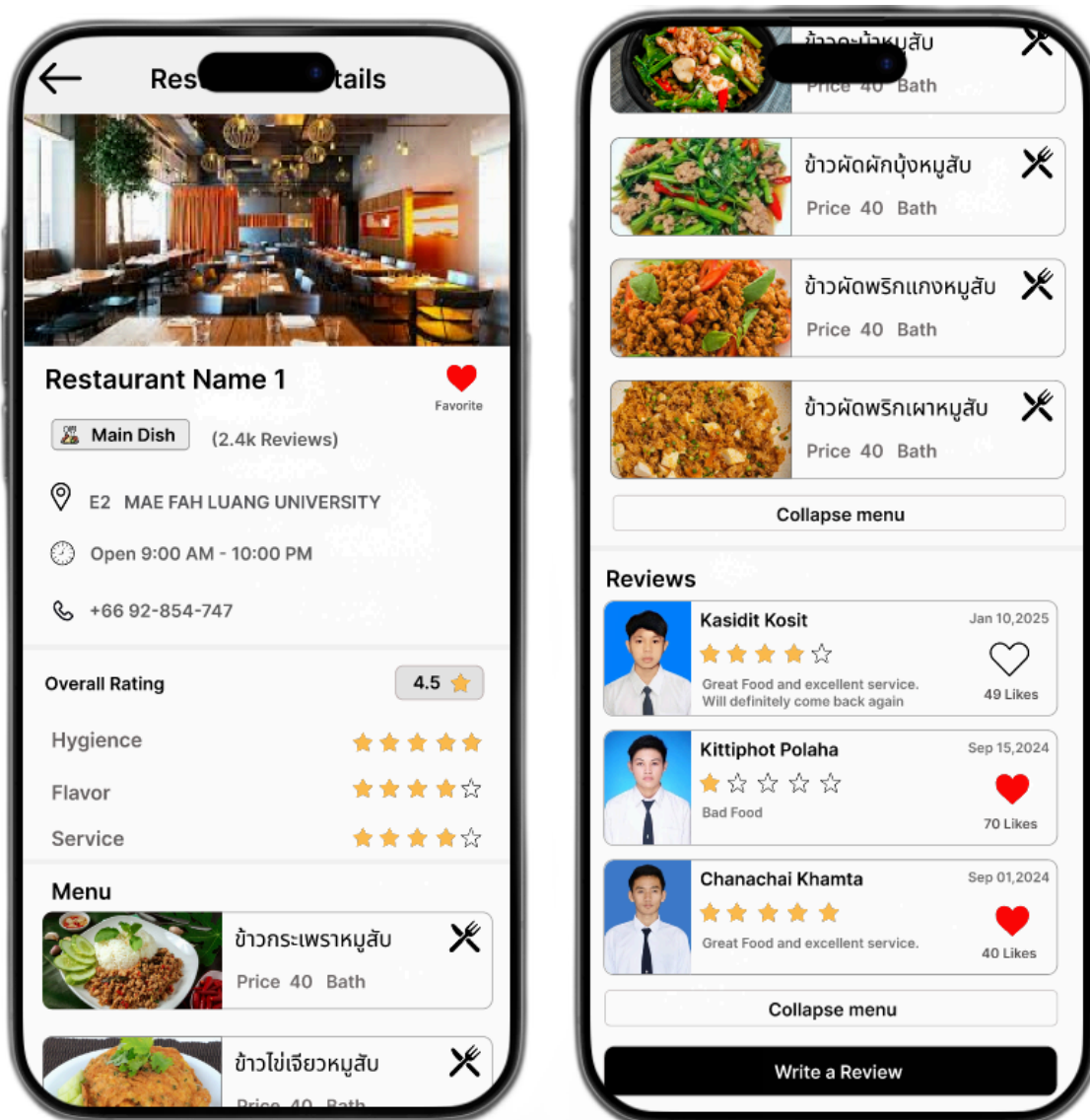


Figure 3.11 Restaurant Information Page (MFU)

When a user finds an interesting restaurant from the homepage and clicks on it, this page displays detailed restaurant information, including name, food type, location, opening hours, contact number, menu with images and prices, and all user reviews with the option for logged-in users to write and submit their own reviews.

3.4.2.4 Reviews Page

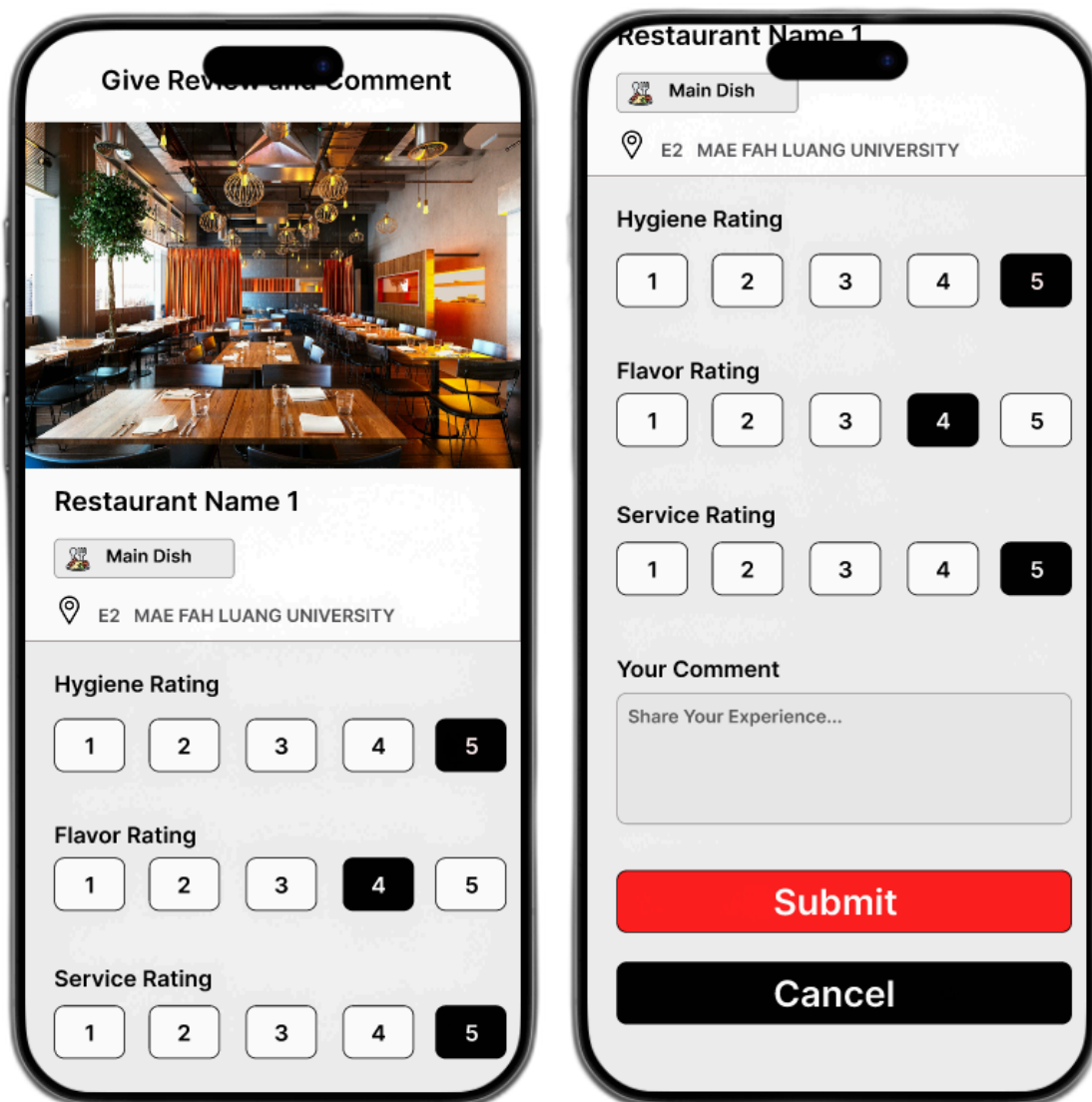


Figure 3.12 Review Page (MFU)

This page appears after a logged-in user clicks the 'Write Review' button from the Restaurant Information page. It allows the user to rate the selected restaurant in three aspects Hygiene, Flavor, and Service and submit a written comment. An AI-powered system is integrated to detect inappropriate language and ensure that reviews remain respectful and aligned with community guidelines before being published.

3.3.2.5 Leaderboard Page

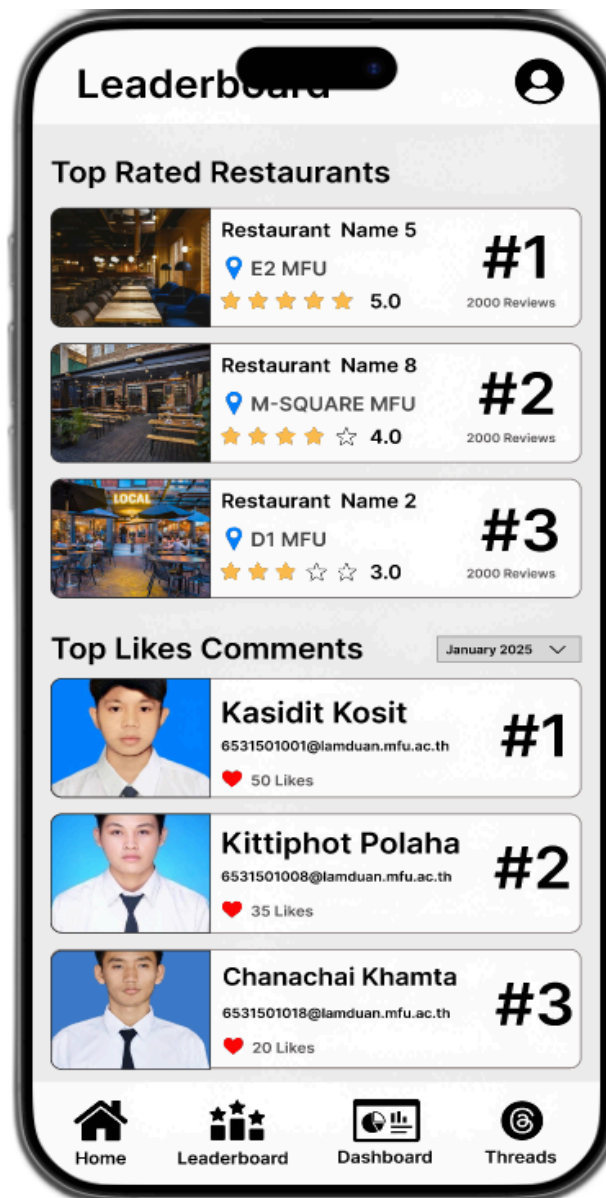


Figure 3.13 Leaderboard Page (MFU)

The Leaderboard page displays two sections: the top 3 restaurants with the highest overall ratings, and the top 3 users with the most total likes on their restaurant reviews for each month rankings reset on the 1st of every month, and users can also view results from previous months.

3.4.2.6 Chatbot Page

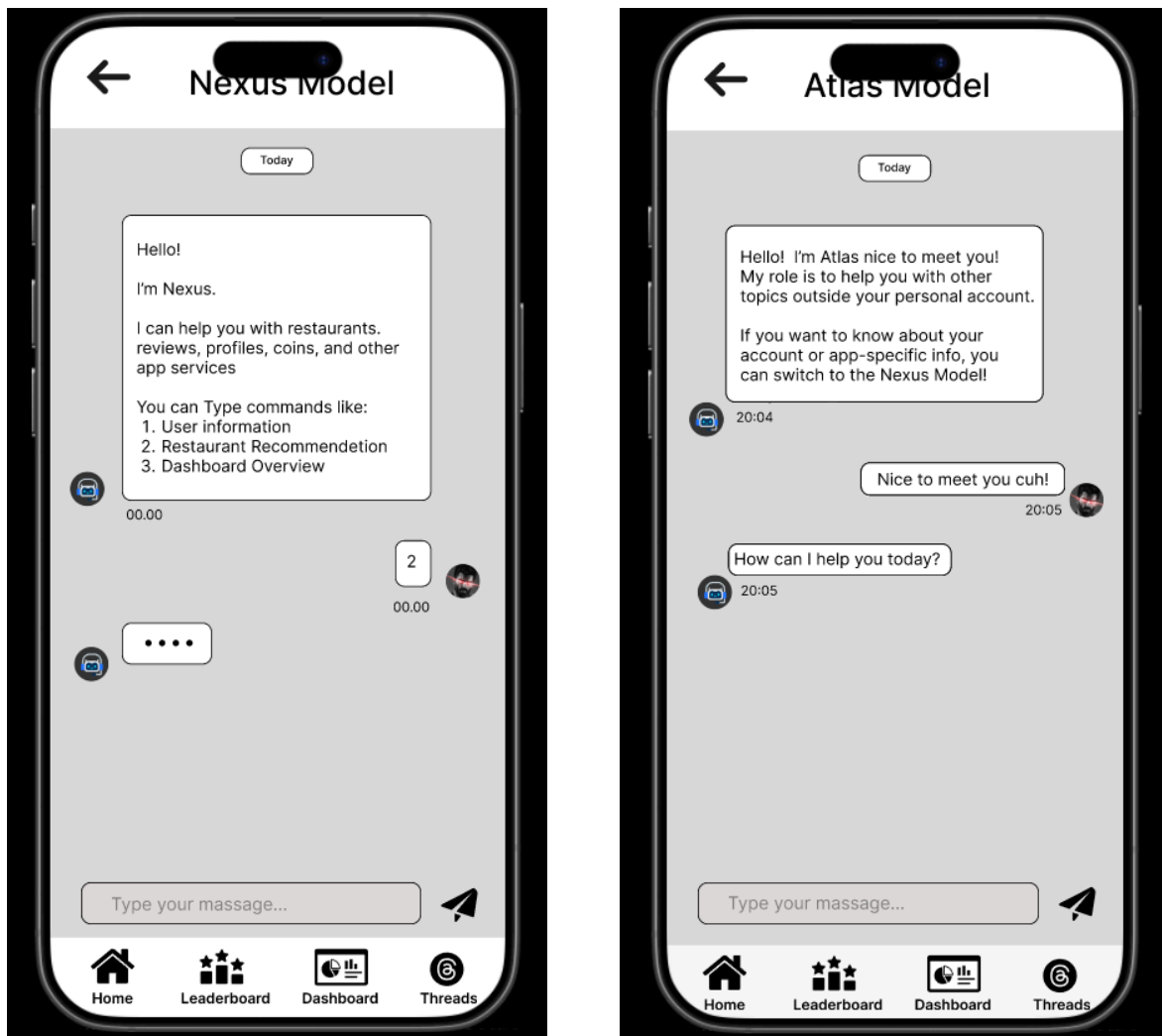


Figure 3.14 Chatbot Page

The Chat type. This helps users compare restaurants based on three main categories: location, rating, and food type, making it easier to make informed decisions.

3.4.2.7 Threads Page



Figure 3.15 Threads Page (MFU)

The Threads page is a community space where users can discuss and share their thoughts about food. It displays all threads posted by users, and allows logged-in users to create new threads through the “Share your thoughts” input box. To maintain a safe and respectful environment, an AI system automatically detects inappropriate words.

3.4.2.8 Threads Reply Page

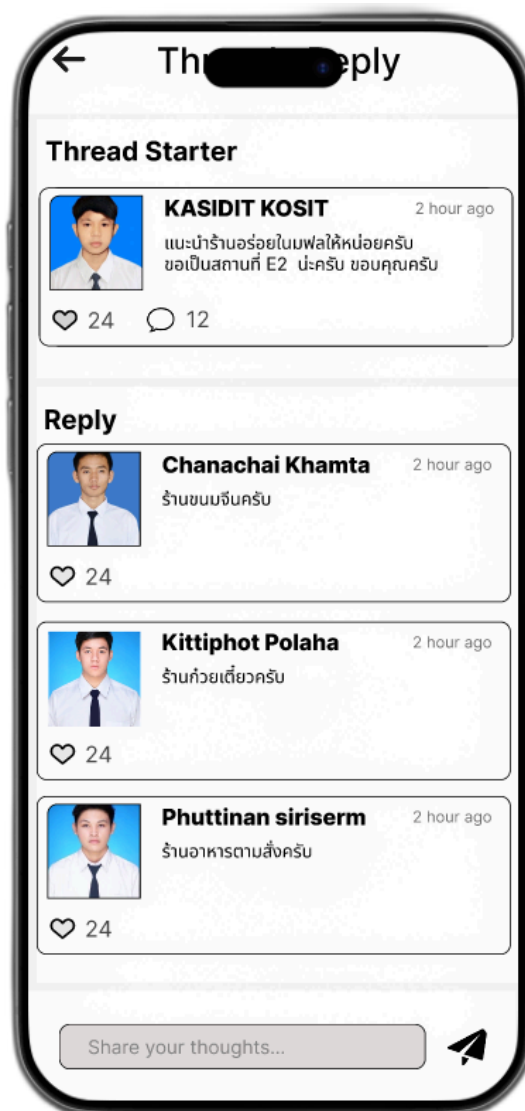


Figure 3.16 Threads Reply Page (MFU)

When a user clicks on a thread they are interested in, this page displays the full content of that thread and all its replies. Logged-in users can contribute to the discussion by writing a response in the "Share your thought" box at the bottom of the page all submitted content is automatically screened by an AI system that detects and blocks offensive or inappropriate language.

3.4.2.9 User Profile Page

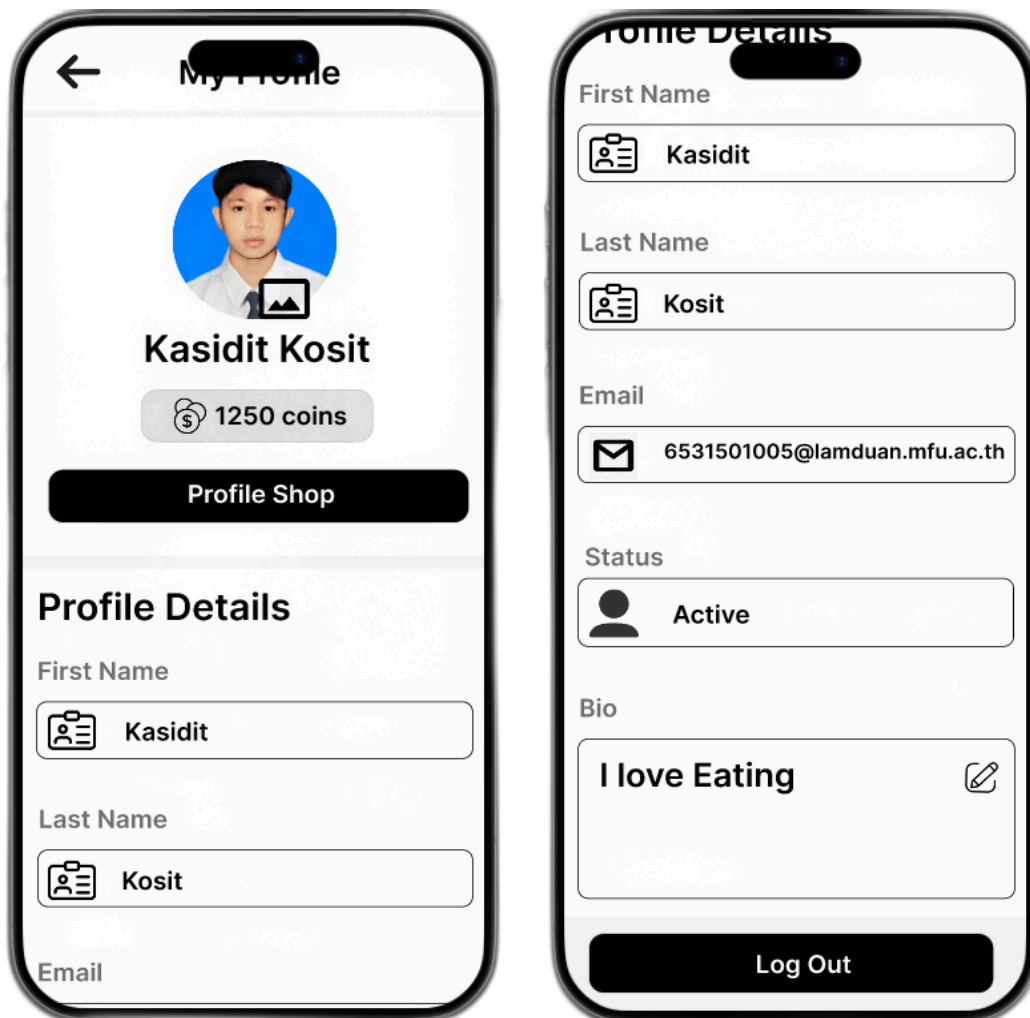


Figure 3.17 User Profile Page (MFU)

The User Profile Page displays the personal information of a logged-in user, including their full name, username (used within the app), email address, user status, a short bio, profile picture, and a 'Log Out' button. Additionally, users can edit their username, bio, and choose a profile picture at any time, allowing them to keep their profile information up-to-date and reflective of their preferences.

3.4.2.10 Profile Image Management Page

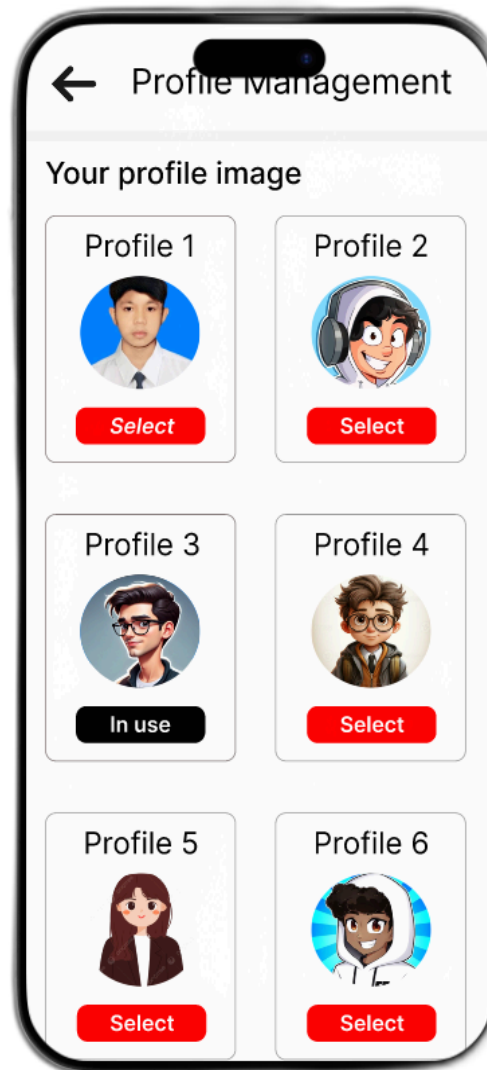


Figure 3.18 User Profile Image Page (MFU)

The User Profile Management page is accessed when the Logged-in user clicks the profile image icon on their User Profile page. On this page, the user can update their profile picture by choosing an image earned through exchanging coins, which adds an engaging experience and encourages user participation in the app's activities.

3.4.2.11 Coin History Page

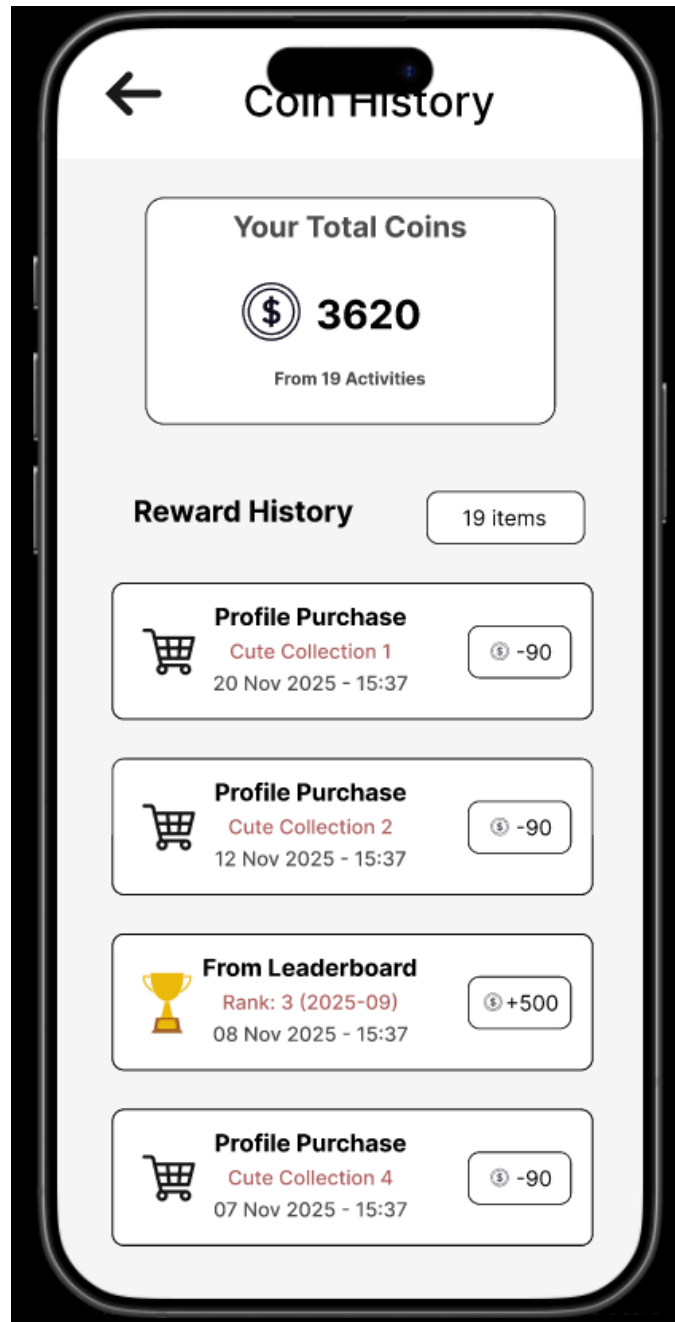


Figure 3.19 Coin History Page (MFU)

This page displays the complete history of all coin-related activities performed by the user within the application. At the top, users can see their total accumulated coins along with the number of activities that contributed to this total.

3.4.2.12 User Management Page

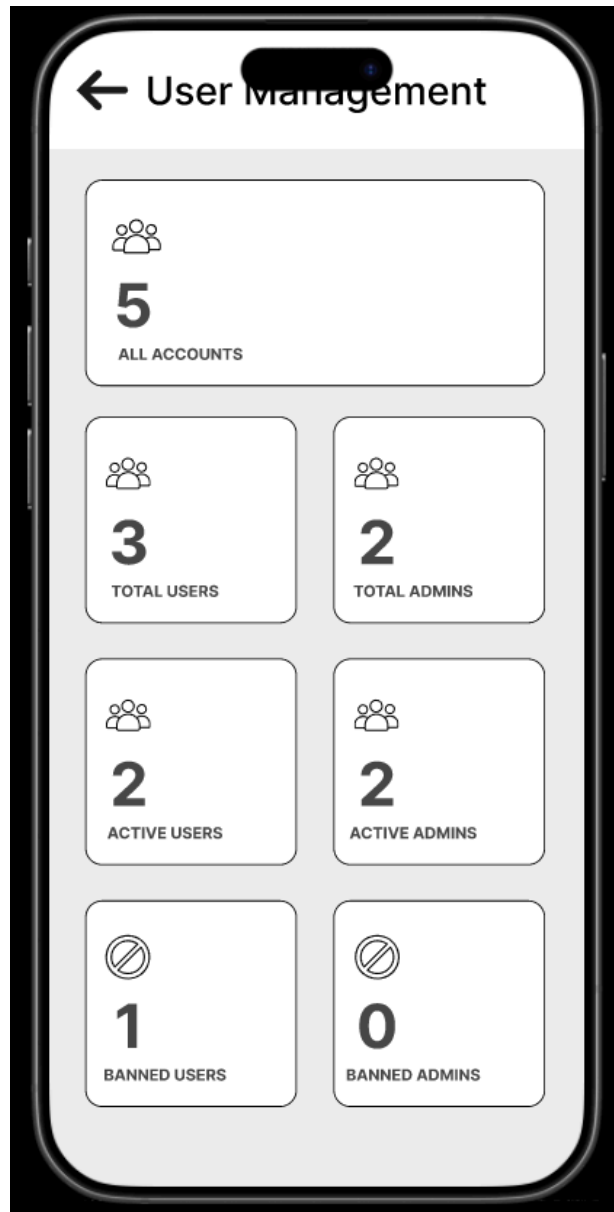


Figure 3.20 User Management Page (MFU)

The page displays the total number of accounts, along with a breakdown between regular users and administrators. It also shows how many of these accounts are currently **active** and how many have been **banned**. Each category is presented in a separate card for clarity, making it easy for the admin to assess system activity and identify any issues at a glance.

3.4.2.13 Coin Exchange Page

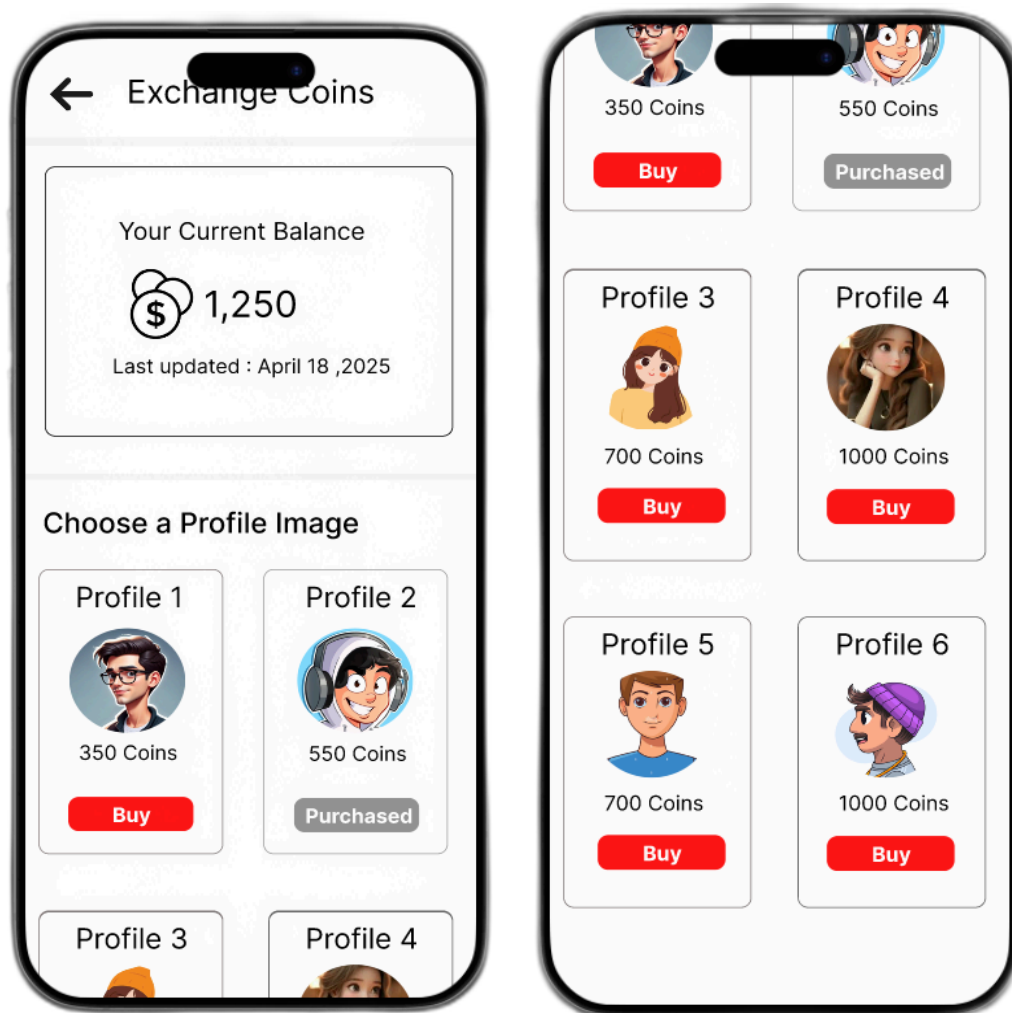


Figure 3.21 Coin Exchange Page (MFU)

When a logged-in user clicks the "Profile Shop" button on their User Profile page, they are redirected to the Coin Exchange Page. On this page, the user can purchase a profile picture using coins that were received on Leaderboard activity. Once the user buys a profile picture, it will be stored in the Profile Image Management Page, where the user can select and use it as their profile image.

3.4.3 Admin Role (User who Login with Lamduan Email or Personal Email with Admin Role)

3.4.3.1 Login Page



Figure 3.22 Login Page (Admin)

This page provides login access to Admin roles by using an @lamduan.ac.th email or a personal email that already has an admin role in the database. The system verifies email domains before granting access to the platform's full features Admin log in via the "Login with Google" button.

3.4.3.2 Home Page

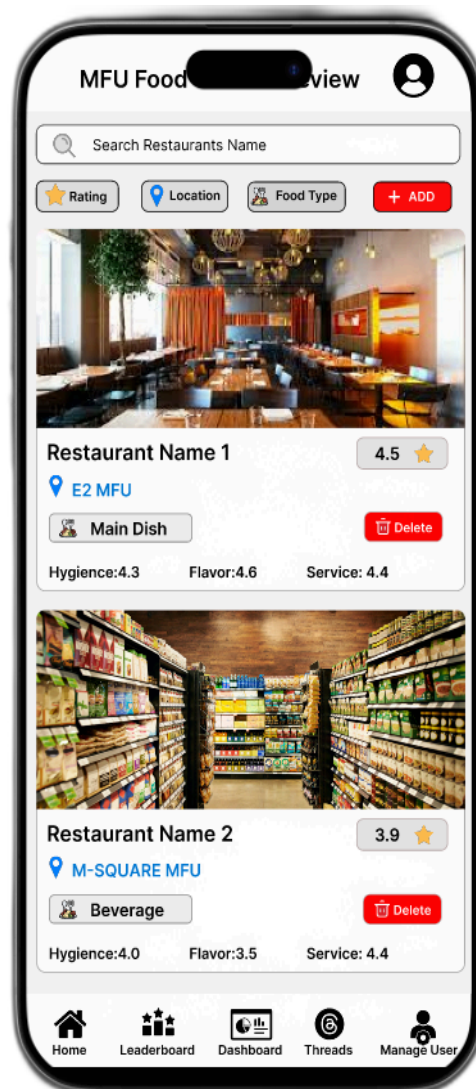


Figure 3.23 Home Page (Admin)

This page displays all available restaurants and allows the admin to add a Restaurant by pressing the red button “ADD” on the top right of the filter bar and delete the restaurants by pressing the red button “Delete” at the restaurant information bar of that restaurant.

3.4.3.4 Restaurant Information Page

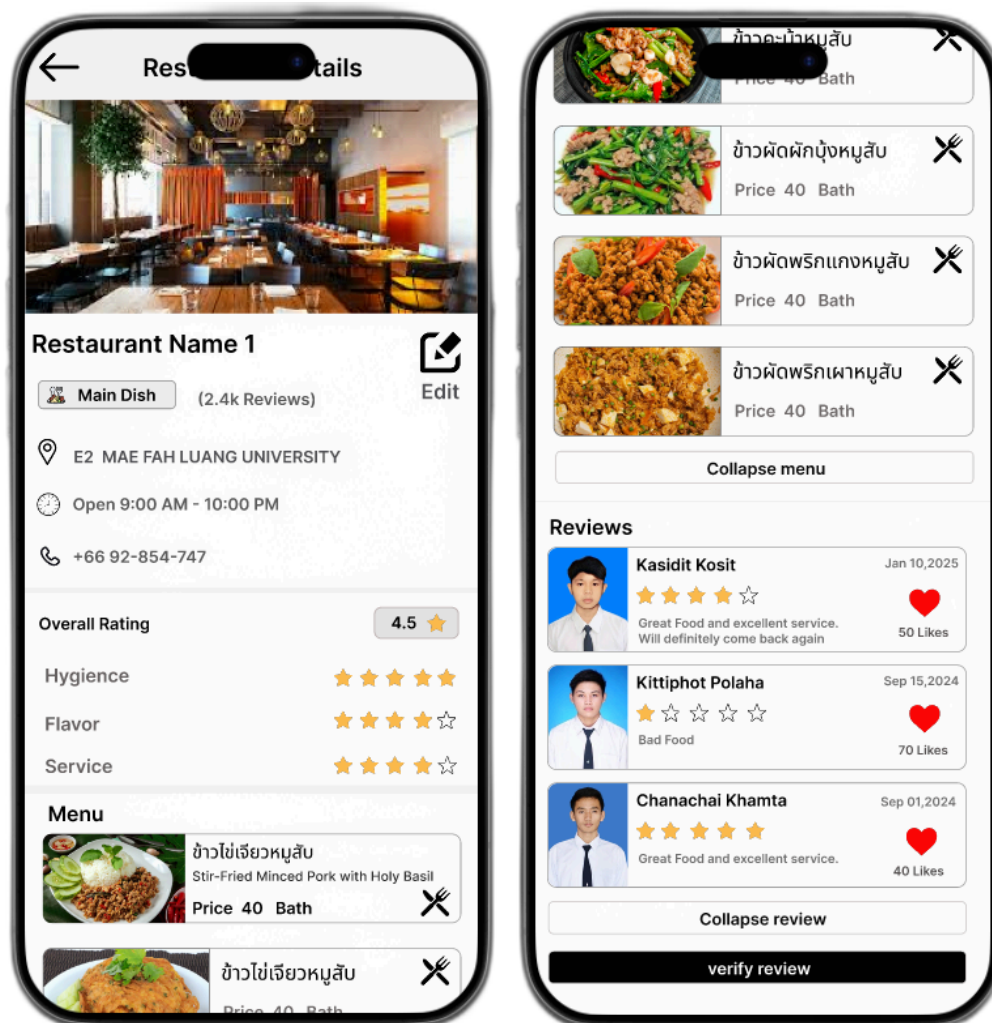


Figure 3.24 Restaurant Information Page (Admin)

From the homepage, when the admin would like to edit or verify reviews of that restaurant, and click on it. It will show this page and display all details of the restaurant information, and with the admin role, this page will have “Edit” and “Verify Review”. “Edit” buttons will stay at the right of the restaurant name, and “Verify Review” will be at the bottom of this page. Both the “Edit” and “Verify Review” buttons are restricted to admin only.

3.4.3.4 Edit restaurant Information Page

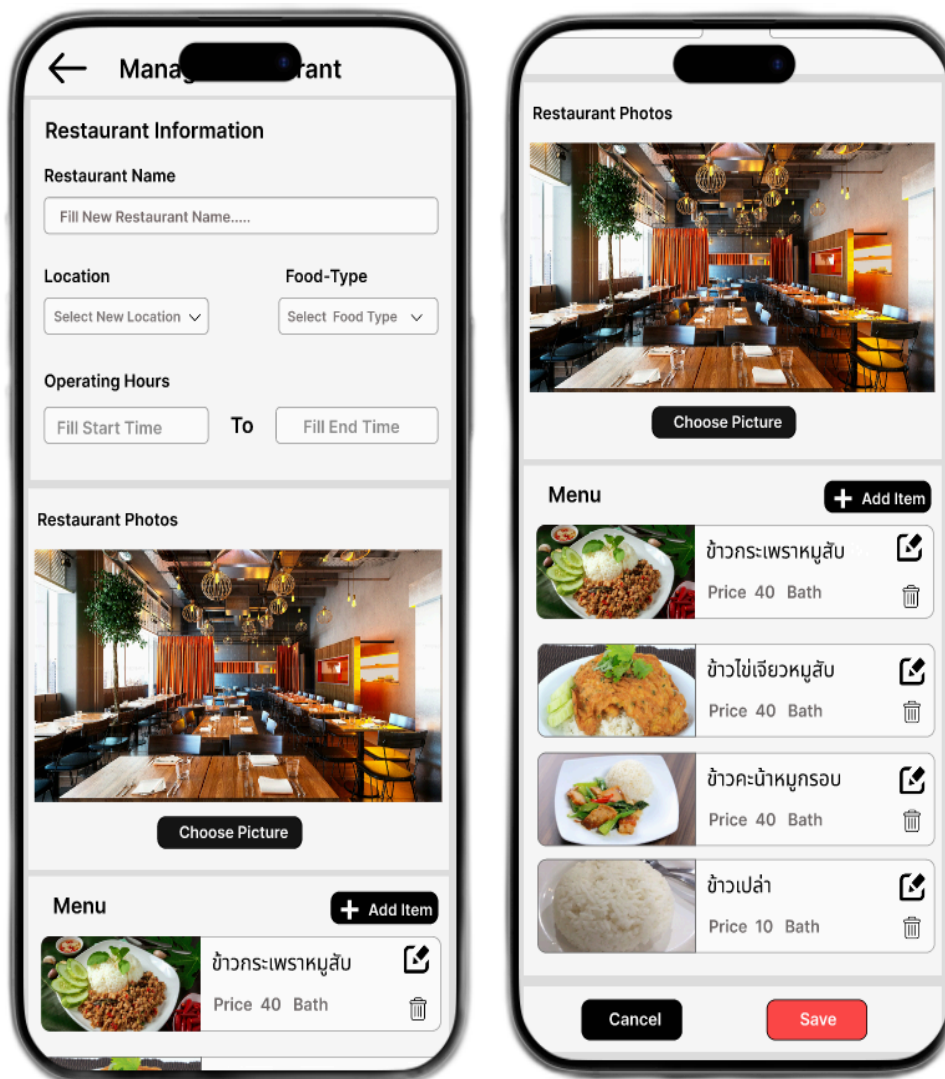


Figure 3.25 Edit Restaurant Information Page (Admin)

If an admin clicks the 'Edit' icon on the Restaurant Information Page, they will be directed to this page. On this page, admins can change restaurant information such as 'Restaurant Name', 'Location', 'Food Type', 'Operating Hours', 'Photo', and they can also add or delete menus and modify the details of each menu item.

3.4.3.5 Verify Review Page

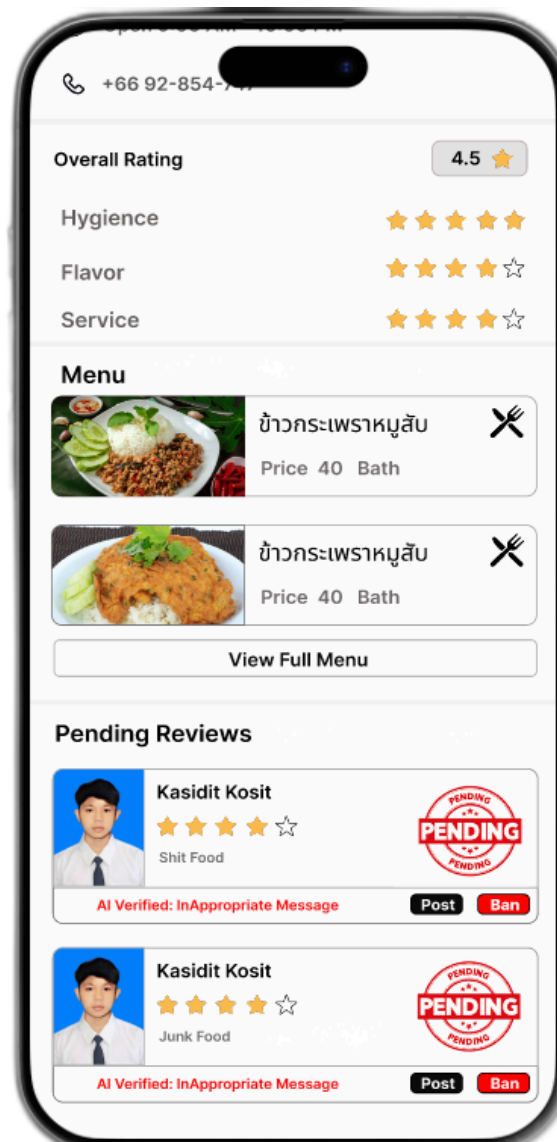


Figure 3.26 Verify Review Page (Admin)

If an admin clicks the 'Verify review' button on the Restaurant Information Page, they will be directed to this page. Admin can view pending reviews of the restaurant and can take action, such as post or ban, by pressing the button on that review. If pressing “Post” that review will post on the restaurant, but if pressing “Ban” that review will get banned instead.

3.4.3.6 Leaderboard Page

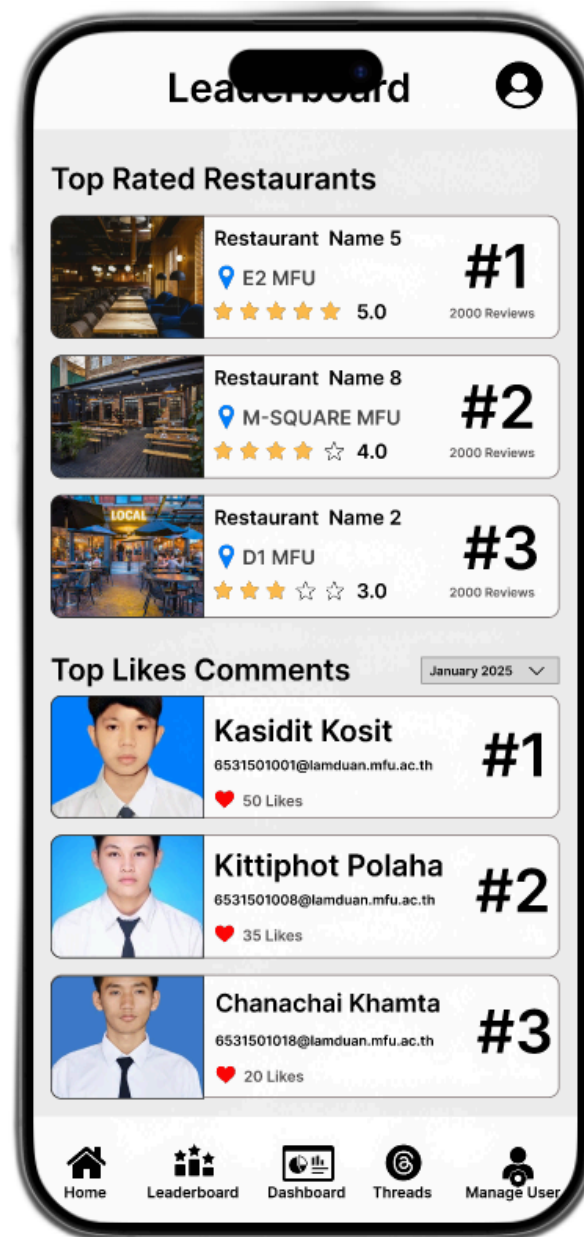


Figure 3.27 Leaderboard Page (Admin)

The Leaderboard page displays two sections: the top 3 restaurants with the highest overall ratings, and the top 3 users with the most total likes on their restaurant reviews for each month. Rankings reset on the 1st of every month, and users can also view results from previous months.

3.4.3.7 ChatBot Page



Figure 3.28 Chatbot Page

The Chat type. This helps users compare restaurants based on three main categories: location, rating, and food type, making it easier to make informed decisions.

3.4.3.8 Banned Review Status Page

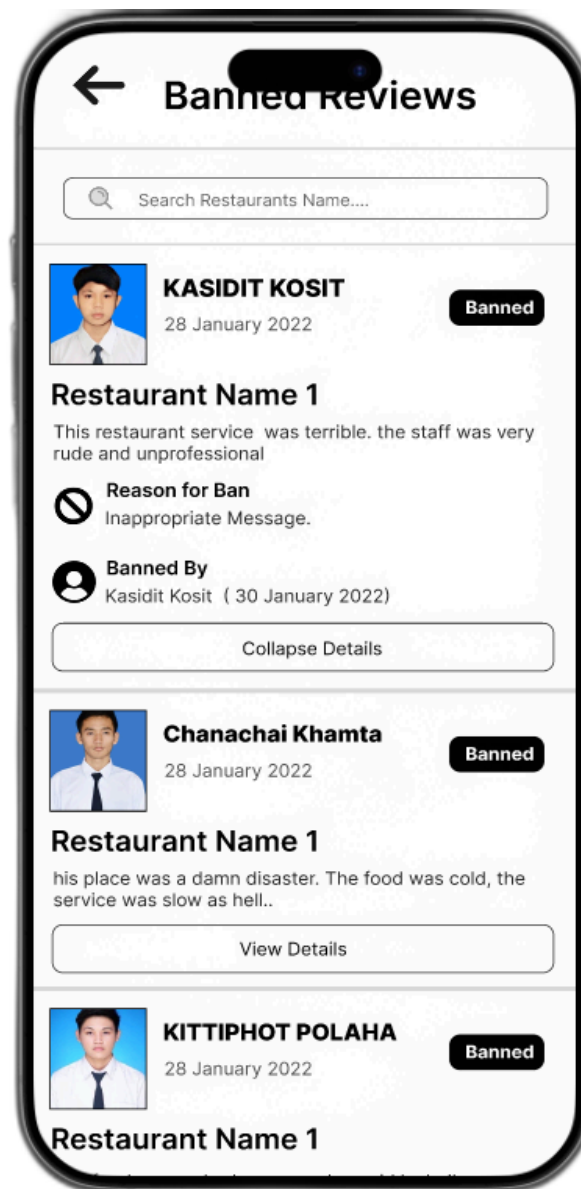


Figure 3.29 Ban Review Page (Admin)

From the Dashboard, after the admin clicks on “Review Overviews” the admin will be brought to this page. On this page admin can see the details of the review that got banned, such as who wrote the review, the review to which restaurant, the reason for the ban, ban by who and when it got banned.

3.4.3.9 Banned Threads Status Page



Figure 3.30 Ban Thread Page (Admin)

From the Dashboard, after the admin clicks on “Thread Overviews” the admin will be brought to this page. On this page admin can see the details of the thread that got banned, such as who wrote the thread, the reason for the ban, ban by who and when it got banned.

3.4.3.10 Banned Threads Reply Status Page



Figure 3.31 Ban Thread Reply Page (Admin)

From the Dashboard, after the admin clicks on “Thread Reply Overviews” the admin will be brought to this page. On this page admin can see the details of the thread reply that got banned, such as who wrote the reply, the reason for the ban, ban by who and when it got banned.

3.4.3.11 Threads Page

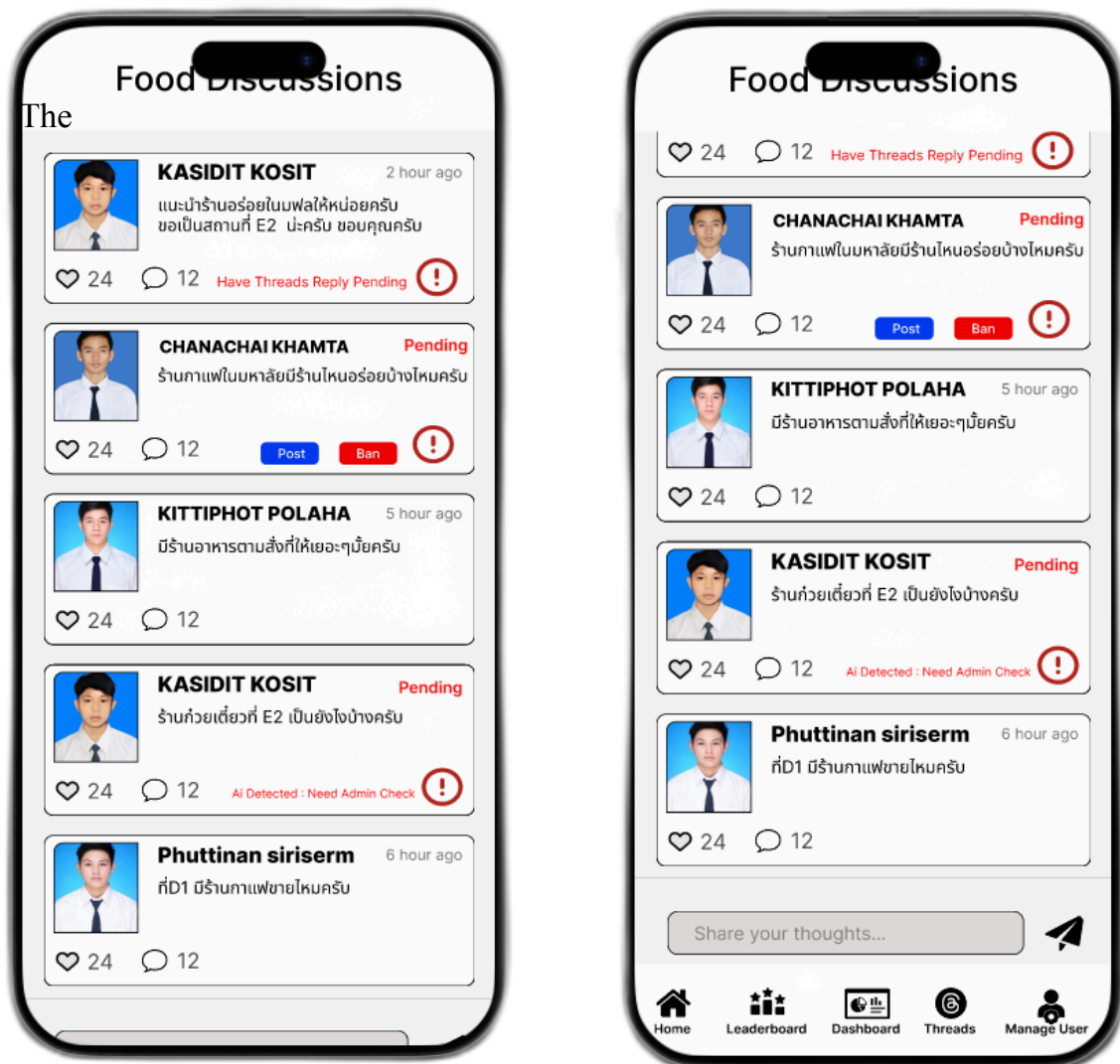


Figure 3.32 Thread Page (Admin)

The Threads page is a community space where users can share thoughts and discuss food-related topics. To ensure a safe and respectful environment, an AI system automatically detects inappropriate language. If a Thread is flagged, it is automatically pending and marked. Admins can review pending Threads and choose to post or ban them.

3.4.3.12 Thread Reply

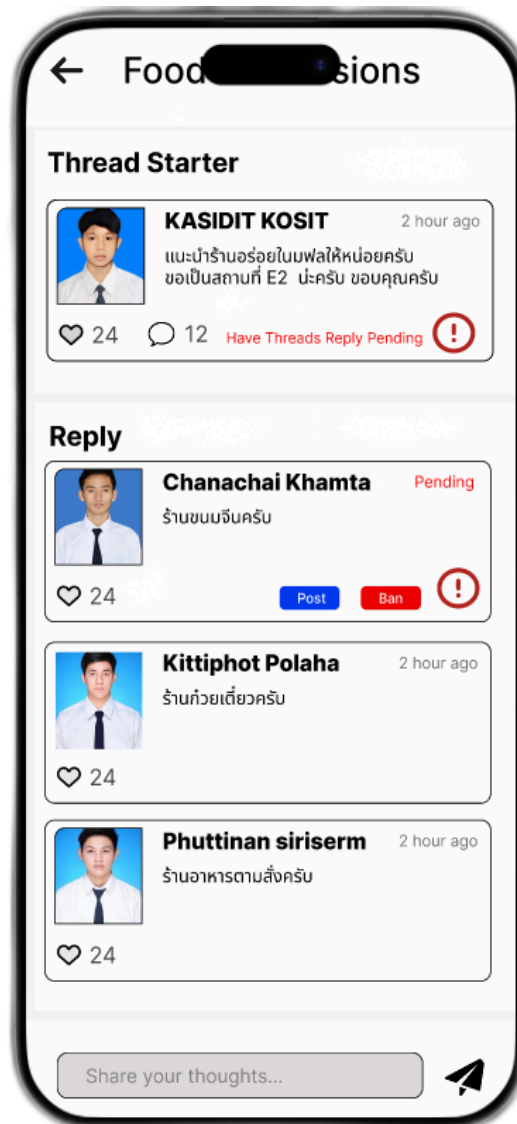


Figure 3.33 Thread Reply Page (Admin)

The Thread Reply page shows all responses to a thread. To keep the discussion respectful, an AI system scans replies for inappropriate content. If a reply is flagged, it is automatically pending and marked. Admins can review pending replies and choose to Post or Ban them.

3.4.3.13 User Management Page

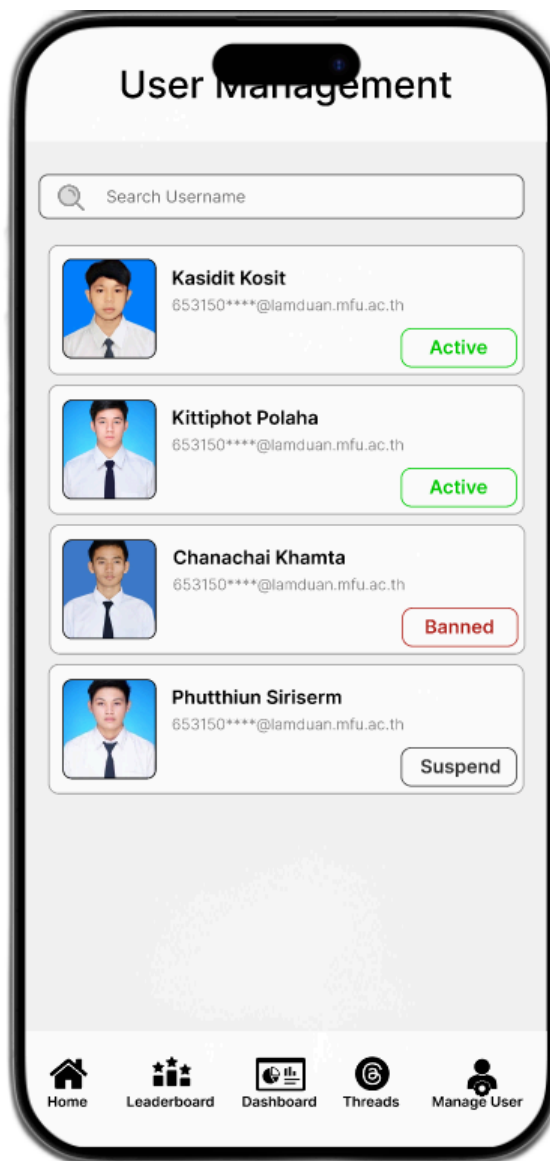


Figure 3.34 User Management Page (Admin)

On this page admin can see all the users with current status such as “Active”, “Banned”, “Suspended”, and can also manage their status from this page.

3.4.3.14 Admin Profile Page



Figure 3.35 Admin Profile Page (Admin)

The Admin Profile Page displays the personal information of an Admin, including their full name, username (used within the app), email address, user status, a short bio, profile picture, and a 'Log Out' button. Additionally, Admins can edit their username, bio, and choose a profile picture at any time, allowing them to keep their profile information up-to-date and reflective of their preferences.

3.4.3.15 Exchange Coin Management Page

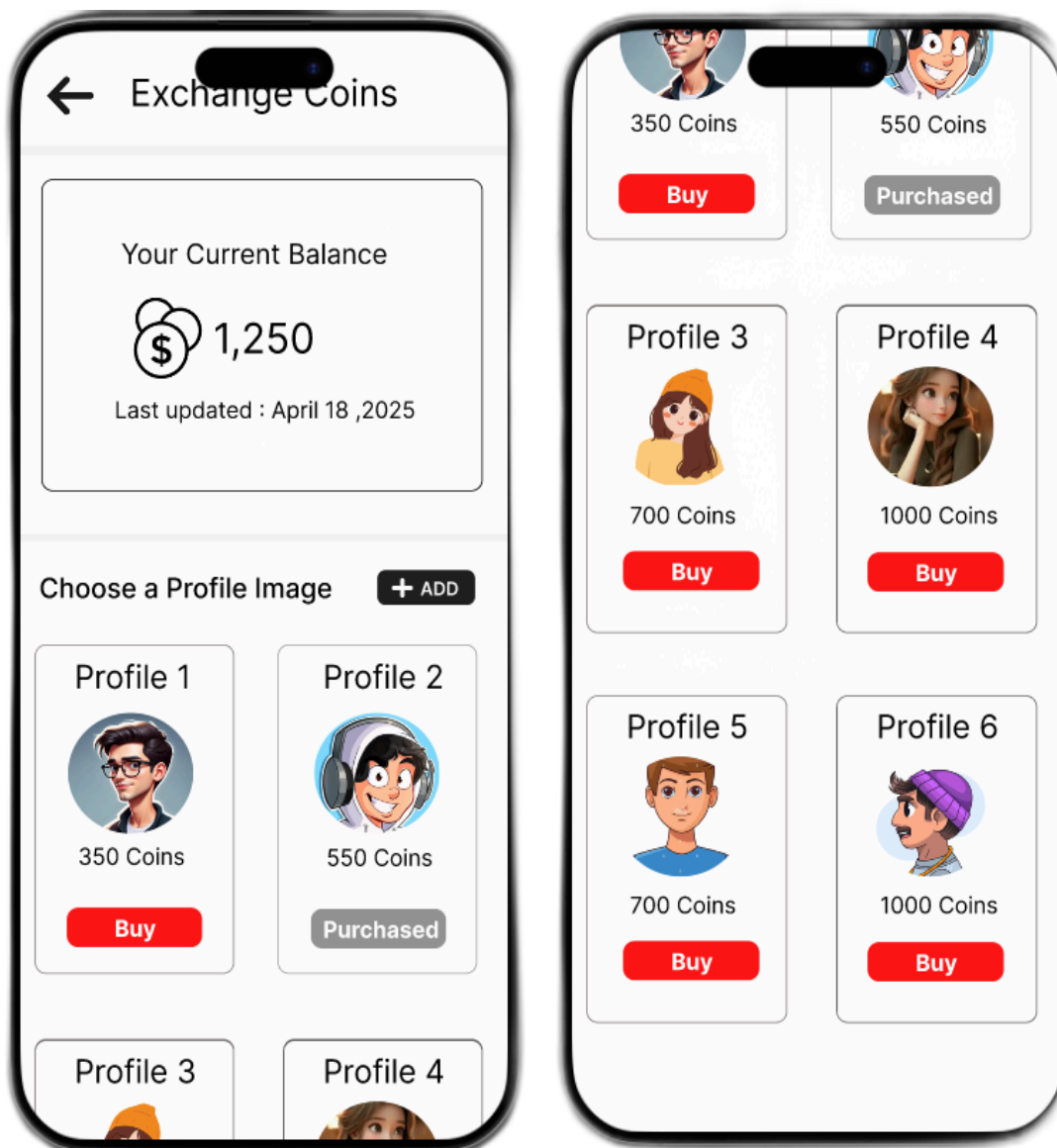


Figure 3.36 Exchange Coin Management Page (Admin)

This page is primarily used by admins to manage profile images available for purchase. Using the “**Add**” button, admins can upload new profile images, set their coin prices, and make them available in the profile shop.

3.4.3.16 Admin Profile Image Management Page

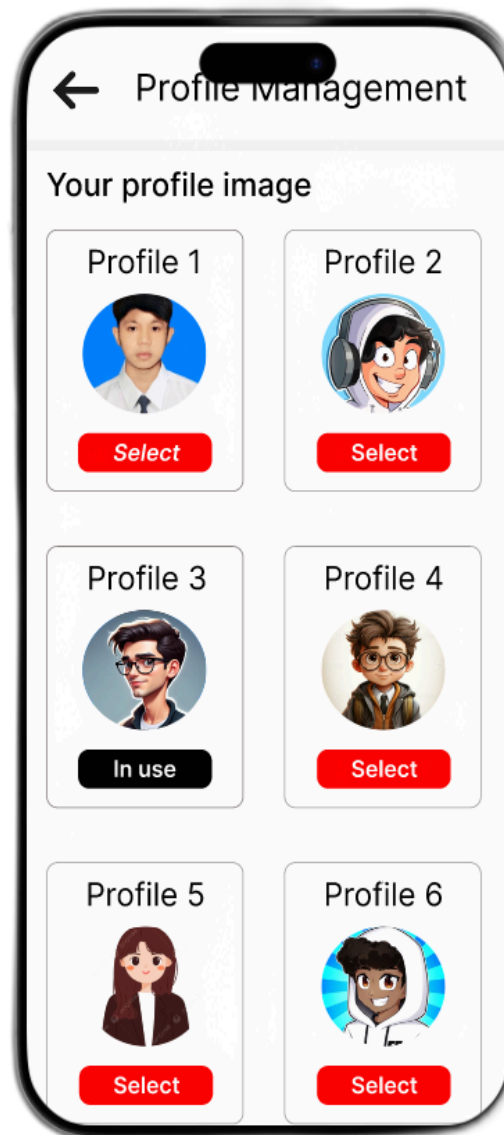


Figure 3.37 Admin Profile Management Page (Admin)

The Admin Profile Management page is accessed when the Admin clicks the profile image icon on their Admin Profile page. On this page, the Admin can update their profile picture by choosing an image earned through exchanging coins.

3.4.3.17 Coin History Page

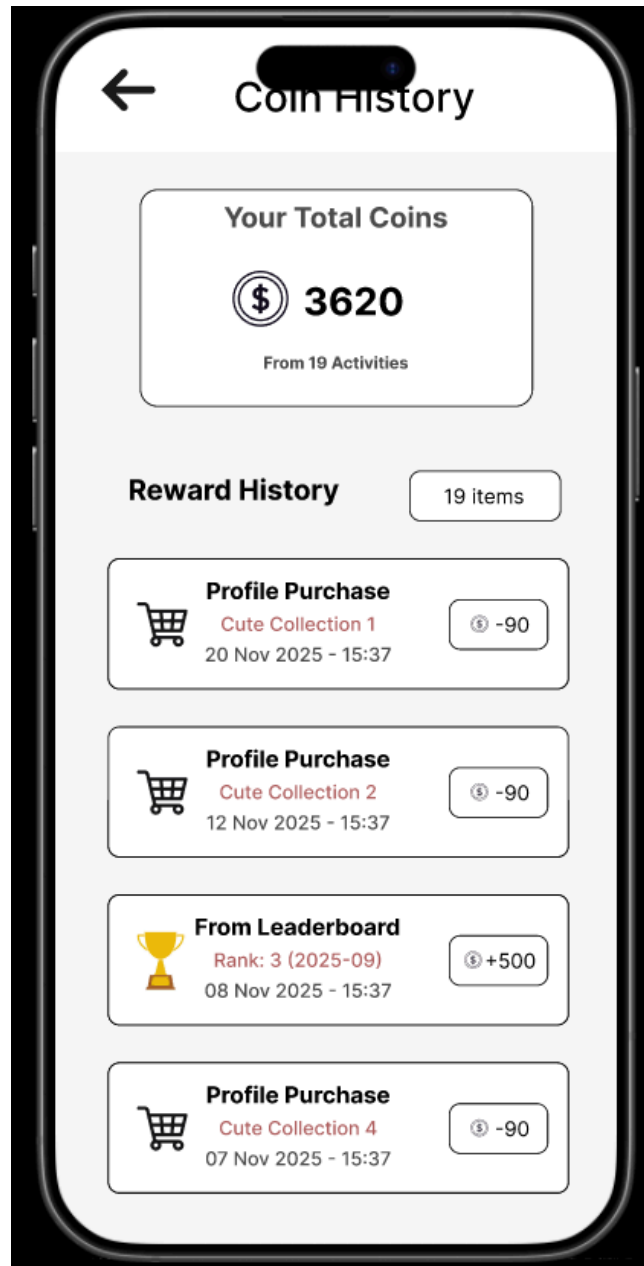


Figure 3.38 Coin History Page (Admin)

This page displays the complete history of all coin-related activities performed by the user within the application. At the top, users can see their total accumulated coins along with the number of activities that contributed to this total

3.4.3.18 User Management Page

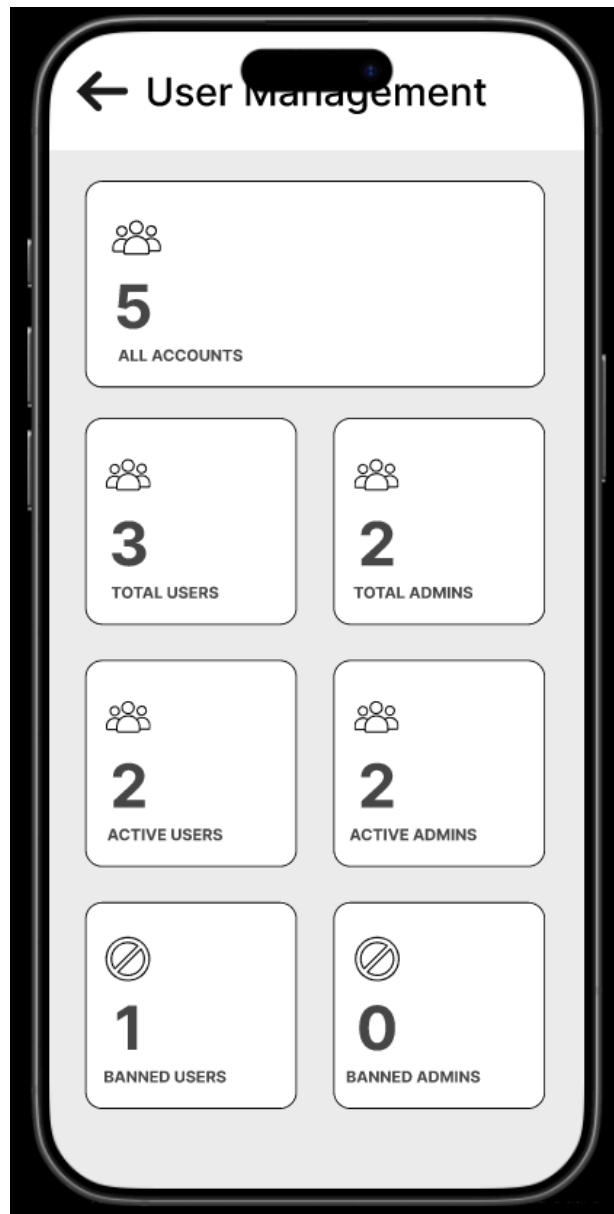


Figure 3.39 User Management Page (Admin)

The page displays the total number of accounts, along with a breakdown between regular users and administrators. It also shows how many of these accounts are currently active and how many have been banned. Each category is presented in a separate card for clarity, making it easy for the admin to assess system activity and identify any issues at a glance.

CHAPTER 4

RESULTS

After completing the application's design phase, we proceeded to the development stage. Our focus was to ensure the application met all user requirements while creating an intuitive and user-friendly experience.

4.1 Application Results

For our application, we decided to have three roles: Guest, User, and Administrator.

4.1.1 Guest role

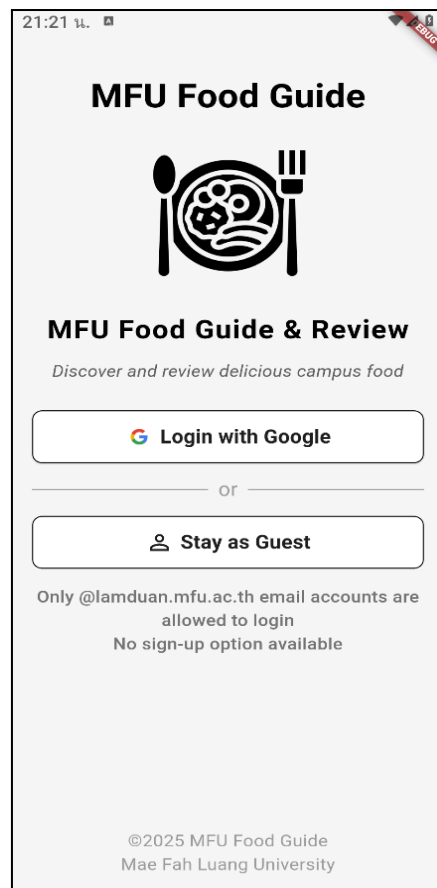


Figure 4.1 The interface of the login page

The login page was the first page of this app, which allows users to choose between MFU email and without an MFU email. These two ways give different authority to users.



Figure 4.2 The interface of the Homepage

The Home page shows all restaurants and lets users filter by rating, location, or food type, making it easy to quickly find places that match their preferences.

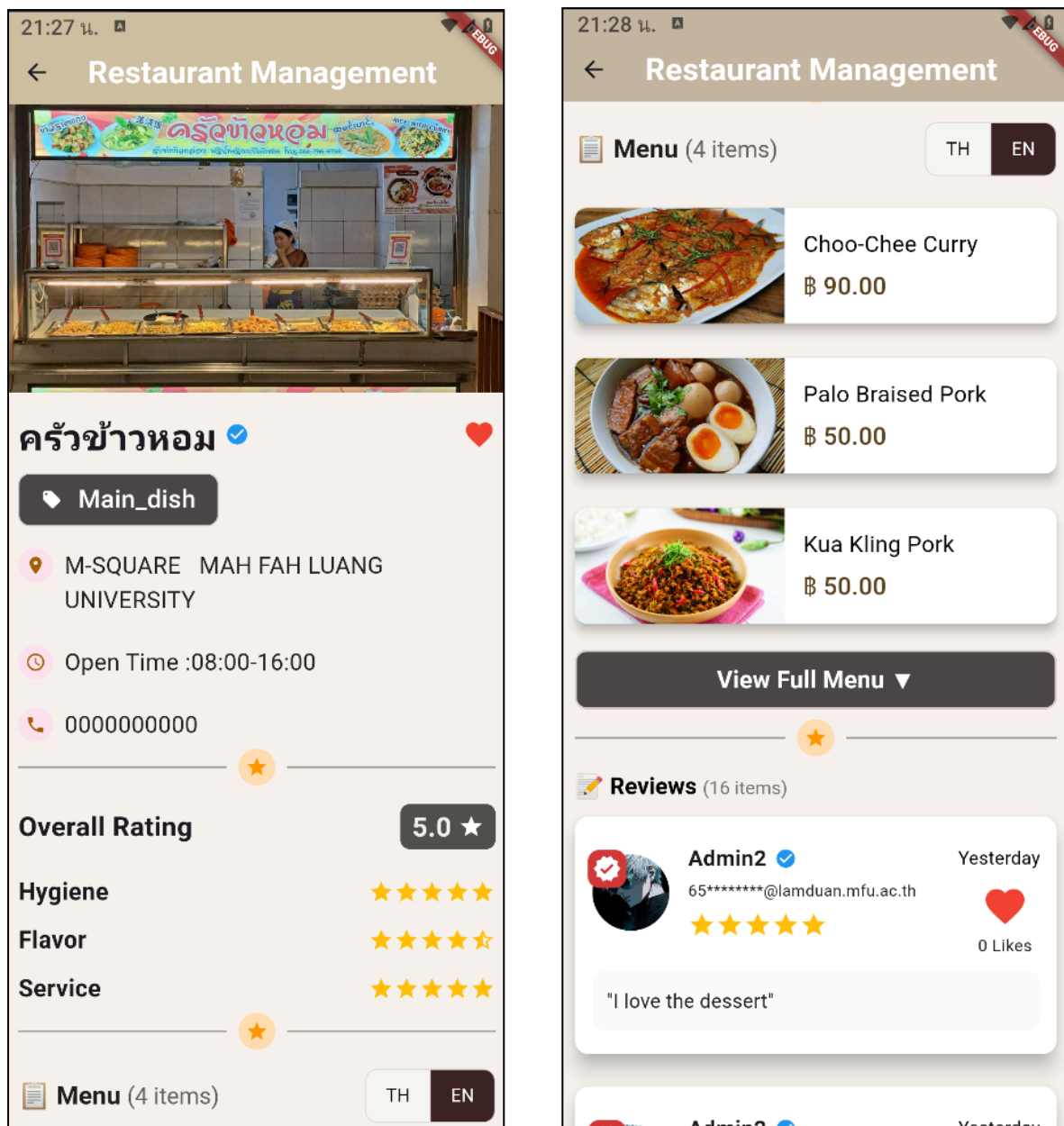


Figure 4.3 The interface of the Restaurant information page

When a user finds an interesting restaurant from the Home page and clicks on it, this page displays detailed restaurant information, including name, food type, location, opening hours, contact number, menu with images and prices, and all user reviews, with review submission restricted to MFU personnel only.

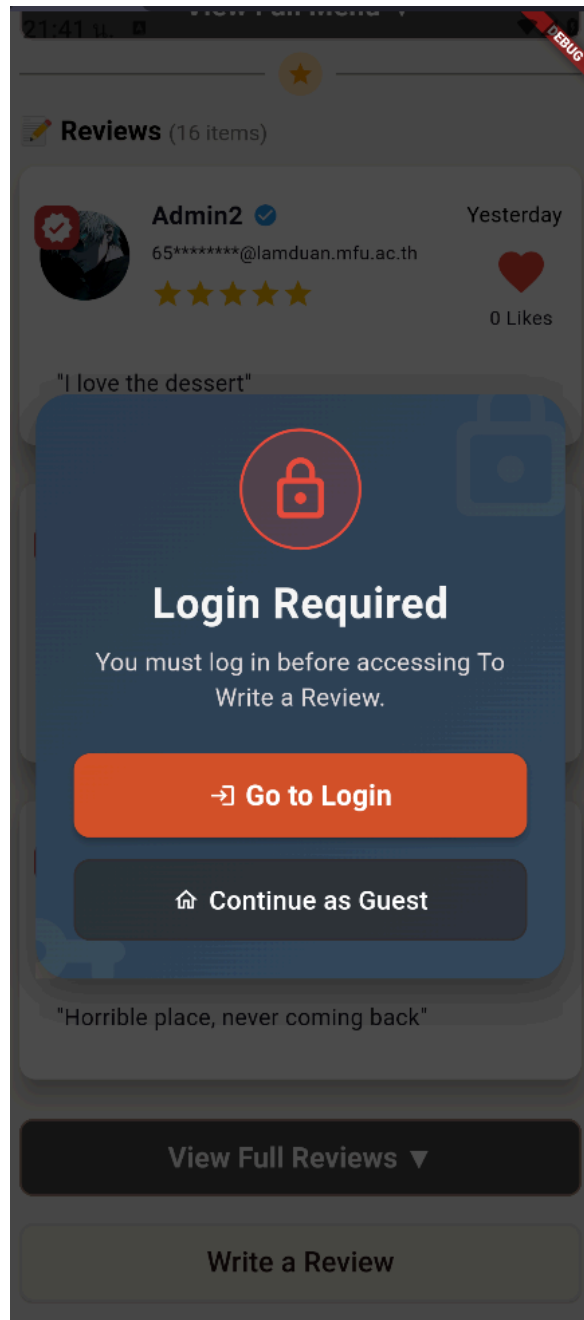


Figure 4.4 The interface of Reviewpage

When a user taps the 'Write Review' button on the Restaurant Information page without being logged in, an alert will appear notifying them that access is restricted users must log in to write a review.

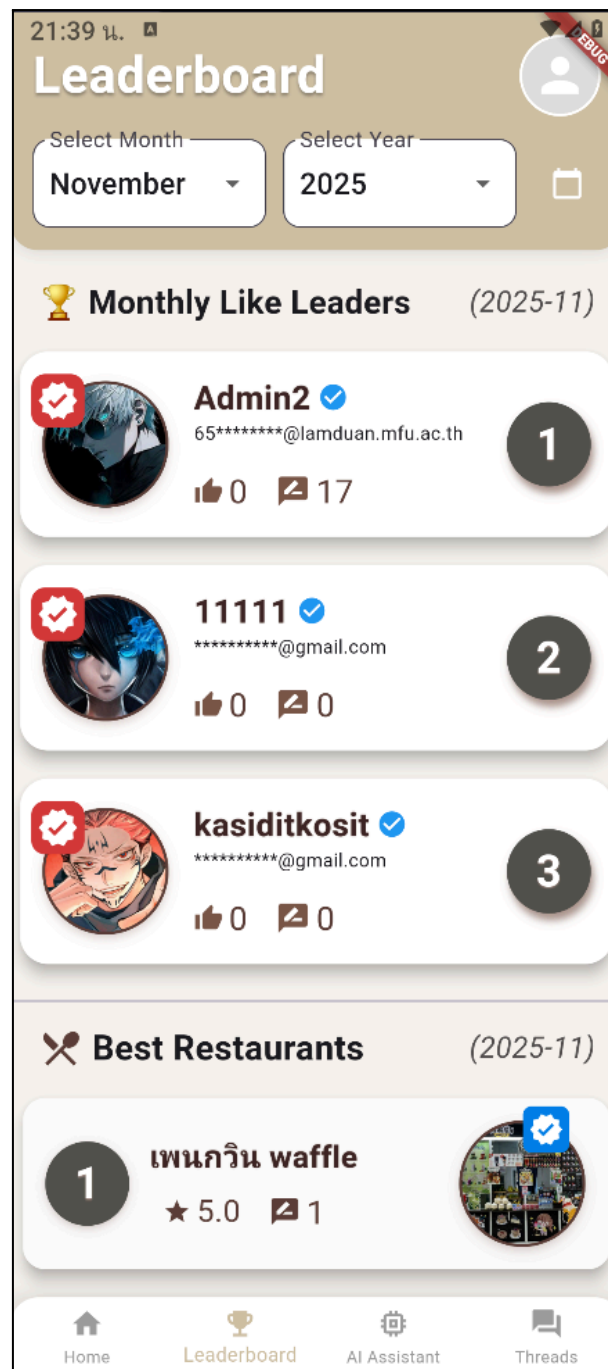


Figure 4.5 The interface of the Leaderboard page

The Leaderboard page displays two sections: the top 3 restaurants with the highest overall ratings, and the top 3 users with the most total likes on their restaurant reviews for each month.

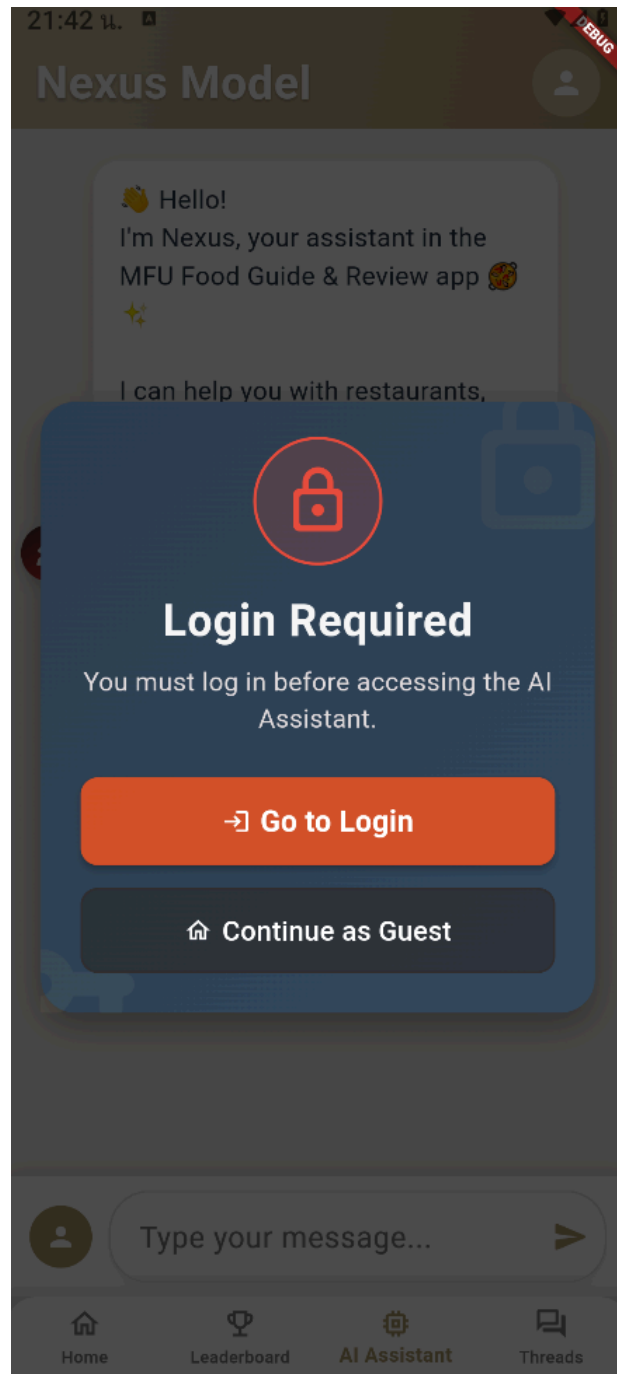


Figure 4.6 The interface of the Login alert for the guest user

When a user navigates to a page that requires authentication, an alert will appear notifying them that access is restricted and users must log in to continue. This ensures that only authenticated members can access protected pages within the MFU community.

4.1.2 User role

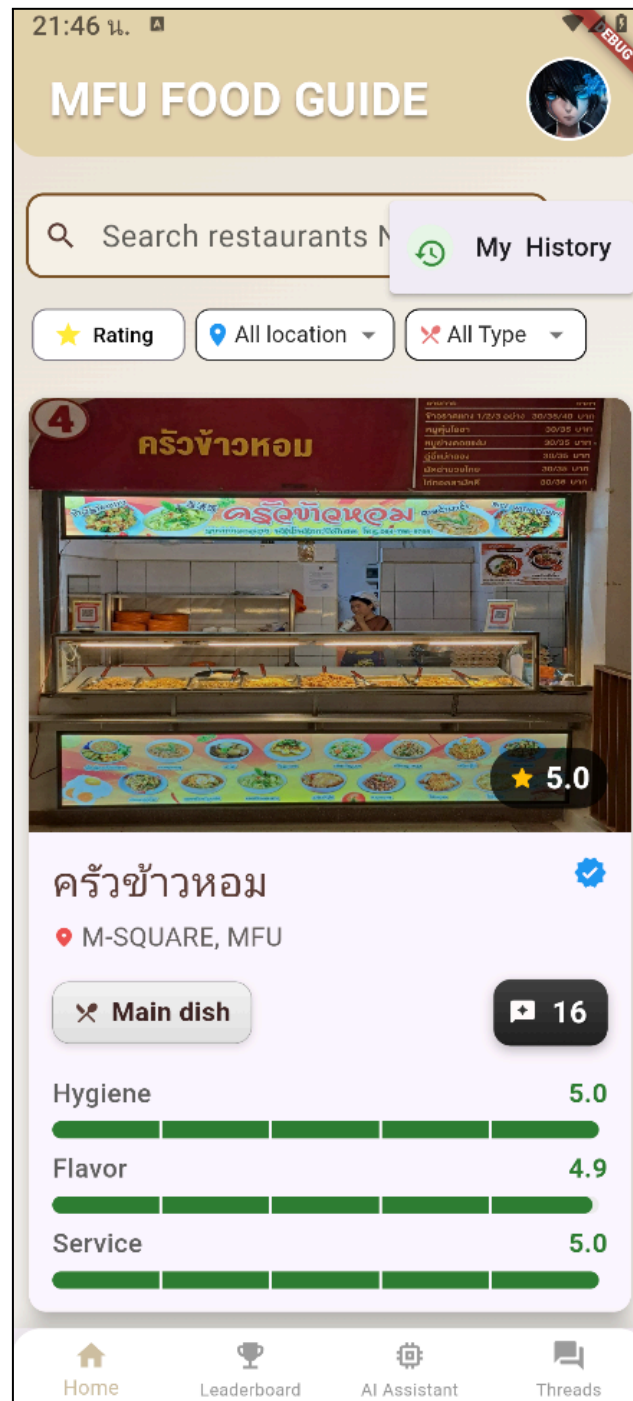


Figure 4.7 The interface of the Home page

The Home page shows all restaurants and lets users filter by rating, location, or food type, making it easy to quickly find places that match their preferences.

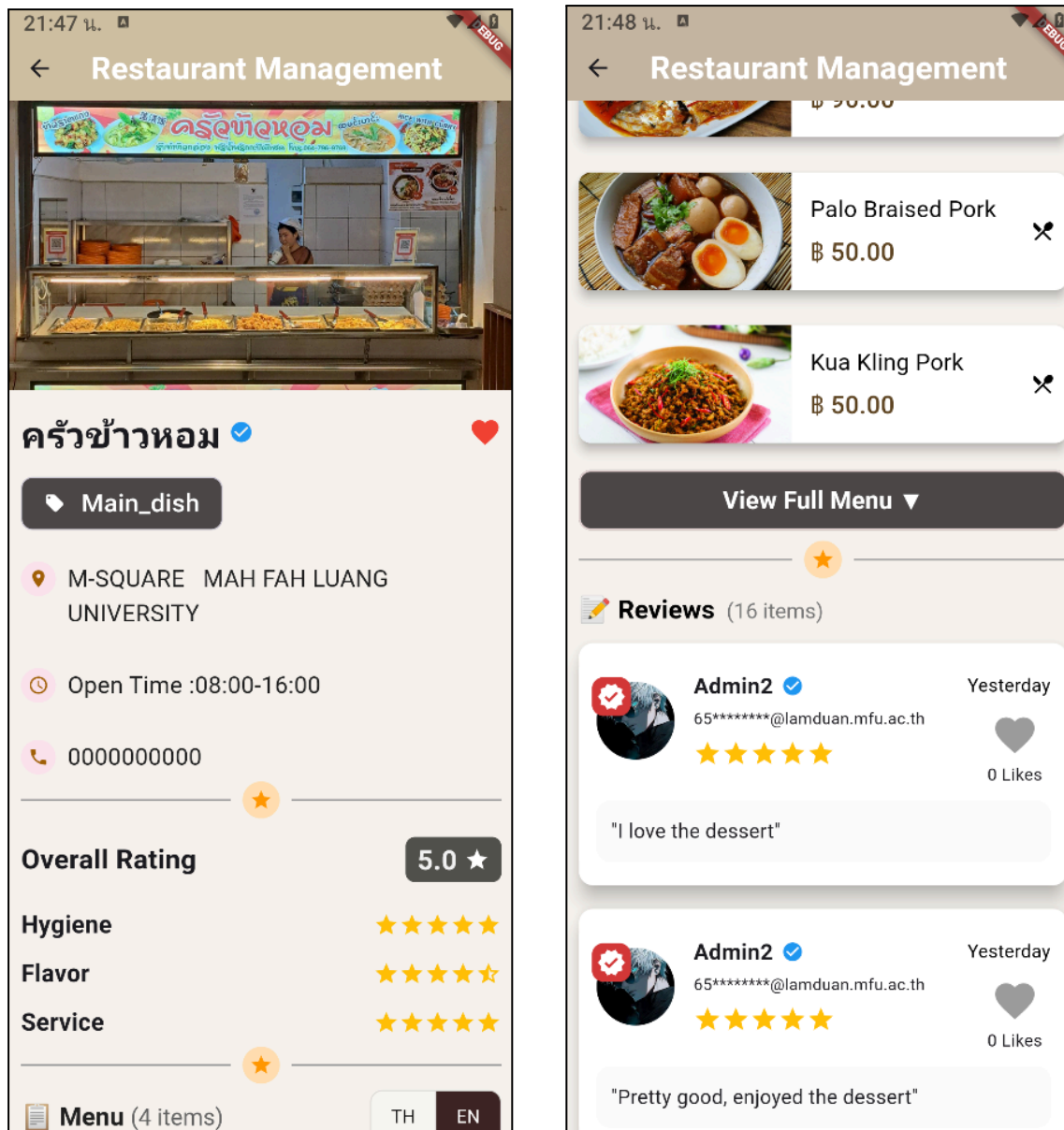


Figure 4.8 The interface of the Restaurant information page

When a user finds an interesting restaurant from the Home page and clicks on it, this page displays detailed restaurant information, including name, food type, location, opening hours, contact number, menu with images and prices, and all user reviews, with review submission restricted to MFU personnel only.

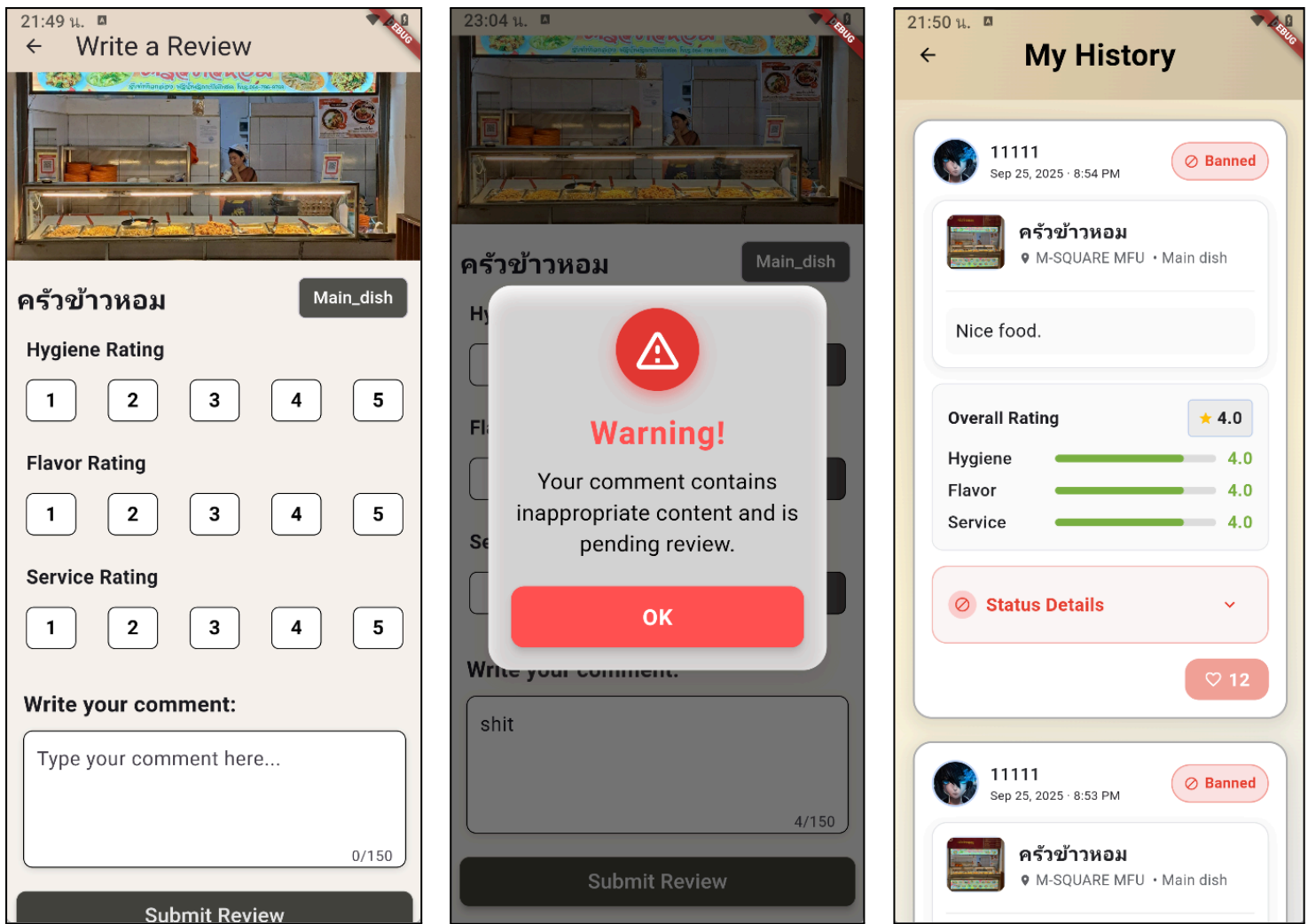


Figure 4.9 The interface of the Review page and My History Page

This page allows users to give a restaurant feedback on their food, service, and hygiene by rating on a scale of 1-5. Users can also write a comment. For this reason, we decided to add AI word detection. This system model can detect inappropriate content and ban it. With this one, we're sure that our app will be clean and Suitable for all ages.

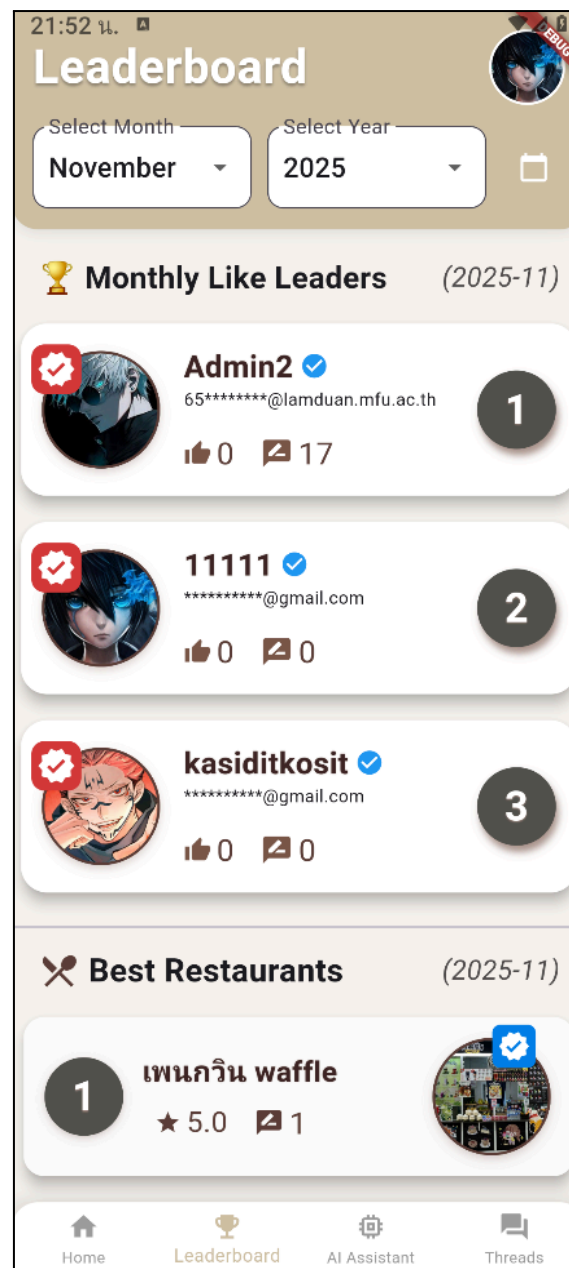


Figure 4.10 The interface of the Leaderboard page

The Leaderboard page displays two sections: the top 3 restaurants with the highest overall ratings, and the top 3 users with the most total likes on their restaurant reviews for each month.

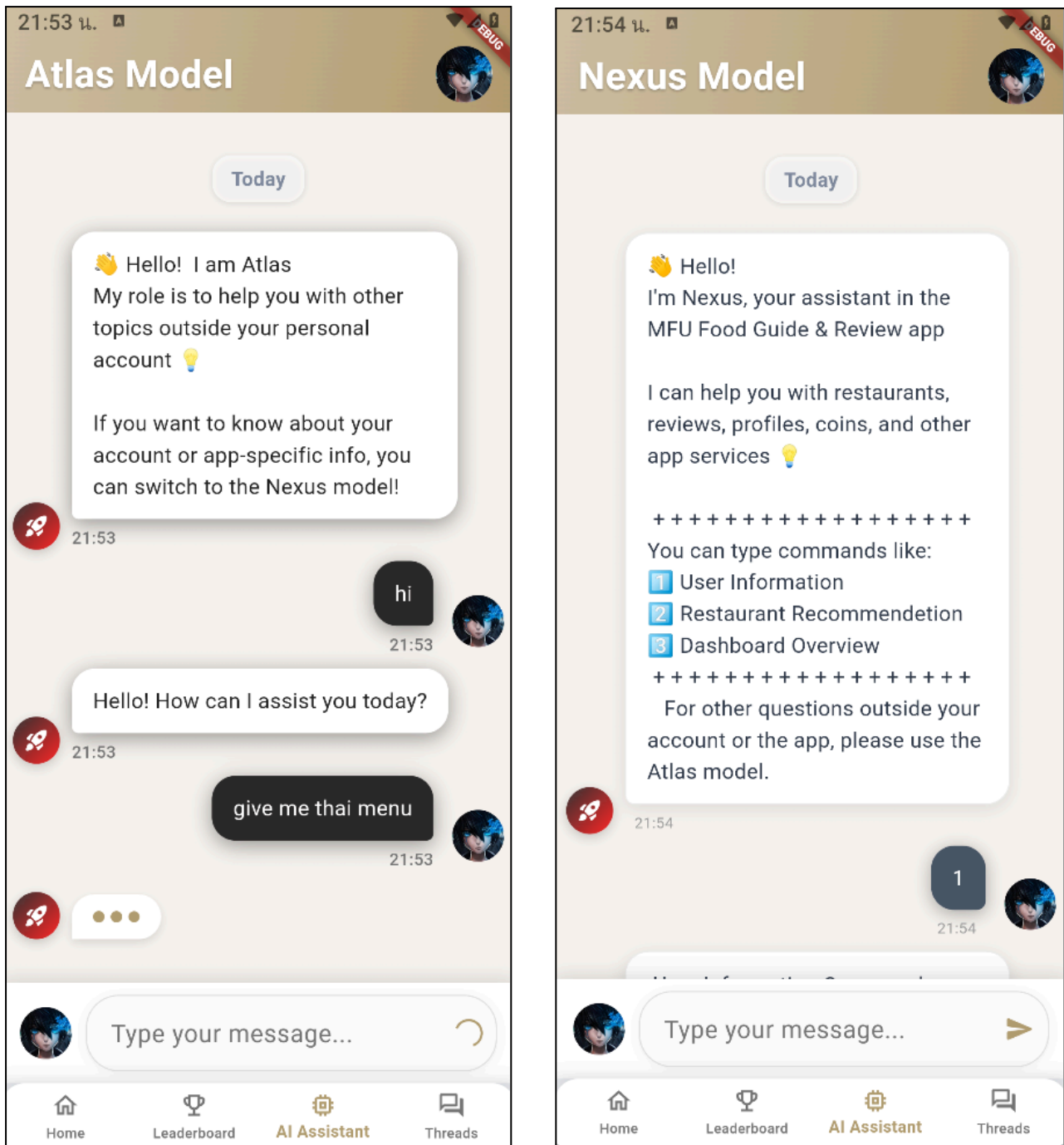


Figure 4.11 The interface of the AI Assistant page

The AI Assistant page provides access to two models: the **Nexus model**, used for answering queries related to in-app data, and the **Atlas model**, used for handling general questions outside the application.

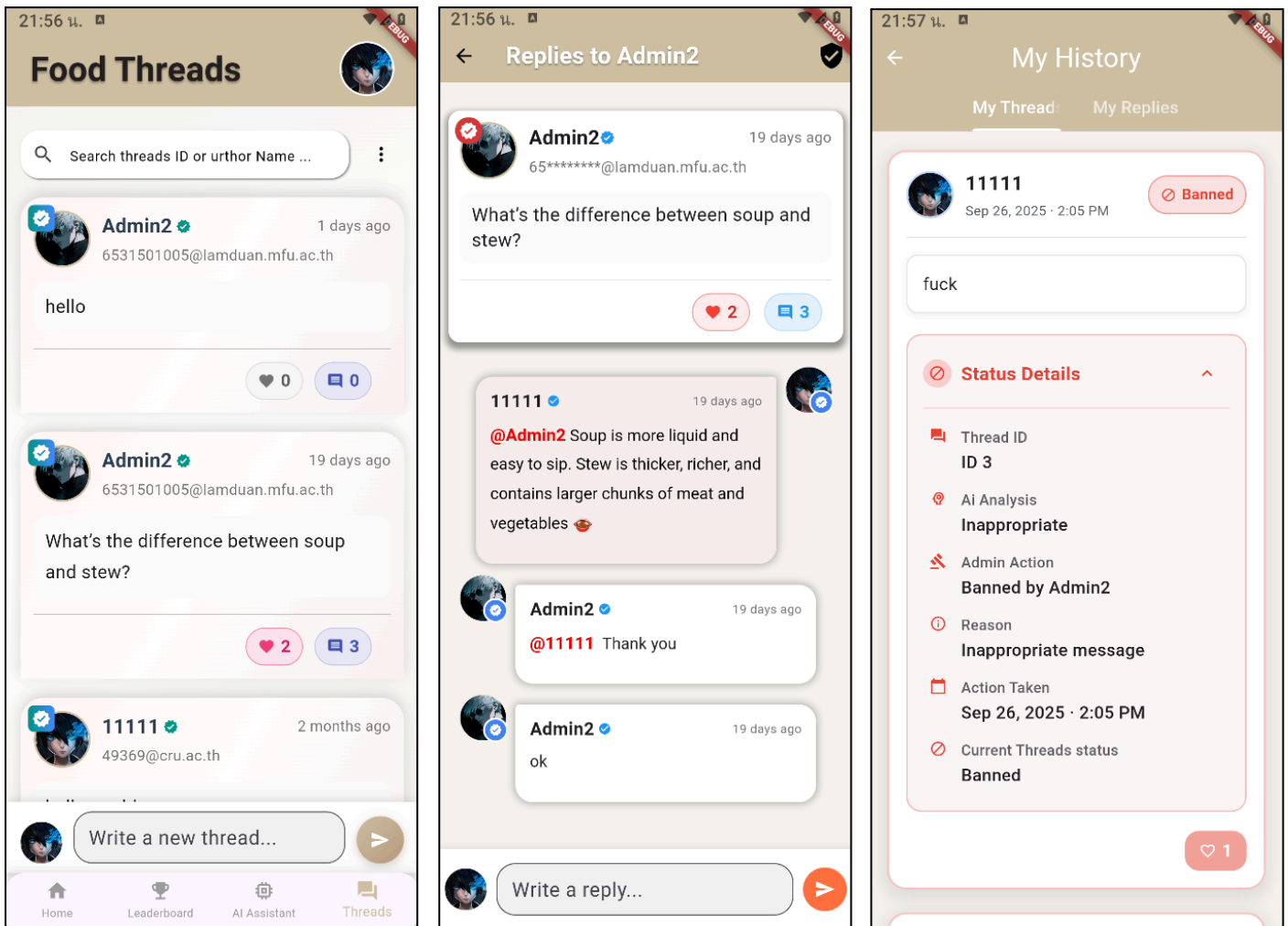


Figure 4.12 The interface of the Thread page

The Threads page allows users to share and discuss food-related topics. Inappropriate language is detected using the Perspective API [14], and any flagged Thread is set to pending. Admins can then review and either approve or ban the Thread.

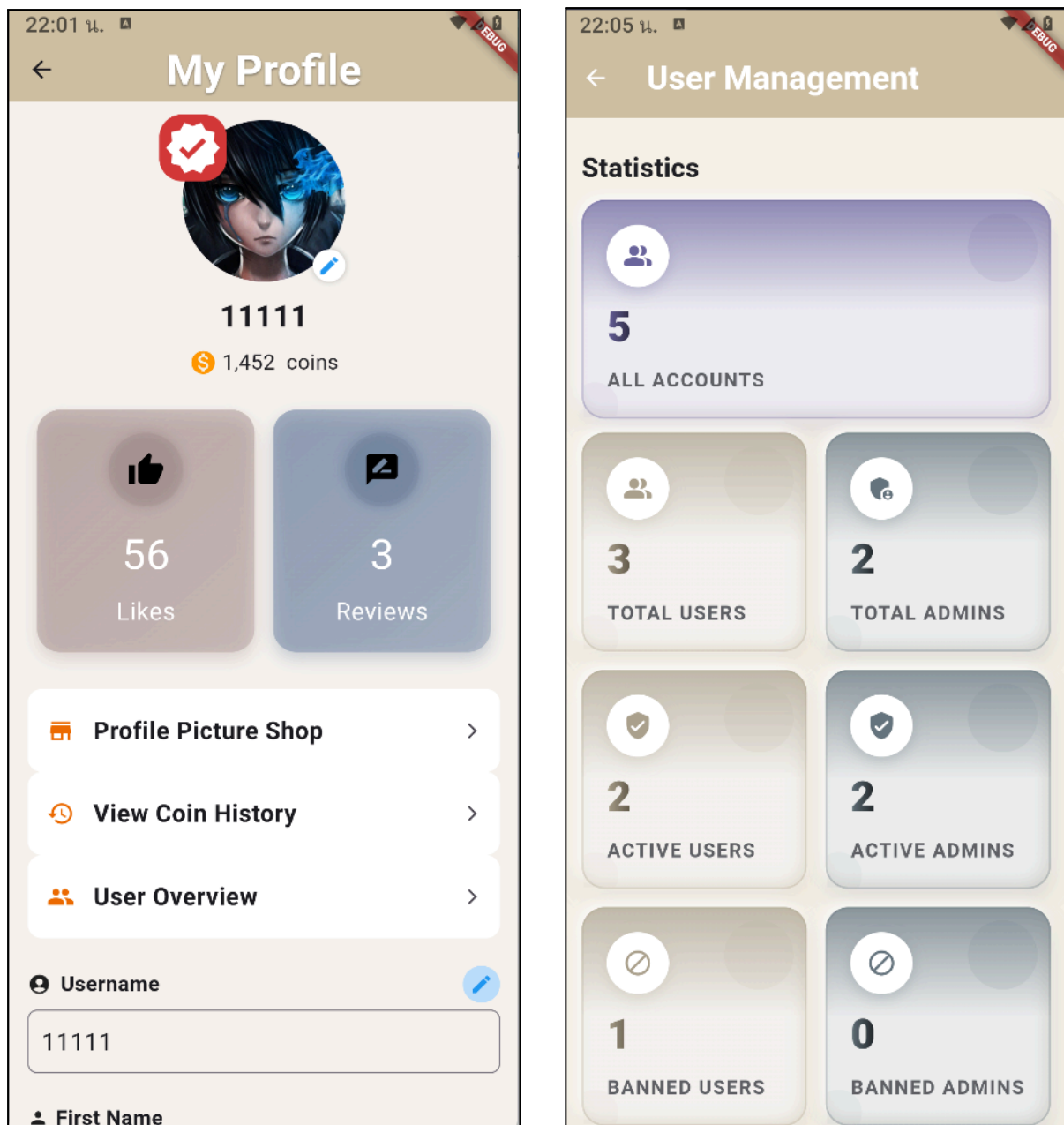


Figure 4.13 The interface of the Profile and the user overviews page

This page displays the user's personal information, including , contact info, and other account-related data.

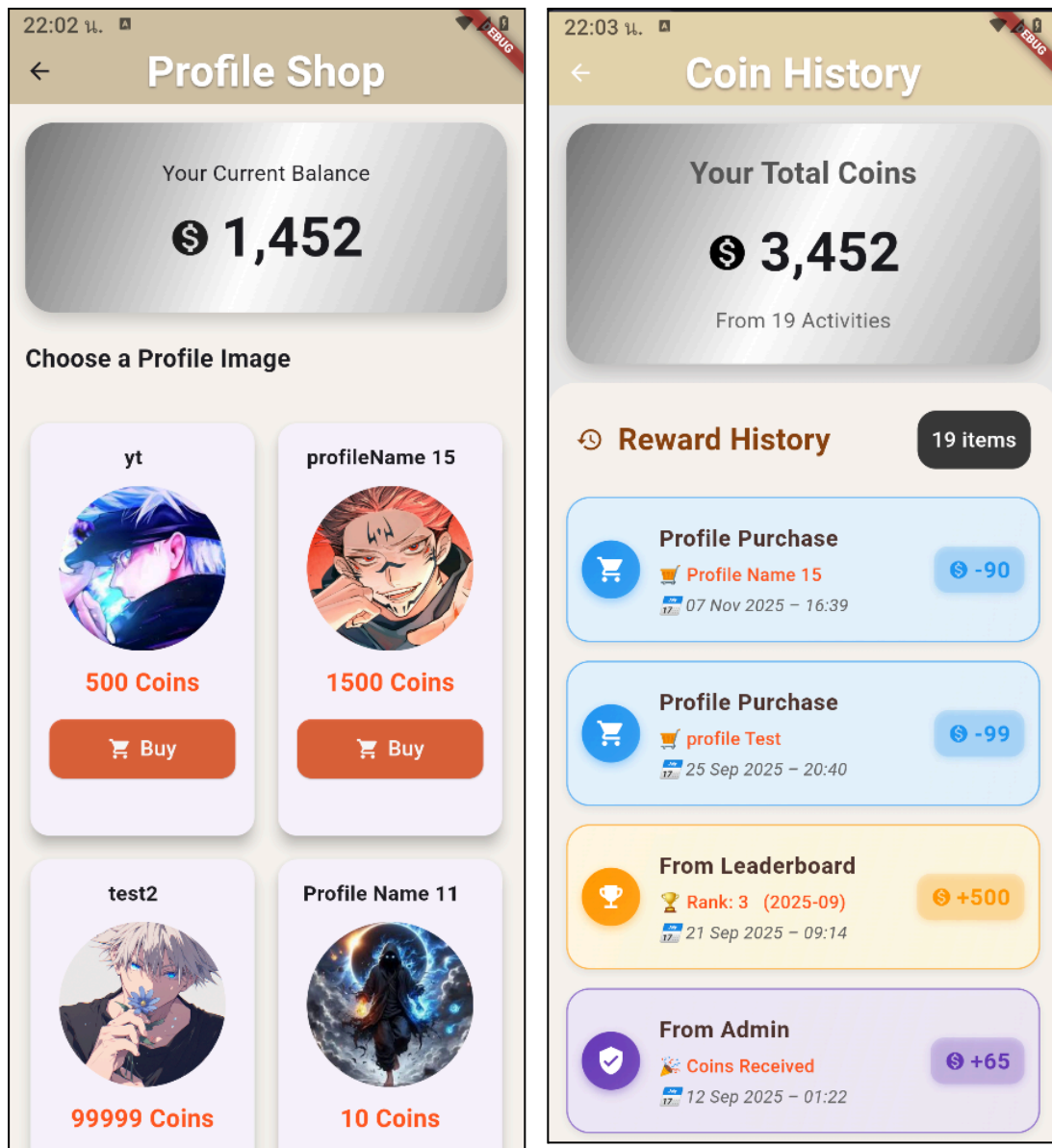


Figure 4.14 The interface of the Profile shop and Coin history page

This page allows users to view and manage their virtual coins, track coin transaction history, and access the in-app shop.

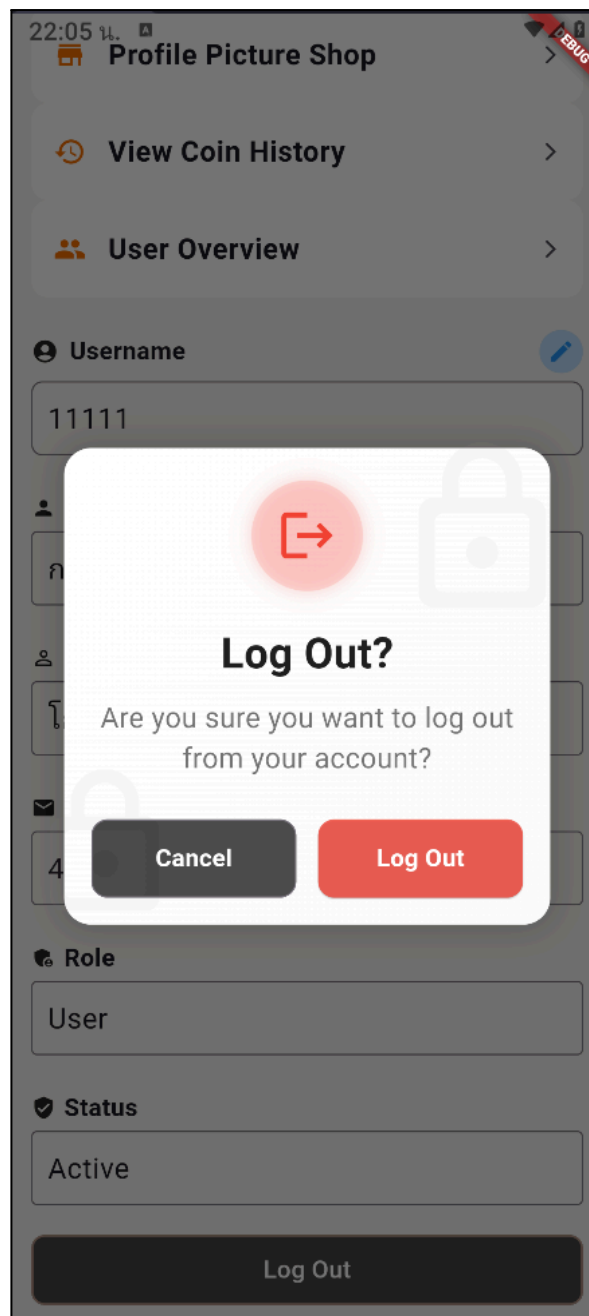


Figure 4.15 The interface of the Logout Page.

This page allows users to securely log out of the application and end their session

4.1.3 Administration role

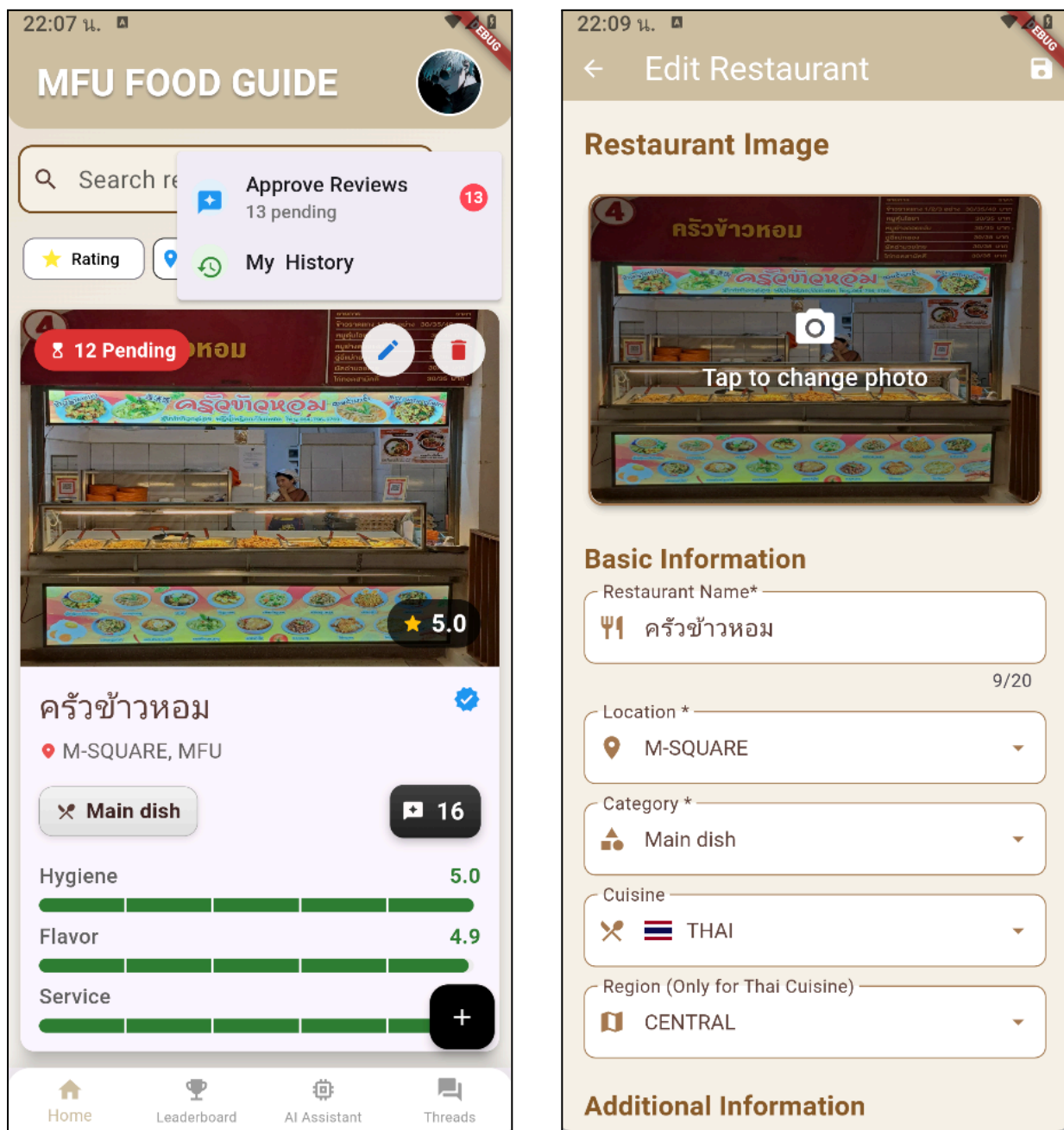


Figure 4.16 The interface of the Home and Edit Restaurant page

The interface for both the Home page and the Edit Restaurant page is designed to be simple, intuitive, and user-friendly, allowing administrators to quickly view restaurant information and make edits efficiently.

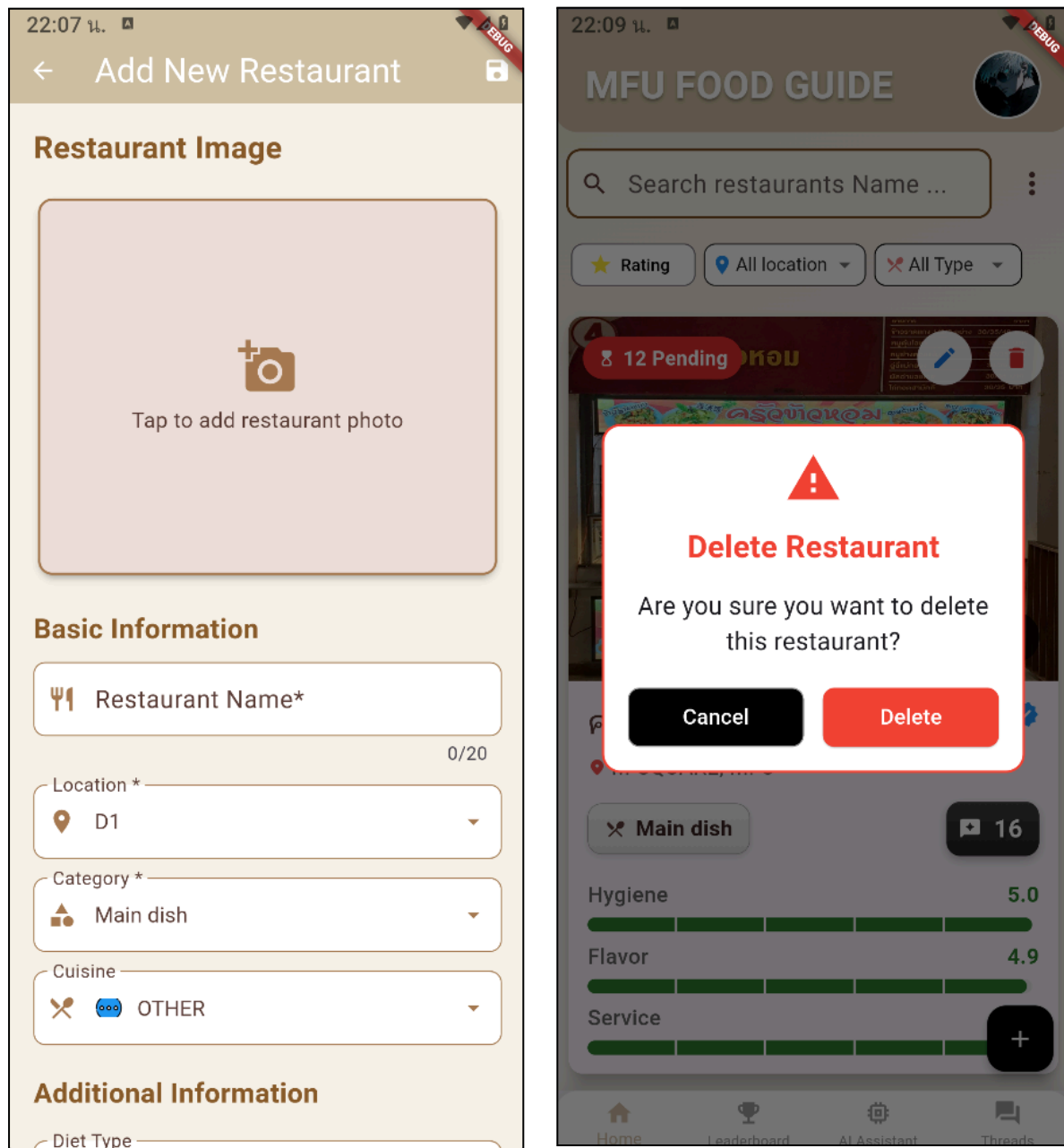


Figure 4.17 The interface of the Add and Delete Restaurant page

This figure shows the interface where administrators can add new restaurants to the system or delete existing ones, providing efficient management of restaurant data.

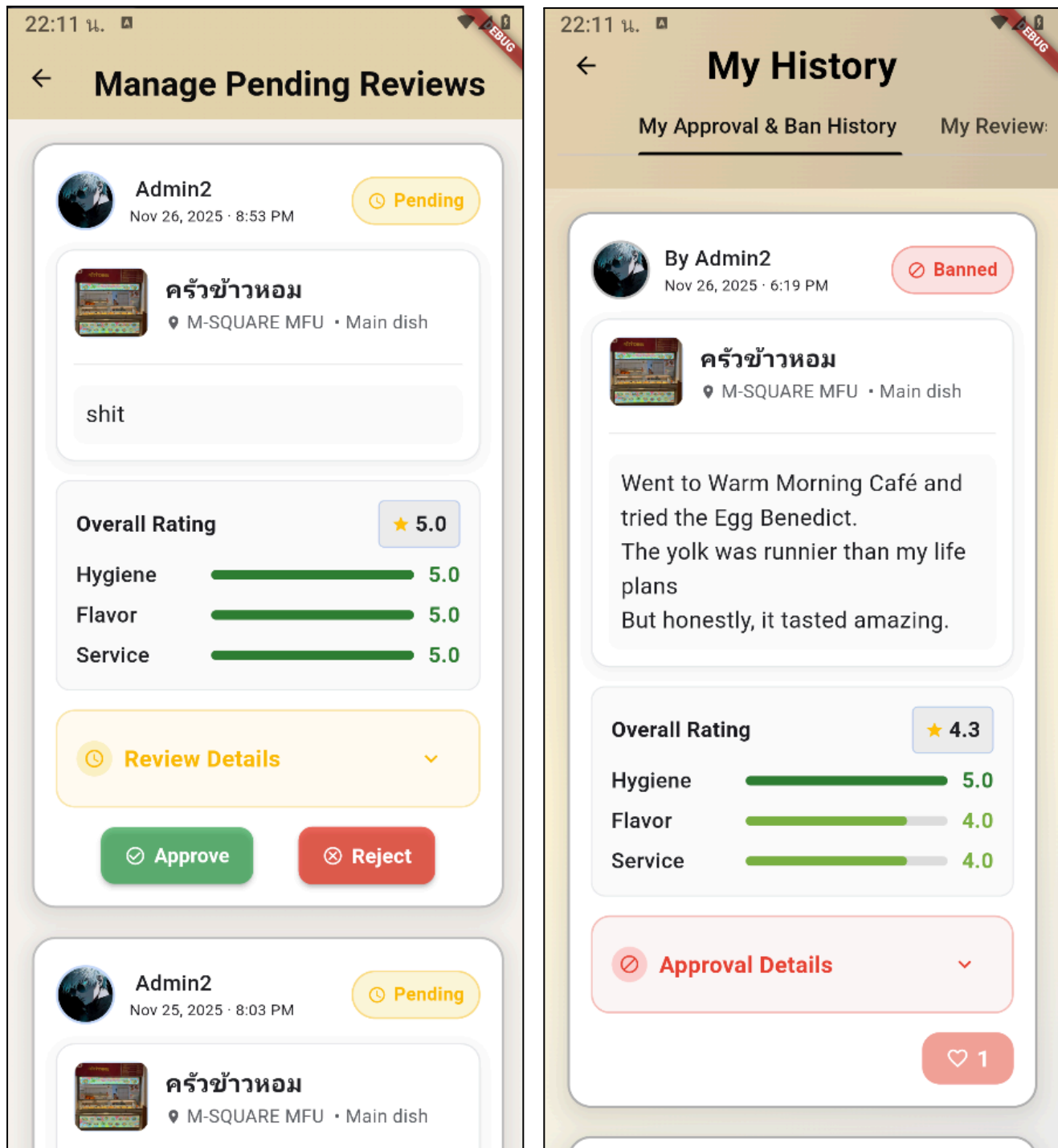


Figure 4.18 The interface of Approve reviews and the History page

This figure illustrates the interface that allows administrators to approve user reviews and view the history of all moderated reviews within the system.

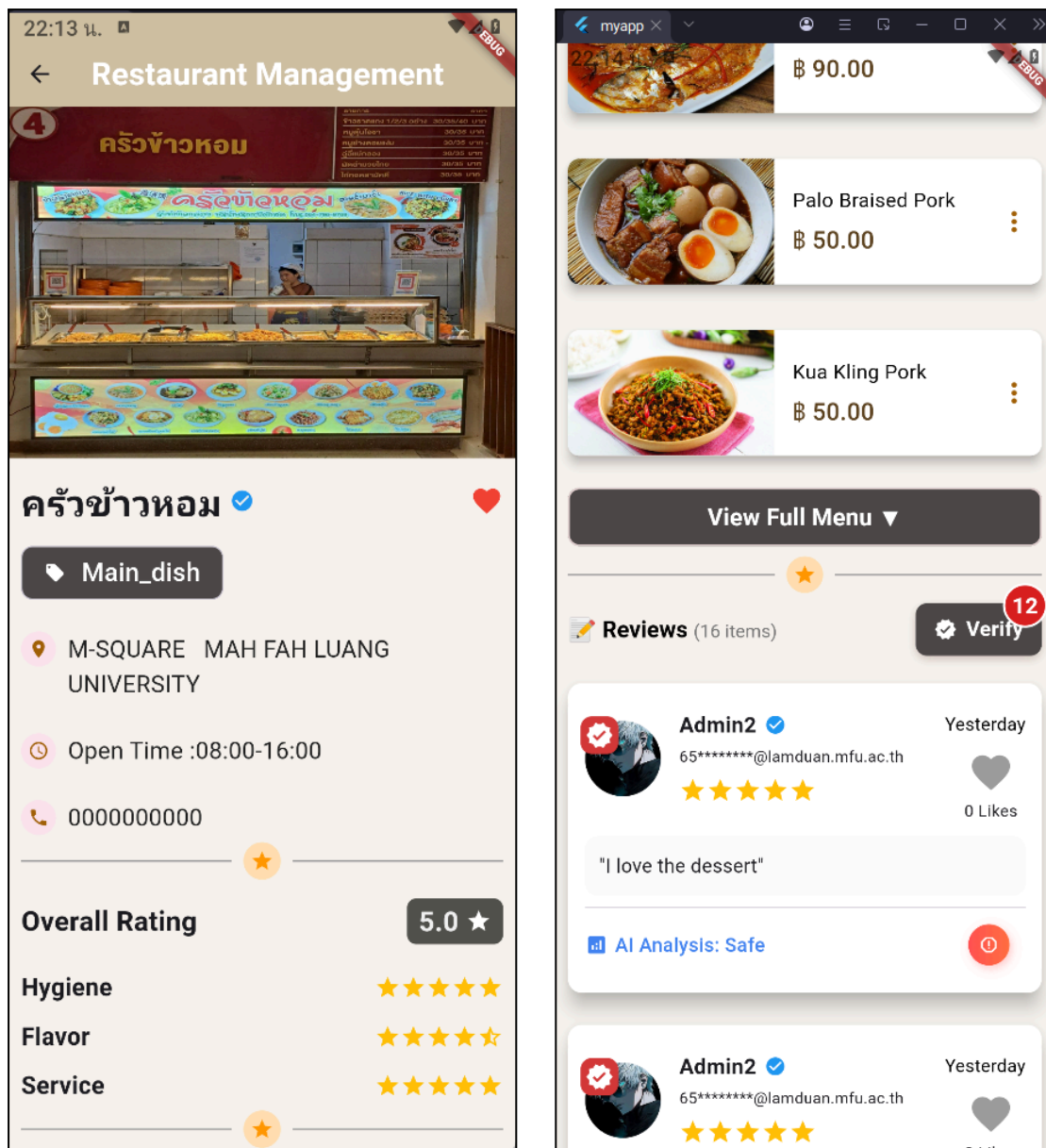


Figure 4.19 The interface of the Restaurants information page

When a user selects a restaurant from the homepage, this page shows restaurant details such as name, food type, location, opening hours, contact, menu with images and prices, and user reviews with an option for logged-in users to submit their own review.

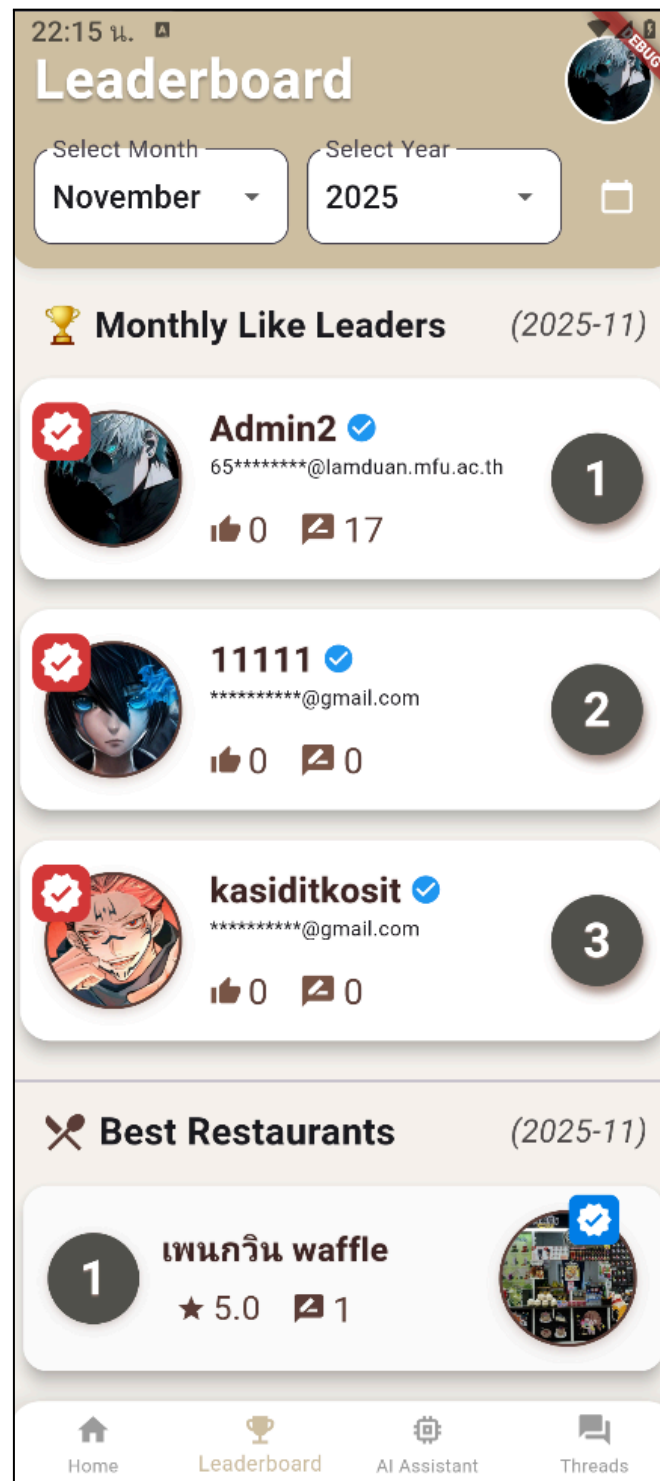


Figure 4.20 The interface of the Leaderboard page

The Leaderboard page displays two sections: the top 3 restaurants with the highest overall ratings, and the top 3 users with the most total likes on their restaurant reviews for each month.

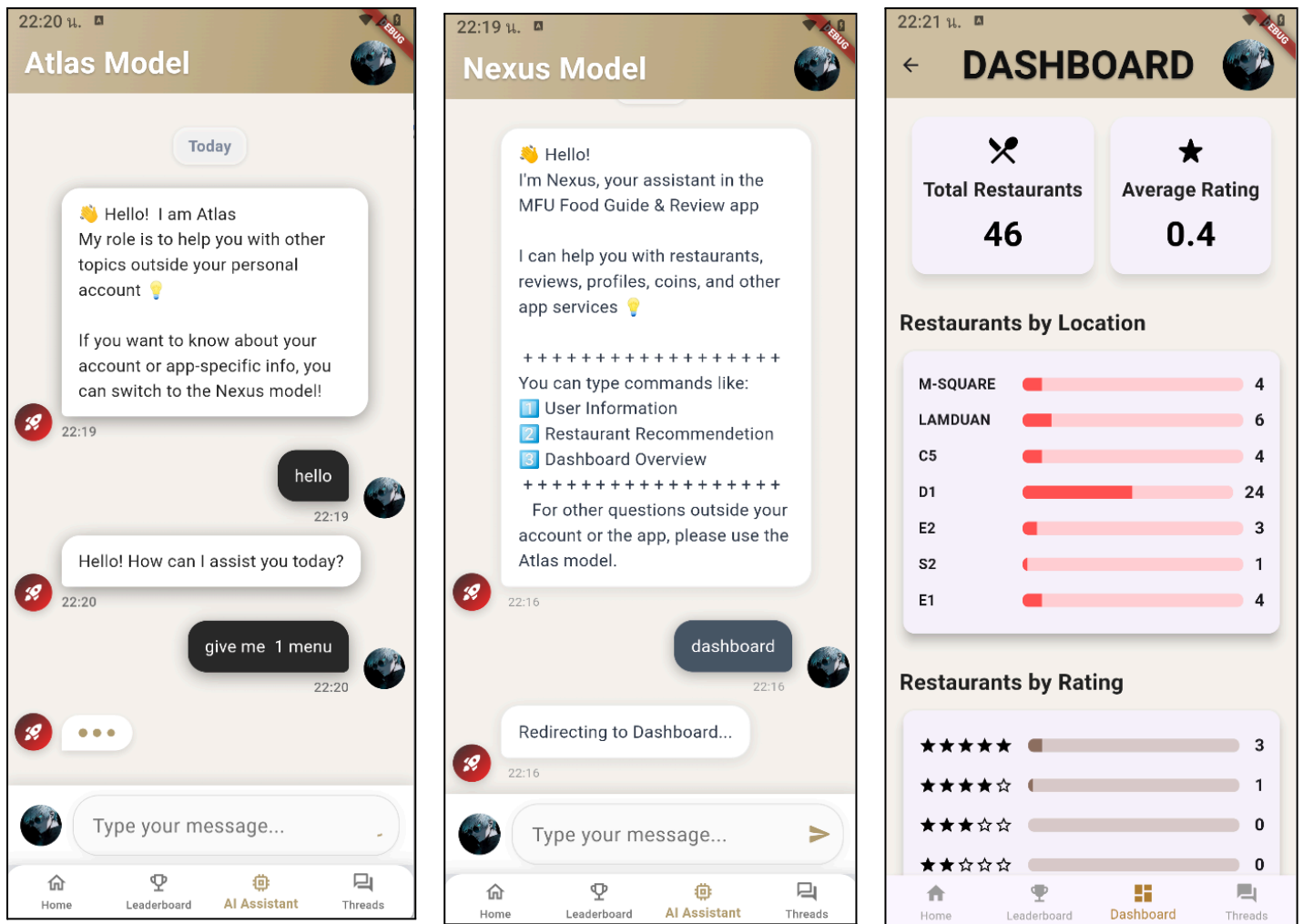


Figure 4.21 The interface of the AI Assistant page

The AI Assistant page provides access to two models: the **Nexus model**, used for answering queries related to in-app data, and the **Atlas model**, used for handling general questions outside the application.

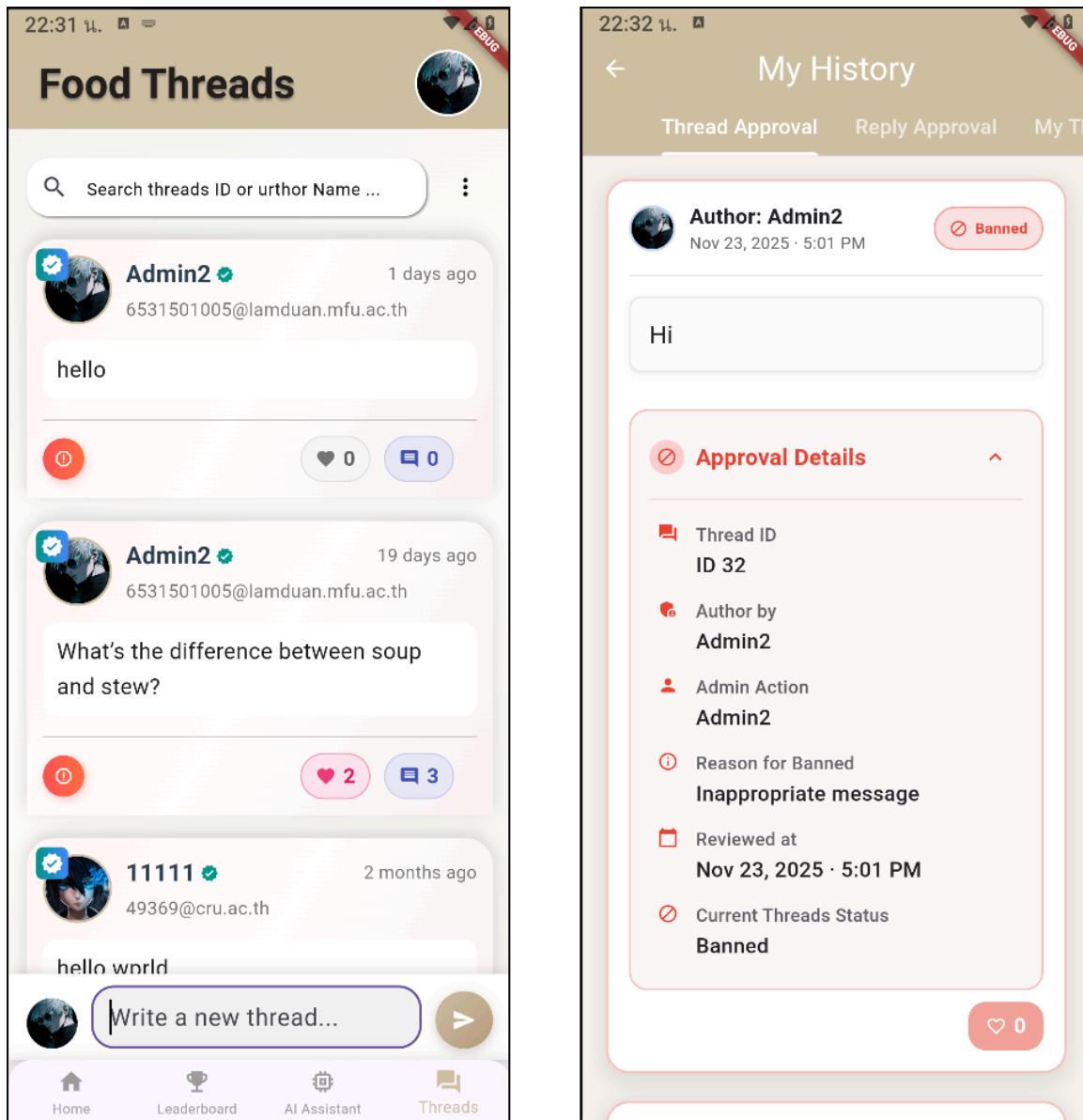


Figure 4.22 The interface of the Thread page

The Threads page allows users to share and discuss food-related topics. Inappropriate language is detected using the Perspective API [14], and any flagged Thread is set to pending. Admins can then review and either approve or ban the Thread.

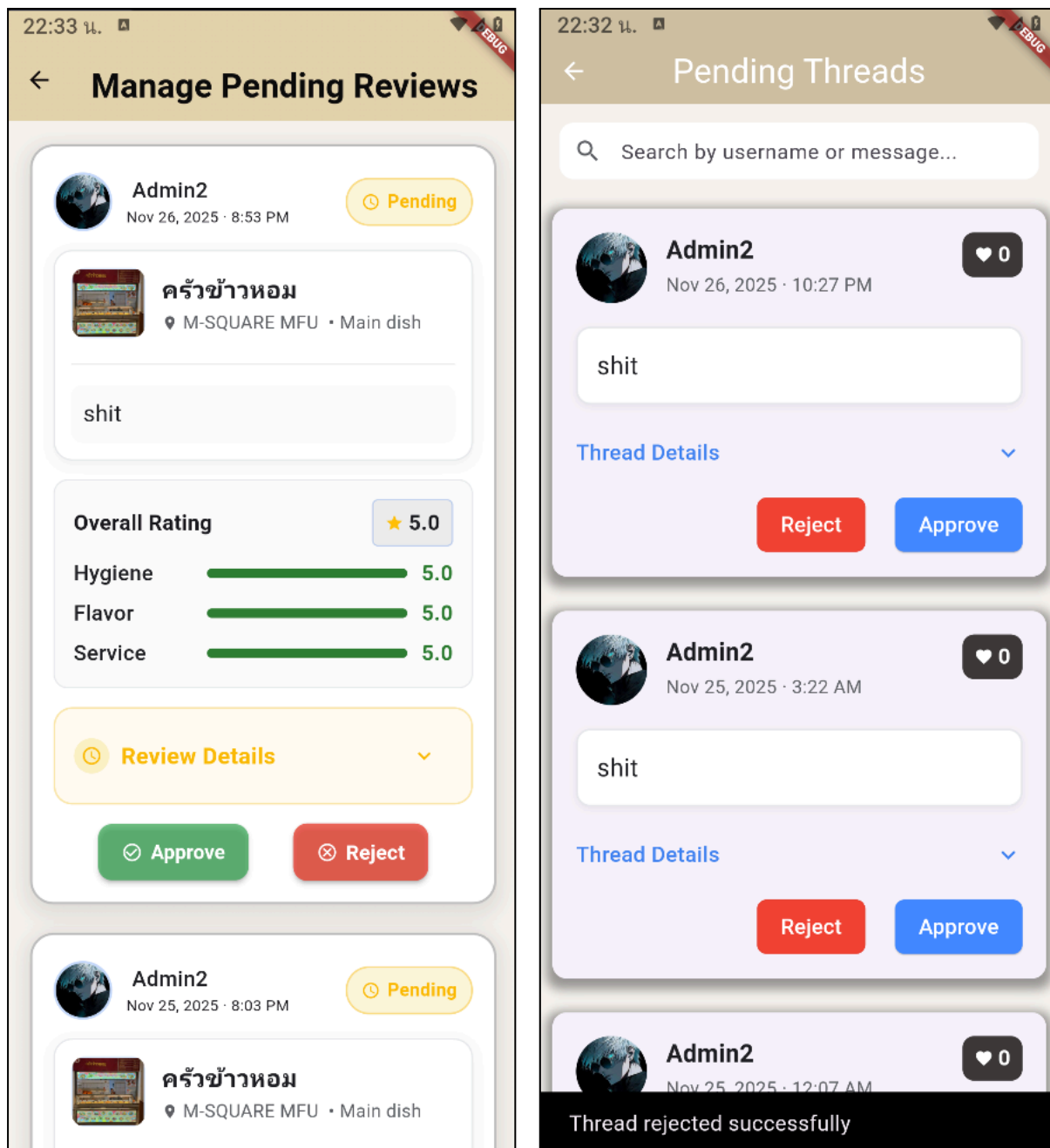


Figure 4.23 The interface of the Verified Threads page

This page is used to manage Threads flagged by the Perspective API [14] for inappropriate language, allowing admins to review, approve, or remove them.

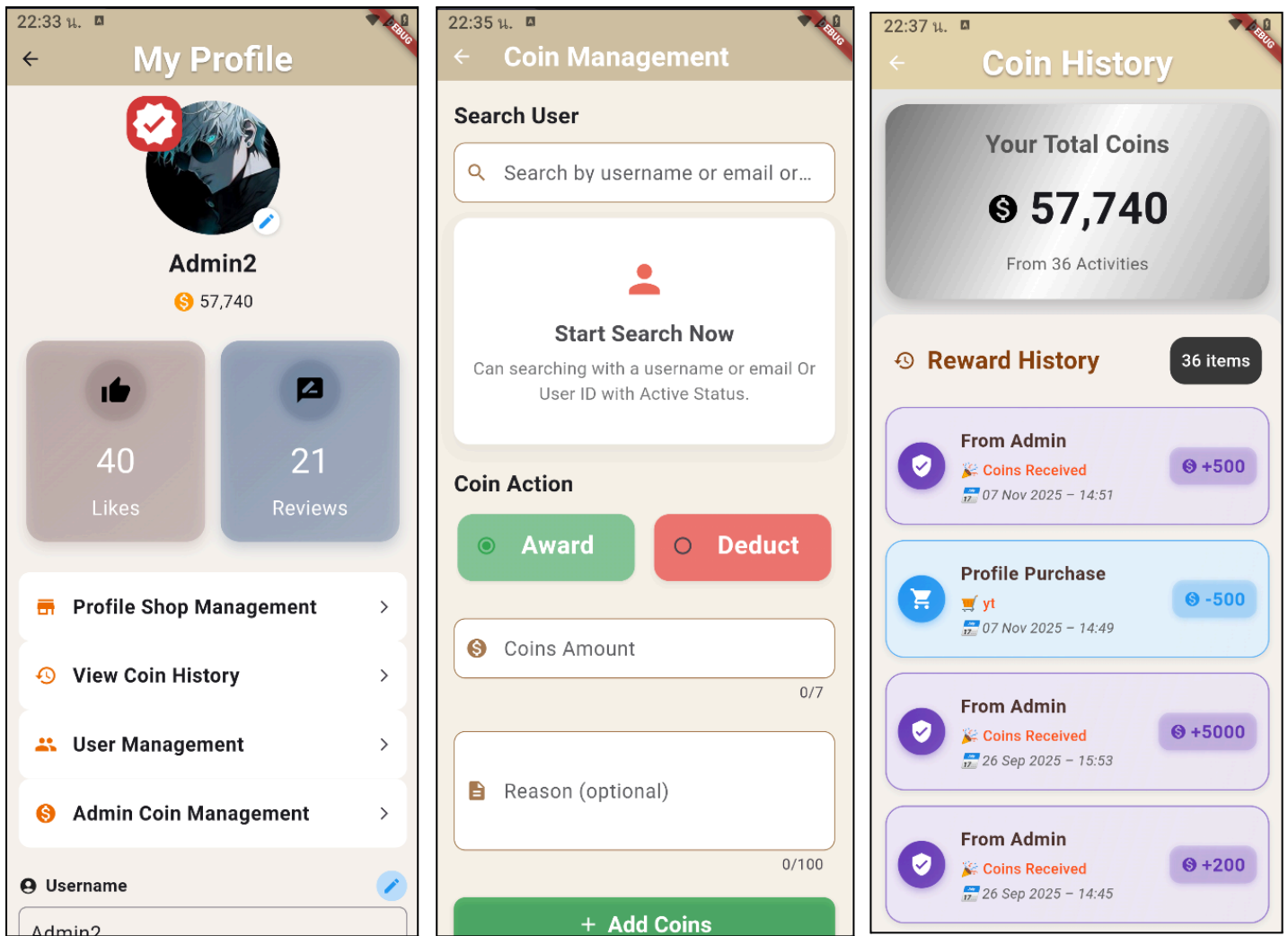


Figure 4.24 The interface of the Profile page

This page displays the user's personal information, including , contact info, and other account-related data.

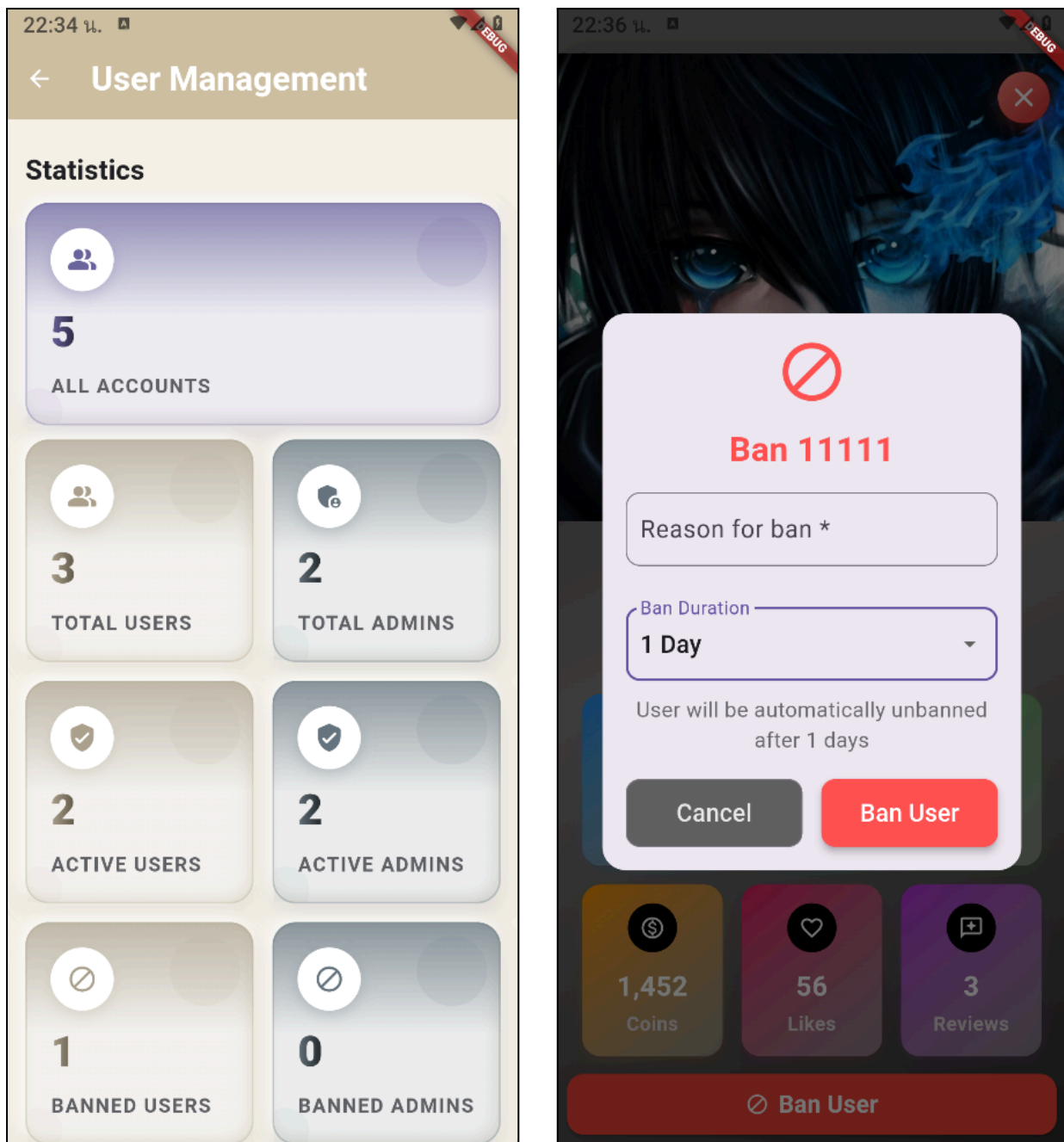


Figure 4.25 The interface of the User management page

This page allows administrators to manage user accounts, including the ability to ban or unban users

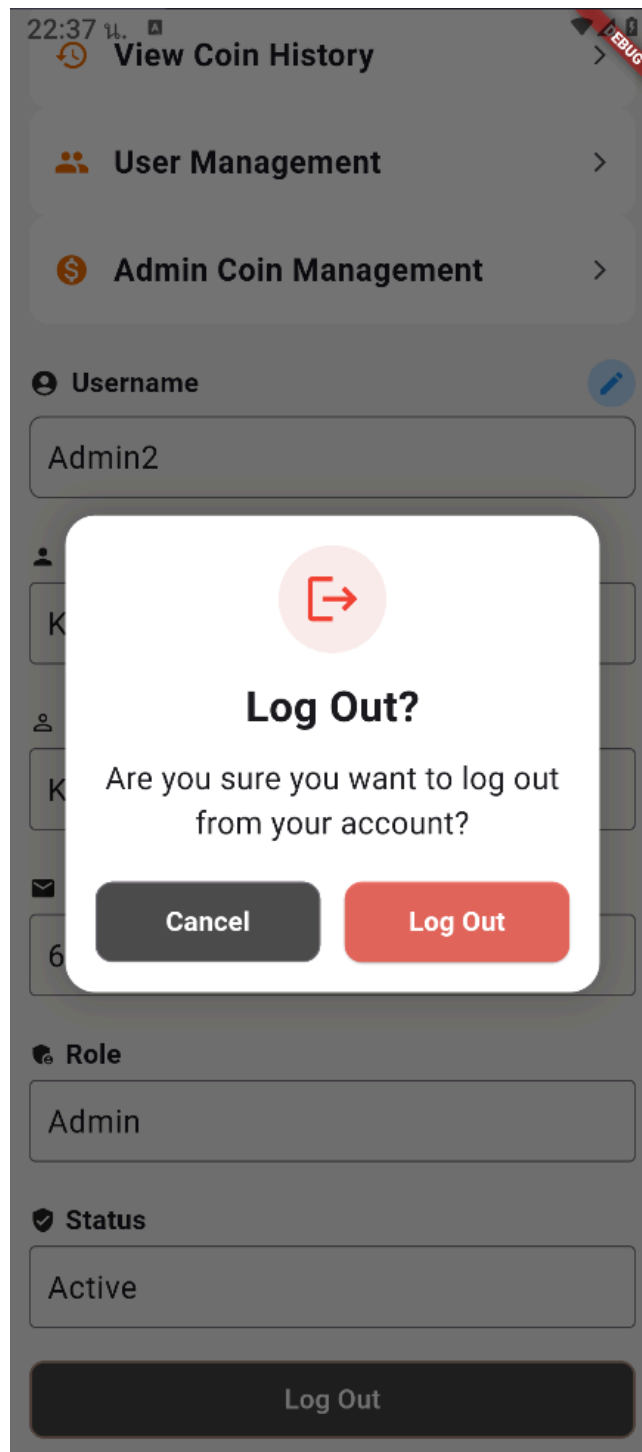


Figure 4.26 The interface of the Logout page

This page allows users to securely log out of the application and end their session.

4.2 Test Case

4.2.1 Guest role

Table 4.1 Guest role Test-case Table

Page	step	Expected Result	Status
Login page	Login in Guest role	Can access to Homepage	Pass
Home page	1. View all available restaurants. 2. Enter the spelling to search.	1. Show all restaurants 2. Show restaurant by search	Pass
Restaurant information page	Write and submit a review	Alert appears: "Please log in to write a review".	Pass
Leaderboard Page	Click Leaderboard page	Shows the top 3 restaurants by rating and top 3 users by review likes each month.	Pass
Ai Assistant Page	Click Ai Assistant page	Alert appears: "Please log in to access this page."	Pass
Threads	Click Threads page	Alert appears: "Please log in to access this page."	Pass
Profile Page	Click Profile page	Alert appears: "Please log in to access this page."	Pass

4.2.2 User role

Table 4.2 User role Test-case Table

Page	Step	Expected Result	Status
Login Page	Using @lamduan.mfu.ac.th or @mfu.ac.th	Successfully logs in and redirects to Homepage	Pass
Home page	1. View all available restaurants. 2. Enter the spelling to search.	1. Show all restaurants 2. Show restaurant by search	Pass
Restaurant information page	Write and submit a review	Review is successfully posted and visible under the restaurant	Pass
Leaderboard Page	Click Leaderboard page	Shows the top 3 restaurants by rating and top 3 users by review likes each month.	Pass
Ai Assistant Page	Ask question using Nexus model	Response returned correctly based on general knowledge outside the app	Pass
Ai Assistant Page	Ask question using Atlas model	Response returned correctly based on general knowledge outside the app	Pass
Threads	Click Threads page	Show information in Threads Page”	Pass
Threads	Enter text in Thread input and submit	If no inappropriate language is detected Thread is posted immediately	Pass
Profile Page	Click Profile page	Show user profile information	Pass
View coin History Page	Click view Coin History button at Profile Page	Coin History page opens and displays all past coin transactions	Pass
Profile Page	Click Log out button	User is logged out and directed to the Login page	Pass

4.2.3 Administration role

Table 4.3 Administration role Test-case Table

Page	Step	Expect Result	Status
Home page	1. View all available restaurants. 2. Enter the spelling to search.	1. Show all restaurants 2. Show restaurant by search	Pass
Home page	Add and edit restaurants	Change the name of the restaurant and add more restaurants.	Pass
Restaurant information page	Add and edit menu	1. Change menu and price. 2. Add more menu.	Pass
Restaurant information page	verifies user reviews by check pending reviews flagged by Perspective API [14], approve or reject	Approved reviews are visible to all users	Pass
Leaderboard Page	Click Leaderboard page	Shows the top 3 restaurants by rating and top 3 users by review likes each month.	Pass
Ai Assistant Page	Ask question using Nexus model	Response returned correctly based on general knowledge outside the app	Pass
Ai Assistant Page	Ask question using Atlas model	Response returned correctly based on general knowledge outside the app	Pass
Profile Page	Click Profile page	Show user profile information	Pass
Threads	Click Threads page	Show information in Threads Page”	Pass
Threads	Enter text in Thread input and submit	If no inappropriate language is detected Thread is posted immediately	Pass
Threads	verifies user threads by check pending threads flagged by Perspective API [14], approve or reject	Approved Threads are visible to all users	Pass
Usermanagement Page	Open User Management page, select a user, tap “Ban” or “Unban”	Banned users cannot access protected pages or features	Pass
Coin management	Open Admin Coin Management page, select user, choose “award”	Coin Added to User Account	Pass

4.3 AI Usage

4.3.1 Perspective API [14] is an AI-powered content moderation tool that detects inappropriate or harmful language. In our application, we use Perspective API to automatically scan user content such as reviews and community threads to identify and filter out offensive, toxic, or inappropriate words. This helps maintain a safe, respectful, and user-friendly environment by ensuring that all posted content meets community standards.

4.3.2 Cohere API [15] is an AI-powered text generation model that we integrated into our application to enhance the chatbot's ability to communicate naturally with users. This API allows the chatbot to understand user input, generate human-like responses, and answer general questions effectively. By leveraging Cohere's language model, our AI chat can interact smoothly with users, provide helpful information, and maintain natural, conversational dialogue. It can also use external knowledge to formulate accurate and context-appropriate responses, improving the overall user experience.

4.3.3 Nexus Model is a custom-built model that we developed to handle specific, predefined questions related to information within our application. Unlike a general AI model, the Nexus Model does not generate open-ended responses. Instead, it answers only the questions and scenarios that we explicitly programmed it to handle. This ensures that it provides accurate, reliable, and consistent answers based on the internal data stored in our database. The Nexus Model is designed to respond strictly to application-related queries such as information about users, restaurant details, or dashboard statistics.

CHAPTER 5

CONCLUSION

5.1 Conclusion

The MFU Food Guide and Review Application was developed with the intention of improving how students and staff at Mae Fah Luang University discover, evaluate, and interact with campus food options. Throughout the project, the team designed key features that support restaurant browsing, review submission, community discussions, and AI-assisted moderation. These functions create a platform that not only helps users make better dining decisions but also encourages meaningful engagement within the MFU community.

The project successfully delivered a complete UX/UI prototype along with a functional system capable of running core features such as restaurant reviews, threaded discussions, and AI moderation for inappropriate content. The integration of multiple AI models including Perspective API [14] , Cohere (Atlas)[15], and the Nexus backend model enhanced system intelligence, safety, and accessibility.

However, several limitations were identified. The system currently relies on free or low-capacity servers, resulting in slower performance and limited scalability. AI moderation also struggles with slang, newly created words, and non-English languages like Thai, which occasionally reduces accuracy. Additionally, multi-language support is still missing, even though it is essential for MFU's diverse user base.

Despite these challenges, the overall outcome demonstrates strong potential. The application provides a solid foundation for a scalable and inclusive food-review platform tailored specifically for MFU. With improvements such as better hosting infrastructure, enhanced AI moderation, and multilingual support, this application can grow into a powerful tool that benefits the entire MFU community.

5.2 Discussion

During the development of the MFU Food Guide and Review Application, we successfully implemented a working system that can run all core features, including restaurant browsing, user reviews, community threads, and AI-assisted moderation. Although the application is fully functional, several practical limitations became clear after deployment.

One of the main challenges is performance. Because the system is currently hosted on a free server with limited resources, response times can be slow, and the application can only run reliably on a local environment. This limitation is a result of budget constraints and highlights the need for future investment in a more stable hosting solution to improve speed, stability, and scalability.

AI also presents challenges. The app uses three models: (1) Perspective API [14] for detecting toxic or inappropriate messages, (2) Cohere Atlas Model [15] for general question answering, and (3) Nexus, the internal assistant for in-app information. While effective, Perspective API struggles with Thai language, slang, and mixed text, causing some inappropriate content to slip through. Atlas may generate overly long answers, while Nexus is limited to simple queries based on app data.

These limitations reflect the realistic constraints of student-developed systems with limited financial resources and technical experience. However, these challenges have established a clear and actionable roadmap for future development. Our immediate plan includes (1) Due to limited hosting resources, the current system is restricted to running on localhost. Future plans include deploying the application on a dedicated cloud service to ensure reliable access, and improved performance, (2) The current Perspective API[14] struggles with Thai, slang, mixed Thai-English, and other regional languages. Future development will explore improved multilingual moderation models or hybrid approaches to ensure more accurate detection of inappropriate content, (3) The current Cohere Atlas Model [15] is effective but still limited in accuracy and response quality. Future development will explore newer, higher-accuracy language models either updated versions from Cohere or alternative providers to improve clarity, relevance, and overall user experience, and (4) Some pages, such as the *Write Review* page, still reflect an older UI style. These interfaces will be redesigned to follow modern design principles, improve accessibility, and create a more seamless user experience.

Despite these challenges, the application demonstrates strong potential. It provides a functional platform tailored specifically to Mae Fah Luang University, and with further improvements, it can become a robust, scalable, and inclusive tool for the entire MFU community.

5.3 Source Code

Link : https://github.com/Scallarr/MFU_FOOD_GUIDE_REVIEW.git

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